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Development and Application of a Methodological Framework for Kinetic Modeling of Flocculation in Kaolinite Suspensions

Freddy Ignacio Rojas Rodriguez

Pontifícia Universidade Católica do Rio de Janeiro
Centro Técnico Científico
Departamento de Engenharia Química e de Materiais

Rio de Janeiro, 26 de Setembro de 2025



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Coorientação: João Batista de Paiva Soares, Ph.D.

Tese apresentada como requisito parcial para obtenção do grau de Doutor pelo Programa de Pós-graduação em Engenharia Química, de Materiais e Processos Ambientais, do Departamento de Engenharia Química e de Materiais da PUC-Rio.

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Abstract

Rojas Rodriguez, Freddy Ignacio; Lemette Teixeira Brandão, Amanda (Advisor); Soares, João B.P. (Co-Advisor). **Development and Application of a Methodological Framework for Kinetic Modeling of Flocculation in Kaolinite Suspensions. Rio de Janeiro, 2025. 221 p. Tese de Doutorado – Departamento de Engenharia Química e de Materiais, Pontifícia Universidade Católica do Rio de Janeiro.**

Treated and purified water suitable for human consumption has long been one of the most valued resources throughout global history. The clarity of water is closely associated with health and the development of civilizations. Behind purification processes lie several stages, including flocculation, which enables the aggregation of colloidal particles for subsequent separation by gravity. Often, the term flocculation is overlooked, as people are more familiar with concepts like drinking water, clarified water, or purified water. However, the reality is that every drop of water reaching a city has undergone a flocculation process, which represents a multi-billion-dollar industry worldwide, considering only urban household consumption, which ranges from 50 to 300 liters per capita. These figures increase by 4 to 30 times in contexts such as entertainment, healthcare, and hospitals. Yet even these numbers are small compared to the water consumption in agriculture, energy production, manufacturing of basic goods, and mining. In an effort to improve and understand flocculation, researchers have studied particle aggregation for over 100 years, using limited mathematical models that involve aggregation rates and later breakage rates. These rates have been calculated theoretically using empirical models, which are constantly refined. Many of these models calculate

aggregation and breakage rates internally, without explicitly reporting their values. Quantitatively, flocculation processes can reduce water turbidity by up to 95%, depending on conditions, the type of flocculant, its concentration, and the type of water. It is important to emphasize that, from a social perspective, there is a clear relationship between public health and the quality of drinking water, as well as the industrial need to reduce operational costs and increase efficiency in processes that depend on water purity. This translates into advantages such as less frequent maintenance, greater process efficiency, longer service life of facilities, and many other benefits. From the standpoint of modern society, there is a growing awareness that water is a vital and irreplaceable resource, whose availability is threatened both by contamination and by rising demand driven by population growth, technological requirements, and economic expansion. This perception has led governments, industries, and research centers to invest large amounts of resources in purification and treatment technologies, whether for freshwater or even seawater. In this context, flocculation emerges as a key process, whose continuous improvement is essential to ensure sources of water of acceptable quality and to meet future needs. For academia, challenges remain in the search for more precise models capable of predicting aggregation and breakage results with lower relative error. Indicators such as turbidity removal efficiency, associated energy costs, and the environmental impact of flocculants are increasingly used as evaluation metrics. Thus, the integration of science, industry, and society becomes fundamental so that accumulated knowledge can be translated into practical and sustainable benefits. In this context, kaolinite particles provide a solid framework for study, as they represent a material that has been extensively investigated and is present in various suspensions. In the case of Mature Fine Tailings (MFT), a byproduct of the petroleum

industry, kaolinite is recognized as one of the main solid components of the mixture. Fundamental studies on the flocculation of these particles are therefore essential, as they provide knowledge that can directly contribute to more efficient treatment and management of these wastes, both MFT and other residues containing this aluminum silicate oxide. This study demonstrates that it is feasible to directly measure the aggregation rates of kaolinite particles, obtaining real values. The results establish models for determining the aggregation rate of kaolinite particles during their birth and death stages, correlating predictive aggregation models with chord length, a particle size parameter reported by the FBRM (Focused Beam Reflectance Measurement) technique. Additionally, models are presented that link aggregation rates to the average molecular weights of polyacrylamide (PAM) polymers used as flocculant agents. Kinetic results also show that the agglomeration rate in a polydisperse kaolinite particle system is independent of particle concentration in the dispersion, resulting in zero-order kinetics, where reaction rates and kinetic constants share the same magnitude

Keywords

Flocculation Rate; Aggregation Rate; Aggregate Breakdown Rate; Monte Carlo.

Resumo

Rojas Rodriguez, Freddy Ignacio; Lemette Teixeira Brandão, Amanda; Soares, João B.P. **Desenvolvimento e Aplicação de uma Metodologia Integrada para a Modelagem Cinética da Floculação em Suspensões de Caulinita**. Rio de Janeiro, 2025. 221 p. Tese de Doutorado – Departamento de Engenharia Química e de Materiais, Pontifícia Universidade Católica do Rio de Janeiro.

A água tratada e purificada para consumo humano tem sido, ao longo da história global, um dos recursos mais valorizados. A clareza da água está diretamente associada à saúde pública e ao desenvolvimento das civilizações. Por trás dos processos de purificação, estão envolvidos diversos estágios, entre eles a floculação, que permite a agregação de partículas coloidais para posterior separação por gravidade. Apesar de sua importância, o termo floculação é frequentemente ignorado no vocabulário cotidiano, onde predominam expressões como “água potável”, “clarificada” ou “purificada”. No entanto, a realidade é que cada gota de água que chega a uma cidade passou por um processo de floculação, o qual representa uma indústria multibilionária em escala global, considerando apenas o consumo doméstico urbano, que varia entre 50 e 300 litros per capita. Esses valores aumentam de 4 a 30 vezes em contextos como entretenimento, saúde e hospitais. Ainda assim, essas cifras são pequenas quando comparadas ao consumo de água nos setores agrícola, energético, industrial e de mineração. Com o objetivo de aperfeiçoar e compreender o processo de floculação, pesquisadores vêm estudando há mais de 100 anos a agregação de partículas por

meio de modelos matemáticos limitados, que envolvem as taxas de agregação e, posteriormente, as taxas de ruptura. Essas taxas têm sido calculadas teoricamente com base em modelos empíricos, que são constantemente ajustados. Muitos desses modelos calculam internamente as taxas de agregação e ruptura, sem reportar explicitamente seus valores. Quantitativamente, os processos de floculação podem reduzir a turbidez da água em até 95%, dependendo das condições, do tipo de floculante, da sua concentração e do tipo de água. É importante destacar que, do ponto de vista social, existe uma relação clara entre a saúde pública e a qualidade da água de consumo, assim como a necessidade industrial de reduzir custos operacionais e aumentar a eficiência dos processos em função da pureza da água. Isso se traduz em vantagens como menor frequência de manutenções, maior eficiência nos processos, maior vida útil das instalações e muitos outros benefícios. Do ponto de vista da sociedade moderna, há uma consciência crescente de que a água é um recurso vital e insubstituível, cuja disponibilidade está ameaçada tanto pela contaminação quanto pelo aumento da demanda decorrente do crescimento populacional, das exigências tecnológicas e da expansão econômica. Essa percepção tem levado governos, indústrias e centros de pesquisa a investir grandes quantidades de recursos em tecnologias de purificação e tratamento, seja de águas doces ou mesmo de águas marinhas. Nesse contexto, a floculação se apresenta como um processo-chave, cuja melhoria contínua é essencial para garantir fontes de água de qualidade aceitável e atender às necessidades futuras. Para a academia, os desafios permanecem na busca por modelos mais precisos, capazes de prever com menor erro relativo os resultados da agregação e ruptura

de partículas. Indicadores como eficiência de remoção de turbidez, custos energéticos associados e impacto ambiental dos flocculantes são cada vez mais utilizados como métricas de avaliação. Assim, a integração entre ciência, indústria e sociedade torna-se fundamental para que o conhecimento acumulado se traduza em benefícios práticos e sustentáveis. Nesse contexto, as partículas de caulinita oferecem um marco de estudo sólido, pois constituem um material amplamente investigado e presente em diversas suspensões. No caso dos Rejeitos Finos Maduros (Mature Fine Tailings – MFT), subproduto da indústria do petróleo, a caulinita é reconhecida como um dos principais componentes sólidos da mistura. Estudos fundamentais sobre a floculação dessas partículas tornam-se essenciais, ao fornecer conhecimentos capazes de contribuir diretamente para o tratamento e manejo mais eficiente desses rejeitos, tanto dos MFT quanto de outros resíduos que contêm esse óxido de silício e alumínio. Este trabalho demonstra que é possível medir diretamente as taxas de agregação de partículas de caulinita, obtendo valores reais. Os resultados permitem estabelecer modelos para a determinação da taxa de agregação nas etapas de formação (birth) e ruptura (death) dos flocos, correlacionando as taxas de agregação com o comprimento da corda, parâmetro de tamanho de partícula reportado pela técnica FBRM (Focused Beam Reflectance Measurement). Além disso, são apresentados modelos que relacionam as taxas de agregação com os pesos moleculares médios dos polímeros de poliacrilamida (PAM) utilizados como agentes flocculantes. Também são incluídos resultados cinéticos, nos quais se considera que a taxa de aglomeração em um sistema polidisperso de partículas de caulinita é independente da concentração de partículas na dispersão, resultando em uma cinética de ordem zero, onde as velocidades e as constantes cinéticas compartilham a mesma magnitude.

Palavras-chave

Taxa Floculação; Taxa de Agregação; Taxa de Quebra de Agregados;
Monte Carlo.

Table of contents

1. Introduction	1
1.1. Objectives	6
1.1.1. General objective	6
1.1.2. Specific Objectives.....	6
2. Bibliographical Review & Theoretical Foundation	7
2.1. Suspension Stability	7
2.2. DLVO Theory	13
2.3. Flocculation Fundamentals	18
2.3.1. Mechanisms of Flocculation.....	19
2.3.2. Classification of Flocculation.....	22
2.3.3. Flocculation in MFT.....	23
2.4. Focused Beam Reflectance Method (FBRM)	25
2.5. Flocculation Models	28
2.5.1. Simulation of the dispersed phases	29
2.5.1.1. Balance Population Modeling	32
2.5.1.2. Monte Carlo method in dispersed systems	44
2.6. Polymerization	49
2.7. Polymers in solution.....	51
2.8. Aqueous-Phase Radical Polymerization of acrylamide.....	53
2.8.1. Free-radical polymerization of acrylamide Free-radical polymerization of acrylamide	54
2.9. Aqueous-Phase RAFT Polymerization of acrylamide	57

2.10.	Reaction kinetic modeling methods	58
2.10.1.	Instantaneous Distributions in Polymerization reactions	58
2.10.2.	Monte Carlo method	62
2.10.3.	Method of moments	69
2.10.4.	Final considerations	71
3.	Kinetic Proposal	73
3.1.	Concept of Modeling	73
3.2.	Data Analysis	74
3.3.	FBRM output data preprocessing	76
3.4.	Kinetic approach to the flocculation process	78
3.5.	Flocculation from a chemical reaction perspective	80
3.6.	Least squares method.	82
3.7.	Sample Dispersion by Particle Size	83
3.8.	Parameter estimation.....	84
3.9.	Final considerations	85
4.	Materials and Methods.....	86
4.1.	Materials	86
4.2.	Experimental Section	86
4.2.1.	Synthesis of oligomer RAFT agent	87
4.2.2.	Synthesis of PAM homopolymer	87
4.2.3.	PAM blends samples.	88
4.2.4.	Polymer Characterization by gel permeation chromatography (GPC)	
	88	
4.2.5.	Flocculation Experiments.....	89

5. Results and discussion	90
5.1. Synthesized PAM homopolymer	90
5.2. Prepared PAM Blend Formulations	91
5.3. Analysis of FBRM Raw Data.....	92
5.4. Kaolinite chord length distribution by FBRM.	93
5.5. Particle conversion proposal.....	95
5.6. Flocculation of kaolinite polymer	98
5.6.1. Dosage of 1000 ppm of flocculant agent.....	98
5.6.2. Concentration of 2500 ppm of flocculant agent.....	100
5.7. What does linear response mean at the beginning of flocculation? ...	104
5.8. Kinetic domain regions.....	106
5.8.1. Kinetic modeling in region I.....	112
5.8.2. Kinetic modeling in region III.....	114
5.8.3. Kinetic modeling in region II.....	117
5.9. Modeling the aggregation rates as a function of the polymer average molecular weight.	119
5.10. Simulation of particle concentration in Regions I and III from aggregation rates	124
5.11. Flocculation Results for PAM Blend Samples.....	126
5.11.1. PAM blends samples Kinetic region I.....	126
5.11.2. PAM blends samples Kinetic region III.....	130
5.11.2.1. Response of PAM blend samples with respect to molecular weight.	
132	
5.12. Breakage rate	136

6. Conclusion	140
7. Future Work	142
8. Bibliography	143

List of figures

Figure 1. Global distribution of recorded tailings dam failures between 1910–2020 (Guimarães et al., 2022 [28]).	10
Figure 2 Schematic representation of the interaction energy between two particles, considering the attractive energy (Hamaker) and the repulsive forces of the electrical double layer, as described by DLVO theory.	14
Figure 3. Dynamics of aggregation and breakage for discretized population balance, adapted from [57].	38
Figure 4. Reactions involved in the four terms of the PBM equation, expressed as symmetric and asymmetric equilibria. The particle accounted for in the balance is the one labeled “2”.	40
Figure 5. Schematic representation of the MC algorithm, adapted from Saliakas V. et al. [98].	47
Figure 6. Illustration of free radical polymerization, where P indicates active chains and D inactive chains.	51
Figure 7. Comparison of Polymer Chain Conformations in Solution.	53
Figure 8. Comparison of Polymerization Mechanisms: Normal Growth vs. Chain Backbiting.	56
Figure 9. Monte Carlo algorithm proposed for simulate polymerization reactions.	64
Figure 10. Scheme of the mathematical modeling stage in the present study.	74

Figure 11. Summary of the steps to follow in this methodology (<i>Lsq</i> refers to the method of least square).....	75
Figure 12 Plot of the tangent intersection method.....	80
Figure 13. Block Diagram of Experimental Methodology.....	87
Figure 14. Dynamic analysis of Mean aggregate size of the flocculation process over FBRM modo macro, unweighted option. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1.	93
Figure 15. Kaolinite Chord length Distribution.....	95
Figure 16 Dynamic analysis of particle conversion X_{Pt} of the flocculation process over FBRM modo macro, unweighted option. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1.	96
Figure 17. Dynamic analysis of individual particle conversion X_{Pi} , tof the flocculation process over FBRM modo macro, unweighted option. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1.	97
Figure 18. 3D graph showing the reference changes from mixing time to real-time reaction, with projections of particle concentration behavior per channel (remember $loglci = 0.018 dchi + 0.009$) at the beginning and end, including slope variation in some channels. The trace of channel 1 is highlighted with a thicker continuous line. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1	102
Figure 19. Linear Behavior of [NP] Over Time Assessed by R^2 Across Channels.....	103
Figure 20. Response of the rate behavior with respect to the logarithm of chord length (lc).	107
Figure 21. Kaolinite Chord length distribution (Cld) at 0 and 20 s in Real-time reaction.....	108

Figure 22 Variation in particle volume fraction ($\Delta Volume Fraction$) between 20 and 0 seconds distributed by particle size.....	109
Figure 23. Superposition of Figure 21 and Figure 22, in which regions I, II, III can be identified.	111
Figure 24. Response of the rate behavior with respect to the logarithm of chord length (lc), including regions I, II, III.....	111
Figure 25 Aggregation rate as a function of the logarithm of chord length l_{ci} for consumed particles (Region I), obtained with PAMs samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.....	113
Figure 26 Logarithm of the aggregation rate as a function of the logarithm of chord length for particles in Region III, obtained with PAM samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.....	116
Figure 27. Simulation of particle concentration variation in Regions II and III using PAMs1, based on the death-birth-aggregation rates during the real-time reaction period.	119
Figure 28. Procedure for developing mathematical models of the slope and intercept of the aggregation rate as a function of the PAM weight-average molecular weight.....	120
Figure 29. Aggregation rate slope as a function of the PAM weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).	121
Figure 30 Aggregation rate intercept as a function of the PAM weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).	122

Figure 31 Simulation of particle concentration variation in Regions I and III using PAMs1, based on the aggregation rates during the real-time reaction period.....	125
Figure 32 Simulation of particle concentration variation in Regions I and III using PAMs1, based on the aggregation rates during the real-time reaction period.....	126
Figure 33. Aggregation rate as a function of the logarithm of chord length l_{ci} for consumed particles (Region I), obtained with PAM blends samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.....	127
Figure 34. Aggregation rate as a function of the logarithm of chord length l_{ci} for consumed particles (Region I), obtained with PAM samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.....	129
Figure 35 Aggregation rate as a function of the logarithm of chord length l_{ci} for consumed particles (Region III), obtained with PAM blends samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.....	131
Figure 36. Aggregation rate slope as a function of the PAMb weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).....	133
Figure 37. Aggregation rate intercept as a function of the PAMb weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).....	134
Figure 38. Aggregation rate slope as a function of the PAM weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).....	135

Figure 39. Aggregation rate intercept as a function of the PAMb weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade). 136

Figure 40. Variation of particle number concentration as a function of mixing time for a dose of 2500 ppm PAMs1, channel 1. 138

Figure 41 Behavior of $-\log Rate$ with $\log lc$ in region I for the agglomeration/breakup phenomenon of kaolinite particles, for PAMs1. 139

List of tables

Table 1. <i>Effects of initiator feeding policies on MWD obtained by Monte Carlo simulation in free radical polymerization</i> [113].....	68
Table 2 Molecular weight averages of the PAM samples used in the flocculation experiments.....	90
Table 3. Molecular weight averages of the PAM blends samples used in the flocculation experiments.	91
Table 4. Linear model for aggregation rate in Region I. Particles evaluated as reactants with PAMs.	113
Table 5. Linear model for aggregation rate in Region _{III}	116
Table 6. Parameters and regression metrics of the linear models describing the intercept and slope of the aggregation rate in Region III	123
Table 7. Linear model for aggregation rate in Region I. Particles evaluated as reactants with PAM blends samples.	127
Table 8. Linear model for aggregation rate in Region I. Particles evaluated with PAM samples.....	129
Table 9 Rate Constants for Kaolinite Flocculation with RAFT-Synthesized PAMs (Zero-Order Kinetics, Units of k: Particles·L ⁻¹ ·s ⁻¹).	130
Table 10. Molecular weight averages of the PAM blends samples used in the flocculation experiments.	132
Table 11. Parameters and regression metrics of the linear models describing the intercept and slope of the aggregation rate in Region III for PAMb.....	134
Table 12. Linear model for aggregation/breakup phenomenon rate in Region I. Particles evaluated with PAMs ₁ samples.	139

List of symbols and abbreviations

$[\]$	Concentration
a	Slope
af, bf	Fitted parameter
agg	Aggregate
$A_{i,j}$	Kinetic terms of aggregation
AgNPs	Silver nanoparticles
AM	Acrylamide
AOS	Amount of shots
ap, bp	Reaction order with respect to monomer for the transfer and propagation reaction
AP	Amylopectin
APS	Ammonium persulphate
$Ap_{v,i}(Dn, t)$	Volume probability diameter density function
b	Intercept
BC	Before Christ
Bf	Birth functions
C8	8-carbon chain length
C10	10-carbon chain length
C12	12-carbon chain length
$CaCl_2$	Calcium chloride
$CaCO_3$	Calcium carbonate
$^{\circ}C$	Graus Celsius
CCD	Comonomer composition distribution

Cds	Numbers and volumes of daughter and satellite drops
Cl _d	Chord length distribution
CLD1	Distributions of chain length
cP	Centipoise
C_p	Parker's constants
C_{tr}	Chain Transfer Constant to Monomer
d	Particle size
d_1	0.27 μm
$\bar{D}$	Dispersity
Da	Dalton
dch_i	Discrete channels
d_F	Mass fractal dimension of aggregates ($1.0 \leq d_F \leq 3.0$)
Df	Death functions
DLS	Dynamic Light Scattering
DLVO	Derjaguin–Landau–Verwey–Overbeek
DMSO	Dimethyl sulfoxide
D_n	Dead polymer chains
Δn	Range of diameter
d_s	0.65 μm
DSD	Droplet size distribution
D_u	Distance
$\frac{du}{dy}$	Velocity gradient of the suspension
D_{XX}	Cumulative distribution percentiles
e	Residual
ECSR	Environmental Corporate Social Responsibility

$E_{Hydrophobic}$	Hydrophobic energy
EPA	Environmental Protection Agency
EPS	Extracellular polymeric substances
$E_{Repulsion}$	Repulsion energy
$E_{Steric\ forces}$	Steric forces energy
E_{Total}	Total energy
$E_{attraction}$	Attraction energy
f	Probability of detecting a particle
fa	Environmental factor
FBRM	Focused Beam Reflectance Measurement
FeNPs	Iron nanoparticles
fg	Initiator Efficiency
FG	Functional group
f_n	Molar instantaneous chain length distribution
-g-	-graph-
G	Average turbulent shear rate (velocity gradient)
GA	Genetic Algorithm
$gf(vp_i)$	Breakage rates
GPC	Gel permeation chromatography
hd	Time dependence of the aggregate size
$^1\text{H NMR}$	Proton nuclear magnetic resonance
HPAM	Hydrolyzed polyacrylamide
I	Initiator
i and j	Index o channel
k	Boltzmann's constant

Ka	Quadratic aggregation constant
k_b	Model parameter
k_{bb}	Backbiting rate constant
k_d	Mass fractal dimension of aggregates
kc	Coalescence rates
k^{exp}	Macroscopic or experimental origin rate
$kf(vp_i, vp_j)$	Coalescence rates
km	kilometer
k^{MC}	Microscopic rate
k_p	Propagation constant
$k_{p,tert}$	Tertiary propagation rate constant
k_t	Termination constant
$k_{t,ss}$	Self-termination rate constant for secondary polymer radicals
$k_{t,st}$	Rate constant for the termination of secondary and mid-chain radicals
$k_{t,tt}$	Self-termination rate constant of mid-chain radical
k_{tr}	Chain transfer constant to monomer
kWh	Kilowatt-hour
l	Particle of size $l = i + j$
L	Liter
lc	Midpoint chord lengths
LCBD	Long chain branching distribution
LD	Laser Diffraction
$\overline{LM}_0$	Unweighted mean
$\overline{LM}_1$	Length-weighted mean
$\overline{LM}_2$	Square-weighted mean

$\overline{LM}_3$	Cube-weighted mean
$\overline{LM}_j$	Size mean
LP	Linear Program
LPCD	Liters per capita per day
Lsq	Least squares method
m	Number of molecules of reactants
<i>MAPE</i>	Mean Absolute Percentage Error
MC	Monte Carlo
MCR	Mid-chain-radicals
MDS	Molecular dynamics simulation
MFT	Mature Fine Tailings
m_k	Mass of kaolinite in the suspension
Mo	Moment order
MW	Molecular weight
MWD	Molecular weight distribution
n	Number concentration of particle
nr	Number of moles of living polymer
N_A	Avogadro number.
NaFS	Hydroxymethanesulfinic acid monosodium salt dihydrate
NaOH	Sodium hydroxide
N_c	Number of unique combinations between reactant molecules inside the control volume
n_i	Particle number per unit volume
$n_i(Dn, t)$	Number diameter density function
n_j	Particle number per unit volume
NLP	Non-Linear Program
NMR	Nuclear magnetic resonance

NMR	Nuclear magnetic resonance
NOS	Number of shots
$[NP_i]$	Particle number concentration
	kinetic method
Ns	Number of sample packages (droplets/particles)
ODE	Ordinary differential equations
P1	Propagating radical of unit length 1
Pab	Probability aggregation/breakage phenomenon
PAM	Polyacrylamide
PAMb	Binary blends polyacrylamide
PAMs	Synthesized polyacrylamide
PAMt	Ternary blends polyacrylamide
PB	Population balance
PBM	Population balance model
PEOMA	Poly(ethylene oxide methyl ether methacrylate)
pH	Pondus hydrogenii
P_n, P_m	Chain protagonist
PSD	Particle size distribution
PSO	Particle Swarm Optimization
PVBTMAC	Poly(vinyl benzyl trimethylammonium chloride)
PVP	Polyvinylpyrrolidone
q_1, q_2	Parameters
$Q_{i,j}$	Aggregation kinetic kernel
Q_{i+j}	Kinetic terms of breaking
Q_n, Q_m	Mid-chain Radical

QSSA	Quasi-steady state approximation
r	Rate
R	Rate (s^{-1}) of coalescence or breakage
R^2	Coefficient of determination
Rab	Total rate of coalescence or breakage event
rad	Radical
RAFT	Reversible Addition-Fragmentation Chain Transfer
Re	Reynolds number
r_n	Number average molar weight
R_p	Propagation rate
R_{sub}^{MC}	Monte Carlo reaction rate
R_{tr}	Transfer to monomer rate
r_w	Weight average molecular weight
s_0	Deviation
s^2	Variance
s	Solid
S	Breakage kinetic kernel
S_1	0.22
S_2	1.73
SEM	Scanning Electron Microscopy
SPR	Secondary propagating radicals
SSA	Stochastic simulation algorithm
sus	Suspension
t	Time
T	Absolute temperature
TOS	Times of shots
$\bar{u}(D_u)^2$	Relative velocity

U.S.	United State
V_b	Volume of the suspension batch
VC	Control volume
Ve	Package viscoelasticity
ν_f	Kinematic viscosity of the fluid
V_f	Volume fraction
ν_g	Growth rate
ν_p	Particle volume
w	Water
ω_b	Breakage frequencies
ω_c	Coalescence frequencies
We	Weber number
w_n	Weight instantaneous chain length distribution
wt %	Percentage by weight
x	Independent variable
x_p	Parker's constants
XP	Conversion particles
XX	Percentage mass or volume the particle
y	Dependent variable
Y_0	Total number of polymer moles
Z	3-dimensional spatial coordinate system
α_c	Model parameter
α_{max}	Model parameter 0.0049
α_{min}	Model parameter 0.0016
$\alpha_{s,s}$	Fraction of termination by disproportionation

$\alpha_{s,t}$	Fraction of cross-termination by disproportionation
$\alpha_{t,t}$	Fraction of mid-chain radical termination by disproportionation
β	Aggregation rate
δ	Mean square velocity constant of the fluid in turbulent flow
Δt	Time interval
$\Delta Volume\ fraction$	Variation of volume fraction
Γ	Breakage distribution function for aggregates
ε_s	Average energy dissipation rate for the dispersion
ε	Average energy dissipation rate
θ_{max}	Maxima surface coverage in the aggregate
λ_{Mo}	living polymer chains
λ_b	Breakage efficiency
λ_c	Recoalescence efficiency
η	Kolmogorov microscale of turbulence
θ	Effective surface coverage
μ	Viscosity
μm	Micrometer
ρ	Density
μ_{Mo}	Dead polymer chains
μ_c	Viscosity of the continuous phase
ρ_c	Density of the continuous phase
ρ_k	Density of kaolinite
ρ_{prs}	Pearson correlation coefficient

ζ	Moment
σ	Death kinetic kernel
σt	Interfacial tension between the dispersed and aqueous phases
τ	Ratio of transfer to propagation rates
χ	Fractional monomer conversion
ϕ	Volume fraction of primary particles in suspension
ϕ_{max}	0.65
ϱ	Capture efficiency of collisions
$\Omega(D_u)$	Ratio of required to available energy

1. Introduction

Flocculation is one of the main tools in the field of water and wastewater treatment. Flocculation contributes to water sanitation by linking small particles together to form larger aggregates, known as flocs[1]. These flocs disrupt the forces that maintain the suspension's stability, leading to destabilization and, consequently, separation through gravitational sedimentation [2]. Historically, the use of plant extracts to clarify water has been documented in India (2000 BC), as well as the clarification of wine using egg whites in Ancient Greece [3]. In the context of wastewater treatment, flocculation is gaining increasing importance, especially considering that at least 80% of global wastewater is discharged into water bodies without prior treatment [4]. This practice leads to significant ecological imbalances, as reported by the U.S. Environmental Protection Agency (EPA), which states that two-thirds of U.S. coastal waters and estuaries are moderately to severely eutrophic outcome to unnecessary nutrient contamination [4].

The terms coagulation-flocculation are commonly found in traditional water treatment contexts, as both processes are responsible for the separation of micrometric and submicrometric particles in solid/liquid dispersions. However, each process exhibits distinct characteristics. In large-scale processing systems, coagulation is a rapidly executed stage (typically in about 10 seconds), involving the destabilization of solid particles through surface charge neutralization, which leads to separation and/or the formation of coagulates, these new particles that are denser and more stable compared to flocs. In contrast, flocculation is a slower process (generally between 20 and 45 minutes), which promotes particle aggregation through bridging mechanisms formed by the flocculant agent, resulting in larger and less

dense structures known as flocs. These flocs overcome gravitational barriers and are subsequently separated by sedimentation [1][2][5].

Despite technological and infrastructure advances, access to safe drinking water is still not guaranteed for 100% of the global population [6]. In regions where water supply is available, it is evident that water consumption is closely linked to population growth, industrial development, and agriculture. According to data focused on domestic water use, consumption levels can vary significantly. For instance, in the United States, the average consumption reaches 560 liters per capita per day (LPCD), as reported by the Environmental Protection Agency (EPA) [7]. In Mediterranean countries, consumption ranges between 175 and 330 LPCD [8], while in the city of Rio de Janeiro, Brazil, the estimated average is 150 LPCD [9]. Globally, the World Health Organization (WHO, 2020) considers 100 LPCD to be the optimal threshold for ensuring adequate access to water, based solely on basic household needs [10].

From an industrial and commercial perspective, water consumption varies significantly depending on the sector. In the energy sector, between 15 and 100 liters are required per kilowatt-hour (kWh) produced [11]. In the hospitality sector, reported usage ranges from 300 to 1200 liters per person per day (LPCD) [12]. In healthcare services, the average reaches 1364 liters per hospital bed per day [13][14]. Agriculture accounts for approximately 8.4×10^{15} liters annually [15], making it the largest consumer of water globally. In the computational services sector, water consumption is estimated at approximately 1/60 of a liter per digital response generated [16][17]. Mining operations show wide variability, with requirements ranging from 87 to 716,000 liters per kilogram of mineral extracted, depending on the type of resource and the process used [15][18]. Meanwhile, the production of natural fiber textiles, beginning with plant irrigation and continuing through to final

fiber processing, demands between 2600 and 22,600 liters per kilogram of fiber [15].

In practice, estimating the proportion of aquatic environments contaminated by wastewater discharge is complex, due to the varying methodological approaches adopted by researchers and the tendency to focus reports on specific water bodies. For instance, some studies emphasize contaminant content, such as in the case of the Potomac River, a major water source on the U.S. East Coast, where the presence of 51 potentially hazardous contaminants for aquatic ecosystems has been reported. [19]. Other articles focus on countries like Mexico, where it is estimated that approximately 73% of water bodies show signs of contamination [20]. Furthermore, a meta-analysis of 121 studies revealed the presence of microplastics in the effluents of wastewater treatment plants, with concentrations ranging from 0.009 to 3862 particles per liter [21]. Although these cases may appear isolated or anecdotal, the reality is that the volume of scientific data is too vast to report comprehensively, and the frequency of new findings continues to rise, driven by factors such as population growth, climate change, and the progressive accumulation of pollutants in aquatic ecosystems.

In the field of wastewater management within the mining industry, one of the most widespread strategies is the construction of containment dams designed to store tailings, a heterogeneous mixture of water, chemical compounds, solids of various sizes, and, in some cases, organic matter [22][23][24]. These structures are engineered to maintain the stability of the waste over long periods, potentially lasting for centuries [25]. In Brazil, a national inventory conducted in 2022 identified 860 dams dedicated exclusively to the containment of tailings from mining activities [25]. However, these infrastructures pose significant environmental and safety risks. Among the most notable are

the unintended attraction of wildlife that mistake these artificial bodies for natural water sources [26], overflow during periods of heavy rainfall [24][27][28], and the release of solid particles into the atmosphere as the waste dries out [29].

One of the largest sources of industrial tailings is probably the fertilizer sector, particularly through the production of phosphogypsum. For every ton of fertilizer produced, at least four tons of phosphogypsum are generated as waste. These residues are stored in containment ponds, which are estimated to hold between 100 and 280 million tons of phosphate tailings per year [22]. Another significant source of tailings is reported in Canada, where oil extraction using Clark's hot water process produces between 2 and 3 million barrels of liquid tailings per day [22]. These residues, which have existed for over sixty years, have lost part of their water content through evaporation, yet they remain stable as a muddy dispersion. They are currently referred to as Mature Fine Tailings (MFT), and their accumulated volume is estimated at approximately 4.8 billion barrels, distributed across a shared mining area of 600 square kilometers [23].

Conditioned by regulatory evolution, concerns about water scarcity and population growth [30], Environmental Corporate Social Responsibility (ECSR) [31] is an initiative from the productive sectors that has evolved with the goal of reducing pollutant emissions and fostering positive impacts that promote ecological regeneration [32]. In this process, the combined efforts of governments and industrial sectors have been key, generating stricter regulatory frameworks that have transformed waste management practices. Although many ECSR initiatives began as voluntary actions beyond corporate marketing, countries in Europe, Asia, Africa, Australia, and the United States have

progressively strengthened their legislation to make these environmental commitments mandatory [33].

This thesis presents a kinetic approach aimed at enabling the direct quantification of flocculation rates, using kaolin particles dispersed in deionized water and polyacrylamide (PAM) as the flocculating agent. The flocculation behavior over time is evaluated based on polymer concentration and molecular weight, using the Focused Beam Reflectance Measurement (FBRM) technique. The proposed methodology is designed to be applicable to a wide range of solid dispersions and flocculant types. Due to its innovative nature, this proposal is conceived as an initial stage in a long-term research effort focused on obtaining quantitative information about the interactions between flocculants and dispersed solids, with the goal of optimizing operational parameters such as dosage, molecular weight, temperature, and other variables that influence the stability of current solid/liquid systems.

1.1. Objectives

1.1.1. General objective

To develop a methodological framework for the experimental and numerical analysis of the flocculation process in kaolinite dispersions in aqueous media, using polyacrylamide (PAM) samples with different molecular weights as flocculants. The goal is to build dynamic predictive models based on particle size distribution, employing the FBRM technique and applying a kinetic approach.

1.1.2. Specific Objectives

To evaluate the influence of polyacrylamide (PAM) polymers with varying molecular weights on the aggregation and breakage rate of kaolinite particles in suspension, using the FBRM technique to measure chord length as an indicator of particle size.

To develop models to estimate the aggregation and breakage rate of kaolinite particles in order to explain the behavior of the flocculation process.

To determine the fundamental kinetic parameters, such as reaction order and kinetic constant, that enable a chemical-based description of the flocculation process of kaolinite particles, using experimental data obtained through direct characterization techniques.

2. Bibliographical Review & Theoretical Foundation

This chapter reviews the literature with the aim of establishing fundamental concepts and describing the current state of knowledge regarding the key aspects addressed in the proposed thesis. The first section provides an overview of suspension stability and how it can be disrupted through the use of polymers. The Derjaguin–Landau–Verwey–Overbeek (DLVO) theory is discussed, explaining how microscopic forces support the stability of colloidal systems, along with its evolution into extended DLVO theories. Next, the phenomenon of flocculation is examined, along with the currently accepted mechanisms, including specific examples related to Mature Fine Tailings (MFT). In particular, the Focused Beam Reflectance Measurement (FBRM) technique is studied as a tool for measuring the size of solid particles in suspension. The modeling of flocculation processes is addressed through population balance approaches and stochastic simulation methods, such as the Monte Carlo algorithm. Additionally, concepts related to polymer synthesis and its modeling are introduced. Finally, kinetic studies are presented using mathematical models, which describe the temporal evolution of flocculated systems.

2.1. Suspension Stability

Almurtaji et al., (2020) [34], emphasize the importance of colloidal system stability when designing a nanofluid, considering three key parameters separately:

Aggregation stability, which describes the ability or inability of particles to grow in size and undergo gravitational separation.

Chemical stability, referring to the reactivity between solids and/or between solids and the surrounding environment.

Brownian motion stability, which relates to the influence of random thermal motion on particle dispersion.

All these parameters are crucial for accurately reproducing and maintaining stable colloidal systems.

Akhatova et al., (2023) [35], point out that zeta potential is a fundamental indicator of electrostatic stability in systems of nanoparticles suspended in aqueous media. This parameter is directly linked to the surface charge of the particles, which varies depending on the pH and the surface chemistry of the system. Changes in this charge generate electrostatic repulsion forces between particles, preventing aggregation and promoting a stable colloidal dispersion. In this context, the case of iron nanoparticles (FeNPs) is studied. These particles exhibit high zeta potential values in acidic media (pH 3 to 5), which results in a reduction in particle size and greater colloidal stability. In contrast, under alkaline conditions, the zeta potential decreases significantly, leading to an increase in particle size and a tendency toward aggregation. A similar behavior is described for silver nanoparticles (AgNPs), which show their highest zeta potential values in slightly acidic to alkaline conditions (pH 5 to 9), also favoring stable dispersion within that pH range. These findings outline how colloidal stability is strongly influenced by both the pH of the medium and the chemical nature of the particles studied. When evaluating the stability of nanoparticle systems, it is essential to consider characteristics such as particle size and shape, as they play a decisive role in maintaining stable colloidal suspensions.

Yuan et al., (2024) [36], studied the modification of kaolinite hydrophobicity using triethoxyvinylsilane and 3-mercaptopropionic acid, applied through calcination reactions to fix the modifying agents. They demonstrated that reducing the hydrophilicity of kaolinite particles leads to stable suspensions with average particle sizes of 1100 nm in polar solvents such as ethanol, dioxane, ethyl acetate, and N,N-dimethylformamide. In contrast, in nonpolar solvents like cyclohexane and petroleum ether, stability decreases, with average particle sizes reaching up to 5000 nm. These findings highlight the potential stability of kaolinite particles depending on the chemical environment and underscore the influence of hydrophobic forces when evaluating the stability of colloidal dispersions.

The bibliographic study by Bahniuk et al., (2022) [22], reveals real cases of wastewater storage generated by the mining industry or its processing activities. The authors mention an estimated 100 to 280 million tons of phosphate tailings per year, as a consequence of fertilizer production based on phosphogypsum.

Leão and Santiago (2022) highlight that Brazil stands out for its strong presence in the mining sector. Although there are no precise records of the volumes of wastewater generated by this activity, an inventory conducted in 2020 identified 860 tailings dams designed to contain wet waste from mining operations. The study also emphasizes the potential risks associated with storing waste in these structures, including:

- Wildlife may mistake the dams for natural water sources [26]
- Overflows during rainy seasons, posing significant risks to nearby communities and the environment[24][27][28]

- The dispersion of solid particles into the atmosphere as the tailings dry out, contributing to air pollution [29]

Guimarães et al., (2022) [28], using a historical approach, analyzed dam failures reported between 1910 and 2020, identifying a total of sixty events in South America, specifically shared between Argentina and Brazil. The study also recorded forty-six failures in Asia, sixty-two in Europe, ninety-three in the United States, fifteen in Africa, and nine in Australia, demonstrating that the risk associated with tailings dams is truly global, regardless of how advanced a country's science and technology may be. Figure 1 illustrates the geographical distribution of recorded of 239 dam failures across six continents, highlighting the global nature of the issue.

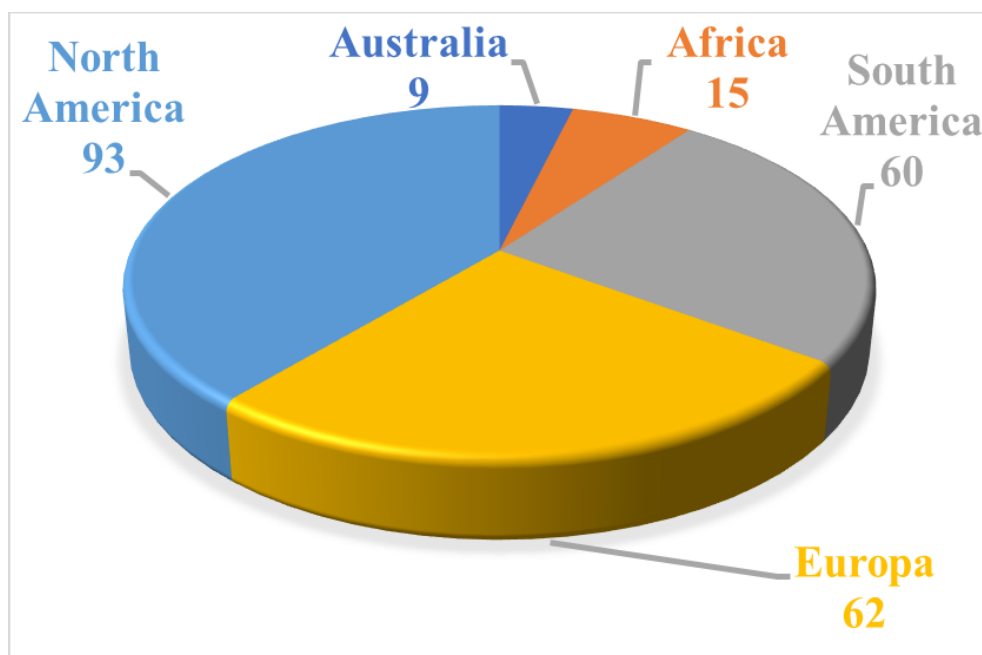


Figure 1. Global distribution of recorded tailings dam failures between 1910–2020 (Guimarães et al., 2022 [28]).

MFT is described as a viscous waste, consisting of a stable solid in liquid dispersion, with approximately one-third solids made up of

particles smaller than 100 micrometers, which remain stable over extended periods. Zhou, j. (2021) [37], agree that a real-world example of a major environmental challenge is the presence of MFT, which currently represent one of the most pressing environmental issues in the Alberta region of Canada [38][39]. Canada is one of the world's leading oil producers, characterized by its predominantly heavy crude, and benefits strategically from its border with the United States and strong trade relations. The oil extraction industry in Alberta is particularly notable for its open-pit mining operations, where subsurface sand deposits containing bitumen are processed to extract heavy oil. This process generates MFT as a by-product in vast quantities [23][40][41].

The viscosity of heavy crude oil serves as one of the driving factors that prompt commercial operators to adopt open pit mining as their preferred method of oil production. To understand the challenges of producing oil through conventional drilling methods in the Alberta region, one can take the viscosity of the Athabasca deposit as a reference, reported at approximately 14,200 cP at 42.7 °C [42]. This value indicates that the oil is about 14,200 times more viscous than water, highlighting its extreme resistance to flow. Moreover, the fact that this measurement is taken at 42.7 °C suggests that, at ambient temperatures, the material would behave even more rigidly, with properties closer to those of a solid. Added to this are the climatic conditions of the region, where temperatures range between -21 °C and 16 °C throughout the year [43], further reinforcing the difficulty of transporting the crude through conventional wellbore production channels. In this context, oil companies prefer open pit mining as the most reliable extraction method, since it allows direct access to the so-called "black gold" under conditions where drilling is impractical.

MFT by-product production maintains a historical estimate of 1.5 barrels of MFT for each barrel of heavy crude (1 barrel = 159 L or 42 US gallons) [23]. MFT can generally be described as stable dispersions of solid particles (30 – 40 wt %) of various minerals and clays in water chemistry (60 – 70 wt %), and bitumen traces (2 – 5 wt %) [23] with stability that can reach decades. For more than 5 decades, the extraction of bitumen by open pit mining in oil sand deposits has been chosen as the preferred method of commercial operators in the Alberta–Canada region [23][37]. A historical record in the growth of the oil sands tailing ponds begins in 2013 with 77 km², in 2017 the oil sands tailing ponds increase tripled to approximately 220 km² [37]. All this progress in the heavy oil extraction industry in oil sand in the Alberta–Canada region was reported in 2020 with mines that cover an area of about 600 km², equivalent to half the size of the city of Rio de Janeiro[23][39][44].

The methodology for the exploitation of heavy crude from oil sands involves several stages. The first stage begins with the open pit extraction of oil sands using large excavators. The extracted material is transported to a processing plant in the second stage. In the separation stage within the plant, oil sand is mixed with hot aqueous NaOH solutions together with an additional portion of steam, reaching temperatures of around 85 to 95 °C. This last stage is known as Clark's hot water extraction process (Clark and Pasternack, 1932; Clark, 1939) [40]. The Clark's hot water process has an oil recovery efficiency of close to 96 %, this method is based on separating the heavy oil from the bitumen–water–sand mixture by differences in density. In Clark's patented process, the viscosity of the bitumen decreases as the temperature increases, and bubbles are created within the oil, placing the black gold at the top of the mixing tank. Most of the sand solid

settles to the bottom with a high-water content [37]. It is estimated that three barrels of the aqueous NaOH solution are used from 8.0 to 8.5 pH for each barrel of oil generated. The tailings produced contain the aqueous phase together with the solid part (55 wt % minerals and clays) and are deposited in dikes where tailings are pumped. In the dike, about 45 % by weight of the tailing's original mass (corresponding to the solid part) is separated by gravity and deposit force. After sedimentation by gravity, the remains are solid content of 9 to 10 % by weight of the original tailings, which are particles with mean sizes of 44 μm [40]. Finally, the supernatant water from the tailings in the dikes is decanted for approximately a period of 5 years, after which the portion of water decreases, transforming the dike into a highly stable muddy deposit and described as MFT [23].

2.2. DLVO Theory

Since the 1950s, the DLVO model (Derjaguin–Landau–Verwey–Overbeek) has been widely used to explain the equilibrium behavior of colloidal dispersions in liquid media. This theory establishes a relationship between the interaction energy of two particles and the distance separating them, allowing for the prediction of dispersion stability through a balance of attractive and repulsive forces[45] (Figure 2).

The DLVO model is based on two main types of forces:

- Van der Waals attractive forces, which tend to bring particles together.

- Electrostatic repulsive forces, generated by surface charges and the structure of the electrical double layer formed in aqueous media.

The energy balance involved in the interaction between particles is represented in equation 1, in accordance with the principles of DLVO theory[46].

$$E_{Total} = E_{attraction} + E_{Repulsion} \quad (1)$$

This theoretical framework has been particularly useful in describing systems such as clay suspensions, pigments, and nanoparticles, where non-contact interactions determine particle aggregation or dispersion. Understanding these forces enables the optimization of industrial processes such as paint formulation, water clarification, and the design of functional materials[45][47].

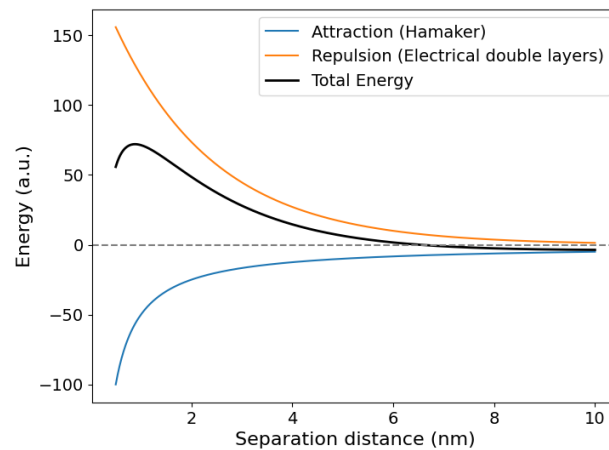


Figure 2 Schematic representation of the interaction energy between two particles, considering the attractive energy (Hamaker) and the repulsive forces of the electrical double layer, as described by DLVO theory.

Bandera et al., (2019) [47] developed a molecular dynamics model to simulate clay particles, applying the classical DLVO theory to describe short-range non-contact interactions. Their findings indicated that this approach offers a more accurate representation of the interaction energy between solid particles, when compared to the model which uses molecular mechanics simulations based on the Gay-Berne potential to characterize particle interactions.

Zhu (2022) [48], describe the rheological behavior of kaolinite colloids through the framework of DLVO theory, emphasizing how chemical factors affect system stability and fluid viscosity. Specifically, an increase in cation concentration in solution or the use of higher-valence cations leads to a reduction in the thickness of the electric double layer, while van der Waals attractive forces remain constant. This shift promotes particle adhesion, resulting in a denser dispersion and a decrease in interparticle separation distance, which in turn causes an increase in the viscous resistance of the fluid.

Although the DLVO model has proven effective in many contexts, it also has limitations. It does not account for effects such as surface hydration, hydrophobic, steric forces, or specific ion interactions, which may be relevant in more complex systems. Therefore, it is currently complemented by extended models that incorporate these additional variables. Equation 2 provides an example of how additional terms accounting for complementary energies can be incorporated into extended DLVO theories.

$$E_{Total} = E_{Attraction} + E_{Repulsion} + E_{Hydrophobic} + E_{Steric\ forces} \quad (2)$$

Zou et al., (2018) [49], investigated the aggregation behavior of mixed coal/kaolinite particle systems in the presence of polyacrylamide (PAM), drawing on the principles of extended DLVO theory. Their findings revealed attractive interactions among coal particles, promoting the formation of stable aggregates. However, in coal/kaolinite systems, the interactions were found to be predominantly repulsive, suggesting lower affinity between these solids in the PAM medium and a potential inhibition of aggregation.

Sögaard et al., (2018) [50] point out that DLVO theory can be a useful tool for predicting the destabilization of silica nanoparticles suspended in aqueous media at pH levels above 8. Under these conditions, the dominance of repulsive forces described by the Poisson–Boltzmann theory, which considers the surface potential of charged particles, contributes to maintaining dispersion stability. In practice, the zeta potential is commonly used as a representative parameter in DLVO-based calculations. On the other hand, the attractive forces described by Hamaker theory are responsible for the tendency of particles to aggregate. Therefore, the stability of silica nanoparticles at high pH depends on which type of force predominates. At lower pH levels, however, DLVO theory alone is insufficient to predict particle stability. This suggests the occurrence of non-DVLO forces, such as a reorganization in the structure of water molecules surrounding the solid particles, which may alter interfacial interactions. Another possible explanation for the observed stability of silica nanoparticles at low pH is the formation of a poly(silicic acid) gel layer coating the surface of the solids, which enhances their stability by providing a protective and steric barrier.

Guo et al., (2014) [51], correlate the mechanical properties of nanoparticles with the energetic interactions between particles and

their interparticle separation distance. The article discusses how both DLVO forces, particularly those associated with the Stern layer, and non-DLVO forces, such as solvation, hydrophobic interactions, ion-correlation, and steric effects, can have notable consequences on the mechanical properties and stability of colloidal systems.

Jin et al., (2023) [52], report that non-DLVO interaction models provide a better fit for describing the flotation and flocculation performance of cassiterite in the presence of alkyl hydroxamic acids with varying carbon chain lengths. Their results show that the minimum concentration required to induce flocculation decreases as the carbon chain length increases, specifically with C8, C10, and C12 compounds. This trend is ascribed to the enhanced ability of longer-chain collectors to induce hydrophobic aggregation, thereby lowering the energy barrier between cassiterite disperse.

As with any scientific theory, the DLVO model has undergone various updates aimed at enhancing its ability to describe real-world behaviors. In this regard, extended versions of the DLVO theory have emerged, incorporating not only short-range interactions such as van der Waals forces and electrostatic double-layer repulsion, but also hydrophobic interactions, steric effects, and depletion forces[46]. The combined influence of these interactions allows for a more accurate adjustment of the model to explain how the potential energy between particles varies with interparticle separation distance [47].

Kolb et al., (2022)[46], modeled the stability of graphite nanoparticles in aqueous dispersions containing polyvinylpyrrolidone (PVP), using an extended DLVO theory. The study showed that incorporating additional interactions, such as steric and depletion forces, enabled a more accurate representation of the system's behavior. The extended model yielded results that were more

consistent with empirically simulated sedimentation data than those predicted by the classical DLVO model. It is important to note that the dispersion was empirically constructed, and the findings were not validated through direct experimental measurements.

Shi et al., (2022) [53], investigated the interaction energy of a quartz–kaolinite mineral system across different pH levels using a computational approach based on the extended DLVO theory. Their findings showed that the silica-rich tetrahedral face of both kaolinite and quartz is highly responsive in terms of surface interaction. Additionally, a specific attraction was identified between the alumina surface of kaolinite and the octahedral faces of quartz at pH 6, indicating an increased likelihood of particle aggregation under mildly acidic conditions.

2.3. Flocculation Fundamentals

Flocculation is a fundamental physicochemical process in solid–liquid dispersion engineering, widely applied in industries such as mining [54], environmental management, and pharmaceutical manufacturing [55]. Understanding flocculation requires analyzing how particles interact in colloidal media under the influence of external agents, particularly binders that alter the system’s stability. In this context, the following behavior is observed: The formation of solid particle aggregates present in a solid/liquid dispersion under the influence of a binding agent is the main characteristic that defines a flocculation process [56]. The property of bringing particles together in a colloidal mixture allows the solids to grow into flocs, which reduce their contact surface with the liquid, thereby weakening repulsive interactions. This promotes the increase of solids when an aggregation additive is

available, organizing the new structures into fractal clusters that eventually lose the ability to remain dispersed and ultimately undergo gravitational separation[57].

Vajihinejad et al., (2019) [58] , mention more specialized flocculants that include stimuli-responsive macromolecules, which undergo changes in solubility and conformational structure when the appropriate stimulus is applied. These stimuli may include pH, ionic strength, temperature, redox reactions, electromagnetic fields, and light intensity.

The selection of an appropriate flocculating agent depends on several parameters, such as the chemical composition of the particles, pH, temperature, and the dispersion medium, among others. An additional variable to consider in these processes is the concentration of the flocculant species, which may exhibit a maximum that promotes solid aggregation. Above this maximum concentration, particle re-stabilization is expected. Therefore, the literature refers to the concentration range in which flocs are able to grow as the flocculation window [56]. Dixon et al., (2023) [59], reports this value as the maximum effective amount that solid particles can adsorb, which may be related to the resistance to intraparticle mass transfer. From another perspective, it can be inferred that a minimum critical concentration is required for the flocculation process to occur, and that complete coverage of the solid surface by the flocculant is not necessary.

2.3.1. Mechanisms of Flocculation

North et al., (2017) [60], reports that, in a study of silica particle flocculation in the presence of a diblock copolymer composed of poly(2-(dimethylamino)ethyl methacrylate) and poly(benzyl

methacrylate), with block ratios of 71 and 100 respectively, the fractional surface coverage of the solid by the flocculant was approximately 0.42. Shkop et al., (2017) [61], acknowledge the importance of polymer coverage on the solid surface for particle aggregation to occur, although they do not specify the required coverage ratio by the flocculant species. They propose a two-stage mechanism to explain flocculation: (i) adsorption of the flocculant agent onto the solid surface, and (ii) particle bridging through flocculation links. In this context, it is suggested that at low concentrations of the agglomerating substance, partial surface coverage may occur, leading to non-uniform surfaces and resulting only in the adsorption stage of the polymer, without inducing particle aggregation.

Jeldres et al., (2018) [57], cite flocculation mechanisms differentiated by the type of collision between particles. The terms perikinetic and orthokinetic flocculation are frequently cited in flocculation studies, as they refer to the two fundamental mechanisms used to explain the process. In both mechanisms, collisions between dispersed elements are essential to promote particle aggregation. Perikinetic flocculation involves dispersed solid particles that collide or approach each other through Brownian diffusion. This phenomenon is predominant in systems containing particles smaller than 1 micrometer and with movements promoted by temperature. Orthokinetic flocculation is particularly relevant in systems subjected to external agitation, where particle collisions are driven by local velocity gradients in the flow. This mechanism relies on the relative motion generated by turbulence or mixing, which increases the likelihood of contact between dispersed particles, thereby promoting floc formation. A third mechanism discussed in flocculation studies is differential sedimentation, which occurs in systems where particles undergo

gravitational settling. In this context, differences in settling velocities between particles exceed the influence of Brownian motion, leading to collisions and aggregation during the downward transport of sedimenting solids. This mechanism is particularly relevant for larger or denser particles, where gravitational forces dominate over thermal diffusion [62].

Maćczak et al., (2020) [56], state that particle aggregation under the influence of flocculant polymers occurs through two main mechanisms: charge neutralization and polymer bridging. Specifically, charge neutralization results from the partial or complete adsorption of macromolecules with short chains carrying an ionic charge opposite to that of the solid surface. This partial or complete neutralization reduces electrostatic repulsion and may even create localized surface charge regions (known as patches or islands), which convert repulsive forces into attractive interactions in proximity to particles with opposite charge. This reduction in repulsive forces allows Hamaker forces to become dominant, promoting particle aggregation. In the case of polymer bridging, the mechanism involves the partial adsorption of a macromolecule, typically with a long carbon backbone, onto a first solid surface. This generates polymer loops or tails that extend away from the surface. Eventually, the remote segments of the chain can be adsorbed by a second particle, thereby forming interparticle connections associated with polymer bridges.

Walch et al., (2022) [63], focused their review on extracellular polymeric substances (EPS), explaining that these materials can exhibit varied electrical charges or neutrality, as well as different molecular weights. These properties align with mechanisms such as polymer bridging, charge neutralization via patch formation, and reduction of repulsive forces. Additionally, the authors highlight the role of

hydrophobic forces as a mechanism for particle aggregation. This hypothesis is supported by extended DLVO theory, in which hydrophobic interactions are incorporated into the balance of attractive and repulsive forces governing colloidal systems.

Lu et al., (1998) [64], present the kinetic differences between systems undergoing coagulation and those experiencing hydrophobic flocculation. In coagulation, the rates of aggregation and breakup remain constant from the beginning of the process. However, hydrophobic flocculation exhibits two distinct stages: in the first, aggregation occurs exclusively, with no observable breakup due to the high stability of the flocs; in the second stage, aggregation and breakup occur simultaneously, initiating a dynamic equilibrium between both rates.

2.3.2. Classification of Flocculation

Flocculating agents are chemical substances that promote the aggregation of particles in dispersed systems. According to Okaiyeto et al., (2016) [65], flocculants can be classified into three categories based on their origin: inorganic (e.g., polyaluminum chloride, aluminum salts or ferrous sulfates, aluminum or ferric chloride), organic (e.g., polyacrylamide, polyethylene amine, generally synthetic macromolecules), and natural or Bio-flocculant [56] (e.g., chitosan, cellulose, microbial flocculants, among others).

Vajihinejad et al., (2019) [58], present a classification from a chemical perspective, separating particle aggregation agents based on the charge of the polyelectrolyte:

- Cationic flocculants: the polymeric species carries a positive charge distributed along the chain.

- Anionic flocculants: contain negative charges throughout the polymer backbone.
- Neutral flocculants: do not possess any electrical charge along their structure.
- Amphoteric flocculants: can shift their local charge depending on the system's pH.
- Zwitterionic flocculants: have a molecular structure that includes both positively and negatively charged groups.

2.3.3. Flocculation in MFT

Polyacrylamide is a non-ionic and non-toxic homopolymer that can be easily hydrolyzed in a basic medium by a reaction with strong bases, transforming portions of the main chain into anionic sections (with negative point charges). The cationic version of polyacrylamide (with positive point charges) is synthesized by copolymerization where unsaturated quaternary ammonium compound monomers are added. This ease of being prepared as a water-soluble macromolecule that can be modified makes it commercially important for a variety of uses: photographic emulsion, food processing, adhesives, coatings, pulp processing, concrete processing, crude oil production processes, paper, mineral processing, cosmetic additives, textile processing, soil and sand treatment, water and waste-water [66].

Chitin, chitosan, and tannins are natural polymeric flocculants that have been employed in the process of separating solids in suspension in water. Chitin and chitosan are substances basically described as a polysaccharide with biological, cosmetic, and medical applications with a very similar structure to cellulose [67]. On the other

hand, tannin is a secondary product of vegetables found in leaves, stems, and fruits that can be extracted in an aqueous medium [68] and that has shown potential in flocculation processes. Polysaccharides have emerged as promising macromolecules in green chemistry because of their biodegradability and biocompatibility, representing an advantage of being environmentally friendly and less harmful to life [69].

Many of these polysaccharides are abundant in nature with the disadvantage of not being soluble in water [70][71][72]. One of these polysaccharides recognized as a natural polymer and with potential as a flocculating agent is amylopectin (AP), but it is a non-ionic polymer formed by glucose monomers whose macromolecules are insoluble in water. AP can be found in plant-based sources with carbohydrate structures. Alternatively, AP can be modified to be water soluble by chemically grafting a variety of functional group (FG) onto its backbone. Modification of amylopectin by chemical grafting of FG creates a copolymer known as amylopectin-graph-functional group (AP-g-FG) with molecular weight ranges from 50000 to 1000000 Da. The new AP-g-FG macromolecule will contain sections of hydrophilic chain; therefore, AP-g-FG can be soluble in water and additionally it is designed to have cationic and/or anionic sections according to the selection criteria of the FG to be grafted. One of the advantages of grafting FG onto polymers such as AP is to save the synthesis step of a main polymer chain, since AP is found naturally in starch [70].

Dixon D. et al., [73] studied the effect of adding a first flocculant, anionic hydrolyzed polyacrylamide (HPAM), followed by a second cationic poly(vinyl benzyl trimethylammonium chloride) (PVBTMAC) flocculant, both oppositely charged polymeric flocculants in dispersions of synthetic fine particles from mature particle sands (MFT) observing an improvement in the effectiveness of the flocculating agent. The

synergy of the use of the 2 polymers involves more than one flocculation mechanism and the attacked particles in suspension with different ionic charges on the surface was added at a higher rate [73]. The HPAM-PVBtMAC polymer blend (cationic and anionic flocculants) has shown advantages surpassing the results of separate solid content during flocculation in experiments with individual polymeric flocculants. The cationic moieties neutralize the anionic charge on the surface of the Mature Fine Tailings particles [73].

2.4. Focused Beam Reflectance Method (FBRM)

The Focused Beam Reflectance Method (FBRM) is widely recognized for particle size measurement, reporting chord length as a representative parameter of suspended solids. FBRM employs an in-situ probe positioned directly within the system under investigation, enabling real-time data acquisition without the need for sampling, modifying agitation conditions (e.g., *Laser Diffraction* (LD) [74]) diluting the dispersion (e.g., *Dynamic Light Scattering* (DLS) [75]), or drying/fixing particles onto a surface (e.g., *Scanning Electron Microscopy* (SEM) [76]). These preconditioning steps, commonly required by conventional techniques, may distort measurement results, compromise temporal accuracy, or introduce manipulation-induced noise.

Each technique used in particle analysis reports different magnitudes to characterize size. For example:

- LD (Laser Diffraction) reports the equivalent volumetric size, also based on spherical models, and is sensitive to size distribution in heterogeneous populations [74].

- DLS (Dynamic Light Scattering) provides the hydrodynamic size and volume, assuming the particles are spherical [75].

- SEM (Scanning Electron Microscopy) yields the geometric size, based on the surface morphology observed [76].

These magnitudes should be considered approximations of the actual particle size, since each technique relies on different physical principles and may be influenced by particle shape, density, or aggregation state. The most important factor when studying size evolution is to keep the measurement technique constant. This allows tracking under a uniform criterion, avoiding inconsistencies that arise from comparing methods with different foundations.

The Focused Beam Reflectance Measurement (FBRM) system is constructed as an optical probe that performs a high-speed laser scan (approximately 2 m/s). This scanning speed enables an instantaneous sweep, effectively freezing a moment in time across the window at the tip of the probe, where the laser beam is reflected by nearby solid particles. The system detects the time interval during which the light is reflected, and by multiplying this interval by the scanning velocity, it calculates the chord length, the distance between the entry and exit edges of the solid intercepted by the laser [77] [78] [79]. In this way, each chord length is recorded and stored, along with its probability of occurrence, allowing the construction of a statistical size distribution known as the chord length distribution (CLD) [80].

Pearson et al., (2004) [81] and Gao et al., (2022) [82], define several types of weighted chord length averages, which are calculated based on the probability distribution of chord detection. These include:

- $\overline{LM}_0$: unweighted mean,

- $\overline{LM}_1$: length-weighted mean, representing the mean length diameter
- $\overline{LM}_2$: square-weighted mean, representing the mean surface diameter
- $\overline{LM}_3$: cube-weighted mean, representing the mean volume diameter

Each of these statistical measures can be calculated using the following equation 3.

$$\overline{LM}_j = \frac{\sum_i f_i lc_i^{j+1}}{\sum_i f_i lc_i^j} \quad (3)$$

Where (lc_i) is the midpoint de chord length in micrometers $[\mu m]$, and (f_i) is the probability of detecting a particle.

Vajihinejad and Soares (2018) [23] analyzed the temporal dynamics of particle size variation using FBRM. Additionally, they conducted a comparison between the results obtained from the macro acquisition mode of FBRM and the $\overline{LM}_3$ diameters measured by laser diffraction, using a static sample. The authors reported a highly significant linear correlation between both techniques when using mean particle sizes, with a coefficient of determination $R^2 = 0.99$, indicating excellent agreement between the two measurement methods.

Martos et al., (2010) [83], analyzed the particle size distribution of $CaCO_3$ using three techniques: (LD), (FBRM), and (SEM). The study demonstrated that each technique exhibits good reproducibility when compared against itself, meaning that repeated measurements under the same conditions yield consistent results. However, when comparing

results across different techniques, visible differences in particle size distributions were observed, confirming that each method may produce distinct outcomes due to its specific measurement principles and sensitivities.

It is worth noting that the study does not address the experimental conditions of measurement [84], such as:

Pre-dispersion of samples, which is essential to prevent agglomeration that could distort the analysis.

Proper transport of the sample to the measurement cell, which can affect system homogeneity.

Adjustment of solid concentration within the cell, particularly in LD instruments, where achieving an optimal level of laser signal obscuration or saturation is a common requirement to ensure reliable measurements.

Specifically, in the case of FBRM, it is stated that the data were obtained in situ; however, it is not mentioned whether the measurements were carried out under stirring conditions or in their absence.

These operational variables are critical for obtaining comparable results across techniques and should be considered in future studies aiming to correlate particle size distributions obtained by different methods.

2.5. Flocculation Models

Each of these goals is studied kaolinite particle flocculation process in the suspension. In the next sections of developing and

formulating the model, several alternatives will be included that consider deterministic or heuristic models based on mathematical principles of cases studied in literature and we will specify each in the next sections individually for each one of the goals.

The code was developed using Python language which has a wide range of libraries for the study of data science that were packaged from Fortran and C language [85].

All the developed models were validated through the comparison of the model responses, such as conversion profile, average molecular weights, MWD, with experimental data. Some model parameters, which are unknown, must be estimated to adjust the model outputs with the real data.

2.5.1. Simulation of the dispersed phases

The challenge of modeling the flocculation of kaolinite particles under a dynamic conditions and flocculating agent concentration is one of the objectives of the present work. Various authors used simulation models based on liquid-liquid dispersions as a basis for the study of solid-liquid dispersions.

Molecular dynamics simulation (MDS) in flocculation is one of the modeling techniques most found in the literature. MDS usually requires very complex computer systems to assess the temporal variation of each individual atom or molecule in the study volume. In the case of a surface, MDS tries to reproduce the individual loads on the material. In flocculation it is very typical to study the absorption of polymers on the surface of the particles[86][87][88]. Therefore, the level of detail is too high and beyond the scope of this work.

Hripko R. et al., [89] worked with polyacrylamides (PAM) grafted with poly(ethylene oxide methyl ether methacrylate) (PEOMA) generating a grafted copolymer of PAM-g-PEOMA. The study variables PAM-g-PEOMA chain length, PEOMA concentration, synthesis conditions, PAM-g-PEOMA chain length, average molecular weight of PAM-g-PEOMA, and initiator concentration. This study offers significant insights that enable the design of PAM-g-PEOMA flocculants with diverse characteristics by manipulating the variables within the experimental design.

The same experimental design proved its utility by generating a response surface graph, which makes it possible to evaluate the impact of the microstructure of a poly(acrylamide-co-diallyldimethylammonium chloride) copolymer on the dewatering performance of MFT particles [90]. It was found that the average molecular weight of the copolymer within the tested range does not significantly impact aggregation. However, the composition of the copolymer and the dosage of the polymer do affect the capillary suction time and filtration resistance. These visual models, developed using laboratory design tools, prove to be more practical for operational implementation of flocculants. However, the information currently available does not provide a comprehensive understanding of the underlying aggregation phenomenon.

While visual laboratory models provide practical insights, they do not fully explain the underlying aggregation phenomenon. To partially address this limitation, researchers have adopted the population balance (PB) as an alternative approach to modeling flocculation systems. PB often yields good results because it is grounded in solid principles of material balance[23][91]. PB has been implemented in computational frameworks such as MATLAB, Fortran [92] and Python

[93]. offering a more systematic way to represent particle aggregation and breakage. Nevertheless, although PB contributes to a deeper understanding, it still does not capture all the complexities of flocculation, and further refinement of modeling approaches remains necessary.

2.5.1.1. Balance Population Modeling

Since 1965, population balance (PB) models have been mentioned for the study of mixtures in chemical reactors for dispersed liquid-liquid systems. These mixtures can be modeled as a continuous medium where coalescence and redispersion of dispersed droplets occur in parallel. Using the term package as described by Call M. et al., [94] dispersion modeling by population balance does not limit the physical state of the package or the continuous phase, but the examples cited by these authors are applied to liquid-liquid dispersions [94].

The investigation of flocculation as a velocity-dependent process is not new; analytical models such as Smoluchowski's (1917) [95] already provided solutions for the variation of particle concentration over time. In this framework, Smoluchowski introduced the mathematical concepts of perikinetic (equation 4) and orthokinetic (equation 5) flocculation [95], which correspond respectively to particle collisions driven by Brownian motion and by fluid shear forces [57] .

$$\beta_{perikinetic} = \left(\frac{2kT}{3\mu} \right) \left(\frac{1}{d_i} + \frac{1}{d_j} \right) (d_i + d_j) \quad (4)$$

$$\beta_{orthokinetic} = \left(\frac{1}{6} \right) \left(\frac{du}{dy} \right) (d_i + d_j)^3 \quad (5)$$

where i and j are particle subscripts, d is the particle size, T is the absolute temperature of the suspension, μ is the suspension viscosity,

$\frac{du}{dy}$ is the velocity gradient of the suspension, and k the is Boltzmann's constant [95].

In the original Smoluchowski model (equation 4), only the aggregation velocity was considered.

The development of the Population Balance Model (PBM) in scientific literature has been extensive and progressive, evolving over more than a century of research. One of the earliest and most influential contributions was made by Smoluchowski (1917), who proposed dynamic processes of particle aggregation as formation rates, laying the groundwork for the statistical study of particulate systems (equation 6) [95].

$$\frac{dn_k}{dt} = \frac{1}{2} \sum_{i+j=l}^{k-1} \beta(i,j)n_i n_j - \sum_{i=1}^{\infty} \beta(i,l)n_i n_l \quad (6)$$

where i , j , and l are particle subscripts, n is particle concentration, β is the aggregation rate, and t is time. The first term $\frac{1}{2} \sum_{i+j=l}^{k-1} \beta(i,j)n_i n_j$ expresses the increase in particles of size l due to the aggregation of two particles whose total volume equals that of a particle of size $l = i + j$. The second term $\sum_{i=1}^{\infty} \beta(i,k)n_i n_l$ defines the loss of particles of size l due to their aggregation with particles of other sizes [57,95].

Later, Hulburt and Katz (1964) [96] formally introduced the concept of population balance for the agglomeration process under homogeneous batch conditions, linking the study of agglomeration with differential mass and energy balances through an approach to phenomena such as particle nucleation and growth, resulting in a set of ordinary differential equations at different moment Mo_n (equation 7)

[96]. This approach allowed particle populations to be modeled based on their physical properties for the leading moments (equations 9–11) [96]. According to the authors, the developed model is independent of the size of the colloidal particles, but considers the influence of particle concentration and environmental factors [96].

$$\zeta = fa(t) \left[\frac{1}{2} \left\{ \sum_{vg=0}^{Mo} \binom{Mo}{vg} \zeta_{vg} \zeta_{Mo-v} \right\} - \zeta_0 \zeta_{Mo} \right] \quad (7)$$

Where ζ is a moment, Mo is the moment order, vg is growth rate, fa is the environmental factor and $\binom{Mo}{vg}$ is calculated by equation 8 [96].

$$\binom{Mo}{vg} = \frac{Mo!}{vg! (Mo - vg)!} \quad (8)$$

$$\frac{d\zeta_0}{dt} = -\frac{1}{2} fa(t) \zeta_0^2 \quad (9)$$

$$\frac{d\zeta_1}{dt} = 0 \quad (10)$$

$$\frac{d\zeta_2}{dt} = fa(t) \zeta_1^2 \quad (11)$$

Subsequently, Randolph and Larson (1971) advanced the population balance approach by applying concepts from continuum mechanics. In the new framework proposed through continuum mechanics, the suspension is treated as a continuous medium, in

contrast to earlier models that focused on the behavior of individual particles. This approach shifts the analysis from discrete particle interactions to the description of the system as a whole, where aggregation and breakage are represented by empirical birth (Bf) and death (Df) functions (equation 12). The local net rate is then accumulated within a three-dimensional spatial domain (Z), effectively representing the system volume. By considering the volume of the suspension rather than individual particles, this innovation significantly enhanced PBM's ability to describe complex industrial processes. [97]

$$\frac{d}{dt} \int_{Z_1} n dZ = \int_{Z_1} (Bf - Df) dZ \quad (12)$$

While many other researchers have contributed to the advancement of PBM, the authors mentioned above are regarded as the key figures behind the formulation of its fundamental equation.

An example of such individual contributions is the work of Tambo and Hozumi (1979), who proposed the equation 13 to quantify the maximum size of flocs formed before undergoing breakage due to a given shear rate.

$$max. floc\ size = Cp(G)^{-xp} \quad (13)$$

where G is the root-mean-square velocity gradient of a fluid undergoing strain, Parker's (1972) constants Cp and xp [95].

Dixon and Soares [59] refined the PBM framework by proposing a model in which the parameters vary with polymer concentration and

molecular weight, thereby capturing the influence of polymer properties on flocculation dynamics, as shown in equation 14.

$$Q_{i,j} = \beta_{i,j} \left[af((1 - hd)e^{-k_d t} + hd) \left(2\theta(1 - \theta) + \frac{1}{1 + e^{-bf(\theta - \theta_{max})}} \right) \right] \quad (14)$$

where af is a fitted parameter and θ is the effective surface coverage, both of which depend on the polymer dose. Parameter bf is also a fitted parameter, θ_{max} is the surface coverage in the aggregate. Finally, hd accounts for the time dependence of the aggregate size.

A historical example is the Smoluchowski equations (equation 6), which considered only the aggregation phenomenon. In contrast, current PBM models (equation 6) account for both aggregation and breakage phenomena [95]. The evolution of the mathematical terms introduced by Smoluchowski (β , equations 4-5) [57,95] to the aggregation kernels used in more recent models (Q equation 14) [23,59] illustrates the increasing reliance on empirical parameters to correct and calibrate aggregation rates.

A commonly used modeling approach in flocculation is the Population Balance Model (PBM), which describes the temporal evolution of floc size distributions by monitoring particle number concentration over time. When formulated in its discrete form, the governing rates are expressed through aggregation and breakage kernels (equation 15) [23,57,95].

$$\frac{dn_i}{dt} = Aggregation + Breakage \quad (15)$$

The extended form of the PBM, which accounts for both aggregation and breakage phenomena, is expressed by equation 16.

$$\begin{aligned} \frac{dn_i}{dt} = & \sum_{j=1}^{i-2} 2^{j-i+1} Q_{i-1,j} n_{i-1} n_j + \frac{1}{2} Q_{i-1,i-1} n_{i-1}^2 \\ & - n_i \sum_{j=1}^{i-1} 2^{j-i} Q_{i,j} n_j - n_i \sum_{j=i}^{i_{max,1}} Q_{i,j} n_j - S_i n_i \quad (16) \\ & + \sum_{j=1}^{i_{max,2}} \Gamma_{i,j} S_j n_j \end{aligned}$$

where Q is a kinetic term representing the aggregation kernel, S is the breakage kernel, Γ is the breakage distribution function and n number concentration of aggregates.

$$n_i = \int_{v_i}^{v_{i+1}} n(v, t) dv \quad (17)$$

Vajihinejad V. et al., [57] developed a mathematical PBM to simulate flocculation processes of MFT particles at 3% w/w in aqueous medium [23]. Although the PB equations can be written continuously or discretely, the solution of the ordinary differential equations (ODE) by the discrete route reduces the number of equations and the computational time [57]. A scheme that describes the processes of aggregation and rupture of the particles by the PB equation is presented in Figure 3. The PBM equation has the basic principle of molar balance, which is applied in a region of space where the inputs and outputs of particles are eliminated in the studied volume, leaving the aggregation and breakdown rates as the studied phenomena. For a discrete solution it is a common simplification to use a particle distribution that varies

with the volume according to $v_{i+1} = 2v_i$. Where N_i is the number concentration of aggregates containing 2^{i-1} particles, $\Gamma_{i,j}$ is the breakage distribution function for aggregates of the interval j which form species in the interval i , S_i represents the breakage rate, while Q_i is aggregation rate of aggregates in the interval i , [57].

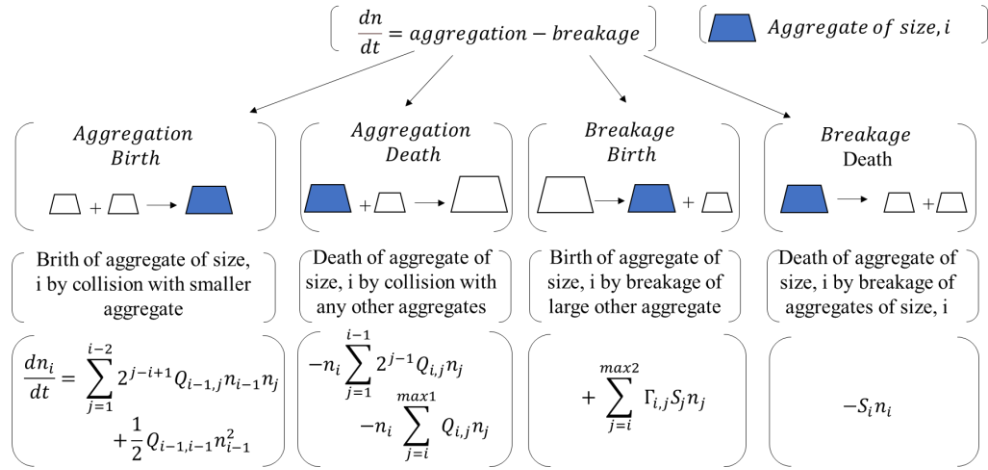


Figure 3. Dynamics of aggregation and breakage for discretized population balance, adapted from [57].

Dixon and Soares (2023) [59] explain the expansion of equation 15 from the Population Balance Model (PBM) in its integral form, as presented in equation 18, where the cumulative contributions of aggregation and breakage terms are events of particle appearance or disappearance (Term 1 to Term 4).

$$\begin{aligned}
 \frac{\partial n(v,t)}{\partial t} = & \frac{1}{2} \int_0^v Q(v-v',v') n(v-v',t) n(v',t) dv' \\
 & + \int_0^\infty \Gamma(v,v') S(v') n(v',t) dv' \\
 & - \int_0^\infty Q(v,v') n(v,t) n(v',t) dv' - S(v) n(v,t)
 \end{aligned} \tag{18}$$

In this formulation, equation 18 can be broken down into four independent reactions, each representing a distinct flocculation mechanism:

Term 1: represents symmetric aggregation, in which two particles of similar size combine to form a new larger particle.

Term 2: describes the formation of aggregates through asymmetric breakage, generating two particles of different sizes; the larger particle is accounted for.

Term 3: quantifies the loss of aggregates due to asymmetric aggregation, that is to say, the collision between particles of unequal size; the count is based on the larger particle.

Term 4: accounts for the loss of particles due to symmetric breakage, quantifying the concentration of particles that fragment into equal parts.

To illustrate these flocculation reactions as equilibrium processes, the corresponding Figure 4 presents two types of interaction symmetric and asymmetric. The particle labeled "2" is consistently used for quantification into PBM.

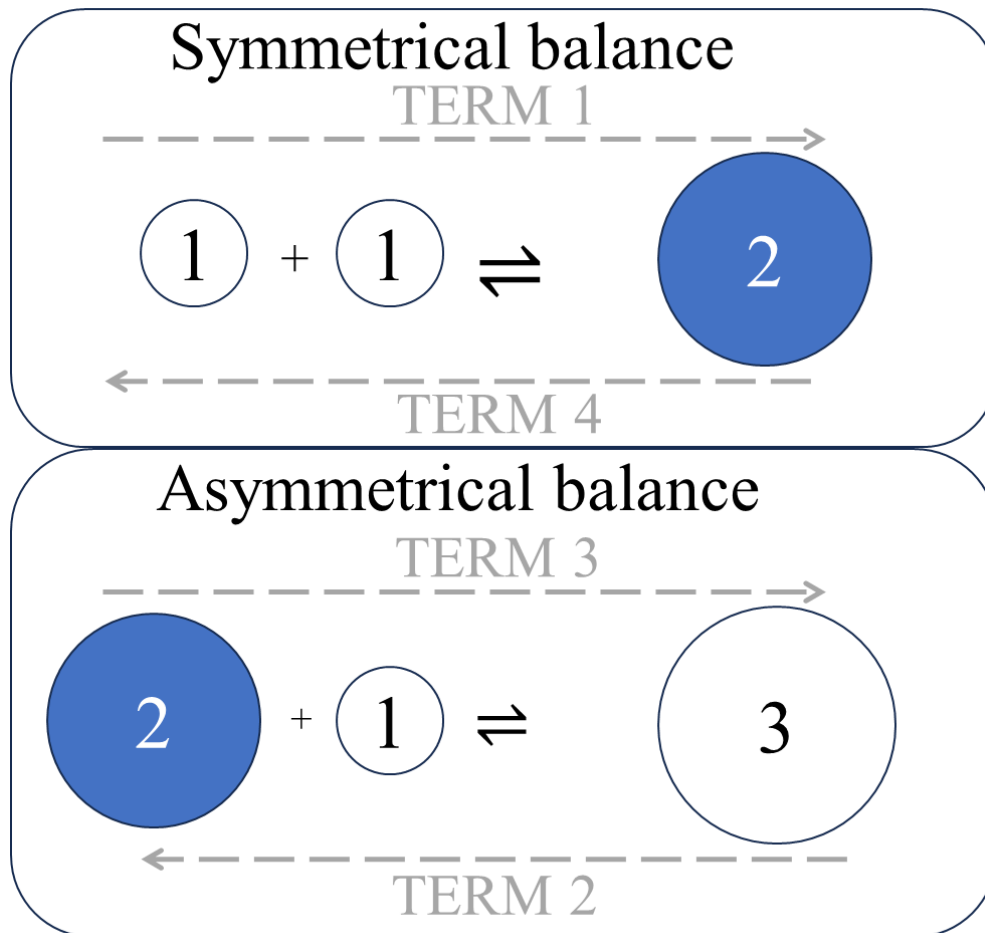


Figure 4. Reactions involved in the four terms of the PBM equation, expressed as symmetric and asymmetric equilibria. The particle accounted for in the balance is the one labeled "2".

The phenomenon of flocculation is characterized by a dynamic equilibrium in which particles remain in constant interaction. This balance arises from two opposing processes: on the one hand, particle aggregation facilitated by the presence of a polymeric agent, which promotes adhesion among them; and, on the other hand, particle fragmentation within an agitated system, where centrifugal forces mechanically overcome the adhesion generated by the polymer on the particle surfaces. Figure 4 'schematically illustrates this dynamic

equilibrium, highlighting the interplay between aggregation and breakage processes.

In the PBM proposed by Vajihinejad et al., [58] three main terms are introduced: the aggregation kernel ($Q_{i,j}$), the breakage kernel (S_i), and the breakage distribution function ($\Gamma_{i,j}$), corresponding to equations 19, 20, and 21, respectively. As observed, these equations depend on the aggregate diameter (d_i), defined in equation 22, as well as on the average turbulent shear rate (G), established in equation 23.

$$Q_{i,j} = \left[0.162G(d_i + d_j)^3 \right] [(\alpha_{max} - \alpha_{min})e^{-k_d t} + \alpha_{min}] \quad (19)$$

$$S_i = s_1 G^{s_2} d_i \quad (20)$$

$$\Gamma_{i,j} = \frac{vp_j}{vp_i} \quad (21)$$

Where vp is the volume the particle.

$$d_i = d_1 (2^{i-1})^{\frac{1}{d_F}} \quad (22)$$

$d = \text{diameter of an aggregate}$

$$G = \left(\epsilon \frac{\rho_{sus}}{\mu_{sus}} \right)^{\frac{1}{2}} \quad (23)$$

ρ_{sus} = density and μ_{sus} = viscosity of the flocculating suspension

G = average turbulent shear rate (velocity gradient)

Where G is average turbulent shear rate (velocity gradient), ρ_{sus} is density of suspension, μ_{sus} is the viscosity of the flocculating suspension and d_F is the fractal dimension.

To complete the model, equation 23 is further refined through the calculation of additional parameters defined in equations 24 to 30.

$$\epsilon = \frac{Ip \, rpm^3 Dv^5}{Vv} \quad (24)$$

Where Ip is the impeller power number, rpm is the rotation speed, Dv is the diameter of the vessel, Vv is the working volume of the vessel.

$$\rho_{sus} = \rho_s \phi + \rho_w (1 - \phi) \quad (25)$$

$$\phi = \frac{\rho_{sus}}{\rho_s} \quad (26)$$

Where ϕ is the volume fraction of primary particles in suspension, ρ_s is $2.60 \, g \, cm^{-3}$, ρ_s is the density of the solid, ρ_w is the density of water

$$\frac{\phi_{eff}}{\phi} = \left(\frac{\overline{d_{agg}}}{d_s} \right)^{3-d_F} \quad (27)$$

$$\mu_{sus} = \mu_0 \left(1 - \frac{\phi_{eff}}{\phi_{max}} \right)^{-2} \quad (28)$$

Where ϕ_{eff} is the effective volume fraction of primary particles, ϕ_{max} is the maximum volume fraction, $\overline{d_{agg}}$ is the average diameter of

the aggregates, $\overline{d_s}$ is the average diameter of the particles, μ_0 is the viscosity of water.

It is important to note that the validity of Equation (29) is assumed throughout the model proposed by Vajihinejad et al., [58].

$$vp_{i+1} = 2vp_i \quad (29)$$

Other models are proposed for example by Lu et al. (1998) [64] used velocity balance equations 30 to 32 to describe the temporal dynamics of flocculation, accounting for both aggregation and breakage mechanisms.

$$\frac{dn_i}{dt} = Ka n_i^2 - \sigma n_j \quad (30)$$

$$Ka = -\frac{2}{3}\varrho \left(\frac{\varepsilon}{\nu f}\right)^{\frac{1}{2}} \quad (31)$$

$$\sigma = \frac{\delta^{\frac{1}{2}} \varepsilon^{\frac{1}{3}}}{\eta^{\frac{2}{3}}} \quad (32)$$

where ε is the energy dissipation rate in turbulence, νf is the kinematic viscosity of the fluid, η is the Kolmogorov microscale of turbulence, δ is the mean square velocity constant of the fluid in turbulent flow, ϱ is the capture efficiency of collisions, and n is the particle number per unit volume[64].

2.5.1.2. Monte Carlo method in dispersed systems

Saliakas V. et al.,[98] proposes the modeling of dispersed liquid-liquid and solid-liquid systems using a stochastic solution with the Monte Carlo method in polymerization reactions of systems in suspension. The authors raise the importance of knowing the droplet/particle size distribution (DSD/PSD) to have a in the system in suspension. Similar to what is proposed in Monte Carlo modeling for monitoring polymerization, the global phenomenon of particle size variation is separated into 2 sub-phenomena that are studied stochastically. The sub-phenomena that they considered were aggregation and rupture. The aggregation phenomenon is defined as a function $kf(vp_i, vp_j)$ because it considers 2 drops or particles of volume vp_i and vp_j that interact to form a new drop or particle. The rupture phenomenon is studied as a function $g(vp_i)$ because it is considered to start with a single drop or particle of size vp_i . The probability (Pab_i) of occurrence of each phenomenon of coalescence or rupture is determined by equation 33. The interesting thing about the proposal is that it bases the probability (Pab_i) of the aggregation/breakage phenomenon occurring on a ratio Rab_i

$$Pab_i = \frac{Rab_i}{\sum_{i=1}^2 Rab_i} \quad (33)$$

$$\begin{cases} i = 1 & \text{for droplet breakage} \\ i = 2 & \text{for droplet coalescence} \end{cases}$$

where Rab_i denotes the total rate (s^{-1}) of coalescence or breakage event in the sample population. To select which event, coalescence or breakage, is going to occur coalescence or breakage phenomenon is used, equation 34 .

$$\frac{Rab_i^*}{Rab_i^{max}} \geq rn_j \quad (34)$$

$$\Rightarrow \begin{cases} \frac{gf(vp_i)}{gf(vp_{max})} \geq rn_j & \text{for droplet breakage} \\ \frac{kf(vp_i, vp_j)}{kf(vp_{max}, vp_{min})} \geq rn_j & \text{for droplet coalescence} \end{cases}$$

where Rab_i^* denotes the rate of breakage or coalescence event and rn_j is a randomly generated number. The time interval (Δt) is calculated by taking the reciprocal of the sum of all coalescence and breakage rates, as expressed in equation 35.

$$\Delta t = \frac{1}{\sum_{i=1}^2 Rab_i} = \frac{1}{\sum_{j=1}^{N_n} (gf(vp_i)) + \sum_{l=j+1}^{N_n} (kf(vp_i, vp_j))} \quad (35)$$

An adaptation of the schematic representation of the MC algorithm implemented by Saliakas V. et al., [98] is shown by Figure 5.

To increase the speed of the calculation, the sum of the rupture collection rates can be calculated discretely as presented in equation.

$$\sum_{i=1}^2 Rab_i = q_1 \sum_{j=1}^{\frac{N_n}{q_1}} gf(vp_i) + q_2 \sum_{j=1}^{\frac{N_n(N_n-1)}{2q_2}} (kf(vp_i, vp_j)) \quad (36)$$

where vp_1, vp_i, vp_j are the volumes of the droplets that are randomly selected from the sample population within each summation. The variables q_1 and q_2 are the percentage parameters.

A discrete representation of the results can be obtained by calculating the number diameter density function $n_i(Dn, t)$ using equation 37, which represents the number of packages per unit volume.

$$n_i(Dn, t) = \frac{Ns_i(t)}{Dn_{(i+1)} - Dn_{(i)}} \quad i = 1, 2, \dots, Ns_n \quad (37)$$

where $Ns_i(t)$ is the number of sample packages (droplets/particles) in the size range of diameter $[Dn_{(i)}, Dn_{(i+1)}]$ at time (t) . Another useful parameter that can be obtained is the respective volume probability diameter density function $Ap_{v,i}(Dn, t)$ obtained by equation 38.

$$Ap_{v,i}(Dn, t) = \frac{\sum_{j=1}^{Ns_n} \frac{Dn_j^3}{Dn_{(i+1)} \leq Dn_j < Dn_{(i)}}}{\sum_{l=1}^{Ns_n} Dn_l^3} \frac{Ns_i}{Ns_n(Dn_{(i+1)} - Dn_{(i)})} \quad (38)$$

$$i = 1, 2, \dots, Ns_n$$

Equations 39 to 41, as proposed by Saliakas V. et al.,[98], incorporate the effects of breakage rates $g(vp_i)$, coalescence rates $kc(vp_i, vp_j)$, and breakage frequencies $\omega_b(v_i)$.

$$g(vp_i) = \omega_b(vp_i)e^{-\lambda_b(vp_i)} \quad (39)$$

ω_b = breakage frequencies, λ_b = breakage efficiencie

$$kc(vp_i, vp_j) = \omega_c(vp_i, vp_j)e^{-\lambda_c(vp_i, vp_j)} \quad (40)$$

ω_c = coalescence frequencies

λ_c = recoalescence efficiencie

$$\omega_b(v_i) = \frac{\bar{u}(D_u)^2}{D_u} \quad (41)$$

$\bar{u}(D_u)^2 = \text{relative velocity}, D_u = \text{distance}$

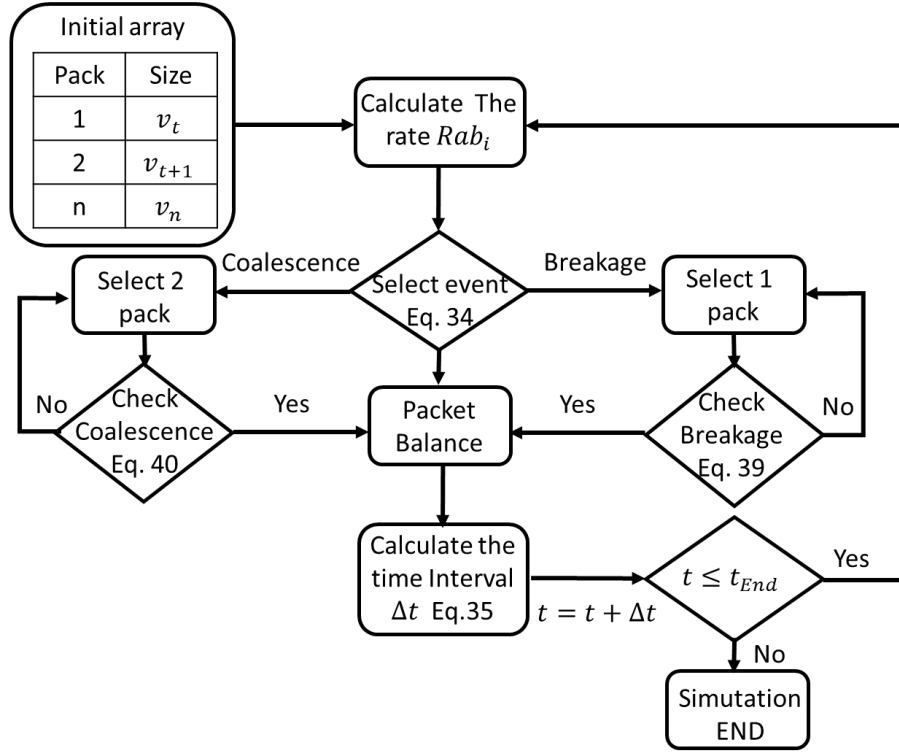


Figure 5. Schematic representation of the MC algorithm, adapted from Saliakas V. et al. [98].

Relative velocity calculations are performed based on equations 42 through 45.

For package in the not viscous dissipation range (42)

(i.e., $\eta < D_u \ll Li$)

$$\bar{u}(D_u)^2 = k_b(\varepsilon_s D_u)^{\frac{2}{3}}$$

$k_b = \text{model parameter}$

$\varepsilon_s = \text{average energy dissipation rate for the dispersion}$

$$\lambda_b = \alpha_b \Omega(D_u) \quad (43)$$

$\alpha_b = \text{model parameter,}$

$\Omega(D_u) = \text{ratio of required to available energy}$

For package in the viscous dissipation range (44)

(i.e., $D_u < \eta$)

$$\bar{u}(D_u)^2 = k_b D_u^2 \left(\frac{\varepsilon_s}{\text{kinematic viscosity}} \right)$$

$$\Omega(D_u) = \frac{2}{Re(1 + ReVe)} + \frac{Cds}{We} \quad (45)$$

$Re = \text{Reynolds number, } We = \text{Weber number}$

$Ve = \text{package viscoelasticity}$

Cds

$= \text{numbers and volumes of daughter and satellite drops}$

The determination of coalescence frequencies is carried out using Equations 46 to 49.

$$\omega_c(vp_i, vp_j) = k_c (Dn_{vpi}^2 + Dn_{vpj}^2) \left(\bar{u}(Dn_{vpi})^2 + \bar{u}(Dn_{vpj})^2 \right)^{\frac{1}{2}} \quad (46)$$

$k_c = \text{model parameter}$

$$\exp\{-\lambda_c(vp_i, vp_j)\} \quad (47)$$

$$= (1 - \chi) \exp(-\lambda_c^{(a1)}(vp_i, vp_j))$$

$$+ \chi \exp\{-\lambda_c^{(b1)}(vp_i, vp_j)\}$$

$\chi = \text{fractional monomer conversion}$

$$\lambda_c^{(a)}(vp_i, vp_j) = \alpha_c \frac{\mu_c \rho_c \varepsilon_s}{\sigma t^2} \left[\frac{Dn_{vpi} Dn_{vpj}}{Dn_{vpi} + Dn_{vpj}} \right]^4 \quad (48)$$

$$\lambda_c^{(b)}(vp_i, vp_j) = \alpha'_c \frac{\mu_c}{\rho_c \varepsilon_s^{\frac{1}{3}} (Dn_{vp_i} + Dn_{vp_j})^{\frac{4}{3}}} \quad (49)$$

where α_c is the model parameter, μ_c is the viscosity of the continuous phase, ρ_c is the density of the continuous phase and σt is the interfacial tension between the dispersed and aqueous phases.

To describe the breakage and coalescence rates (equation 39 – 49) Saliakas V. et al.,[98] use of dispersed packets. The calculation of relative velocity parameters ($\bar{u}(D_u)^2$) in equation 39 is differentiated based on two distinct cases: the first considers a negligible viscosity of the dispersion medium, while the second case assumes a high viscosity (corresponding to equation 44 and 45 respectively).

2.6. Polymerization

Free radical polymerization in aqueous solution is the name given to the process of synthesis of macromolecules using water as a solvent where the addition of mers is promoted after starting the reaction by a free radical that acts as an initiator [99]. Within the great variety of reactions and polymerization techniques, solution polymerization tries to control the propagation speeds by diluting the monomer in the solvent of the system, maintaining a more homogeneous control of the hot spots that occur in reactions in the absence of solvent. Additionally, the changes in the rate coefficients of the reactions are a consequence of the action of hydrogen bonds in the various stages of the reaction: initiation, propagation, chain transfer, and termination. There is interaction monomer, growing radicals, polymers and water at all times through the hydrogen bonds and that can be reinforced by the presence of ionic sections in the elements of the mixture. In the

polymerization reaction of methacrylic acid by free radicals in an aqueous medium, Beuermann S. et al., [100] describe the existence of "transition state complexes of the propagation stage" interactions with water that occur through hydrogen bonds and they negatively affect the reaction speed, decreasing the propagation speed with the increase in the proportion of water in the system. As previously mentioned, the polymers of interest for flocculation phenomena are soluble in polar media, which allows the use of water as a solvent. In addition to the polymerization technique, it is important to know the kinetic mechanism that governs the reaction. In the case of the synthesis of HPAM and (AP-g-PVBtMAC), this mechanism occurs via free radicals, in which four types of reactions can occur simultaneously [99]. The first is initiation, where an initiator decomposes, generating primary radicals. This step promotes polymerization through reaction with monomers, giving rise to an active species.

The second type is called propagation, in which the active species react with other monomers, successively incorporating them into their structure, forming polymeric chains. The third type is the transfer of radicals to other species present in the reaction medium, inactivating the active chains by transferring the individual electron from the free radical site to monomer or dead polymer molecules that are reactivated to a living chain. The fourth type of reaction is termination. In termination reactions, two types of mechanisms are observed: a) termination by disproportion: two active chains react with each other, forming two inactive chains; b) termination by combination: two active chains react with each other, forming a single species of size equivalent to the sum of the two original active chains [101]. Figure 6 illustrates in a simplified way the reactions considered in free radical polymerization.

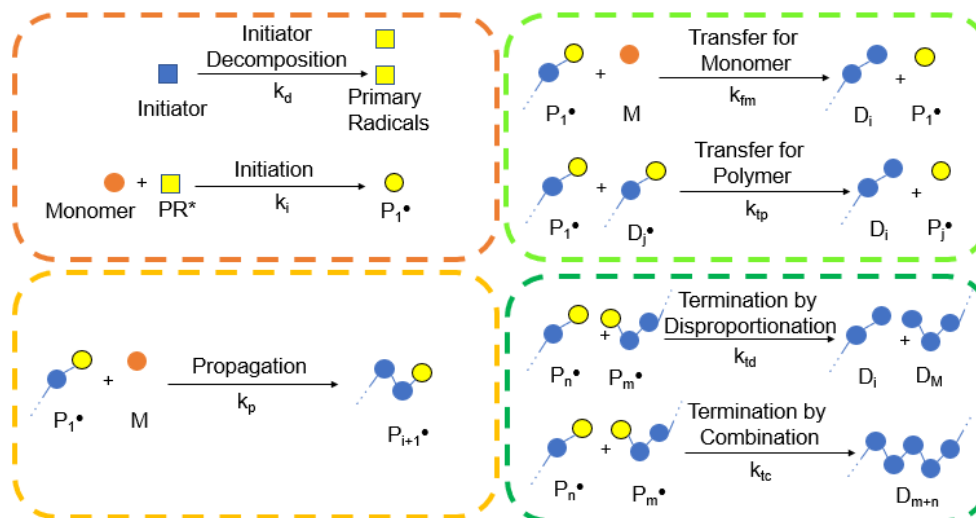


Figure 6. Illustration of free radical polymerization, where P indicates active chains and D inactive chains.

2.7. Polymers in solution

For decades, different groups have studied polymer-solvent interaction in understanding how parameters such as temperature, pressure, solvent quality, pH, polymer-polymer interaction parameters, or liquid-liquid interaction parameters can be quantified to predict the best conditions to solubilize a polymer [102].

In the case of HPAM in aqueous solutions, it is known that the degree of hydrolysis allows the polymer to stretch along the main chain due to the electrostatic repulsions of the sections with hydroxyl groups that repel each other. From the point of view of polymer-solvent interaction in the HPAM-water pair, hydrogen bonds are formed that favor the dissolution of the polymer in the solvent. The presence of monovalent ionic salts (for example NaCl) and divalent (for example CaCl_2) strengthen the repulsion of the hydrophilic sections of the HPAM,

strengthening the ionic repulsion by the formation of a double electronic layer, which contributes to the repulsion of the OH⁻ groups in the main chain of the polymer. As most of the solutes are in a solvent, there is a correlation of good solvation with the variation of pressure and/or temperature, which finally can have consequences in the efficiency of a process [103].

The interaction of the Detrax/water and Detrax/DMSO pair, a known polysaccharide and a polar solvent, was studied, varying the MWD in a range that goes from oligomers to macromolecules of approximately 1,000,000 Da, presenting that the enthalpy of formation of the mixture has a linear behavior for the MWD less than 2,000 Da. For the polymer blends with MWD greater than 2,000 Da, an almost constant enthalpy was observed. These results are attributed to conformational changes of the polymer in solution, which explains that for Detrax oligomers and molecules with MWD less than 2,000 Da, the polymer is in a rod conformation (Figure 7) in which the main chain is stretched in solution. While the case of Detrax with MWD greater than 2,000 Da, the conformation of the macromolecule is expected to be of coils. Parameters such as intrinsic viscosity, the partial specific compressibility repeat the tendency of the Detrax/water interaction by altering the MWD, similarly justifying the result by the conformational changes suffered by the macromolecules in the solvent. These conformational changes of polymer chains ranging from a fully stretched chain conformation to coil-shaped chains have a different density of bound water per monomer unit.

On the other hand, Detrax coils (Figure 7) in water tend to change with the amount of salts dissolved in the solvent. In this opportunity, the increase in the content of salts in the solvent causes an expansion in the coil, increasing the density of bound water per unit of monomer. In

systems where the intrinsic viscosity is studied with solvents with higher salt content, a change in the intrinsic viscosity of the fluid is observed in comparison with systems without dissolved salts [104]. In the case of polymeric flocculants in an aqueous solution, it should be considered that the conformation of a coil-shaped chain limits the efficient use of the macromolecule, and the same results can be obtained with smaller MWD molecules.

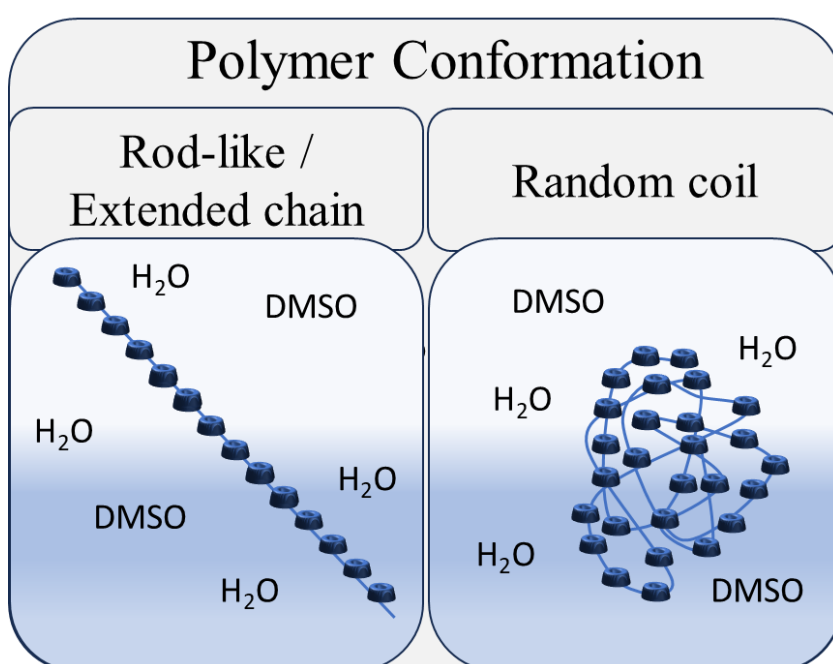


Figure 7. Comparison of Polymer Chain Conformations in Solution.

2.8. Aqueous-Phase Radical Polymerization of acrylamide

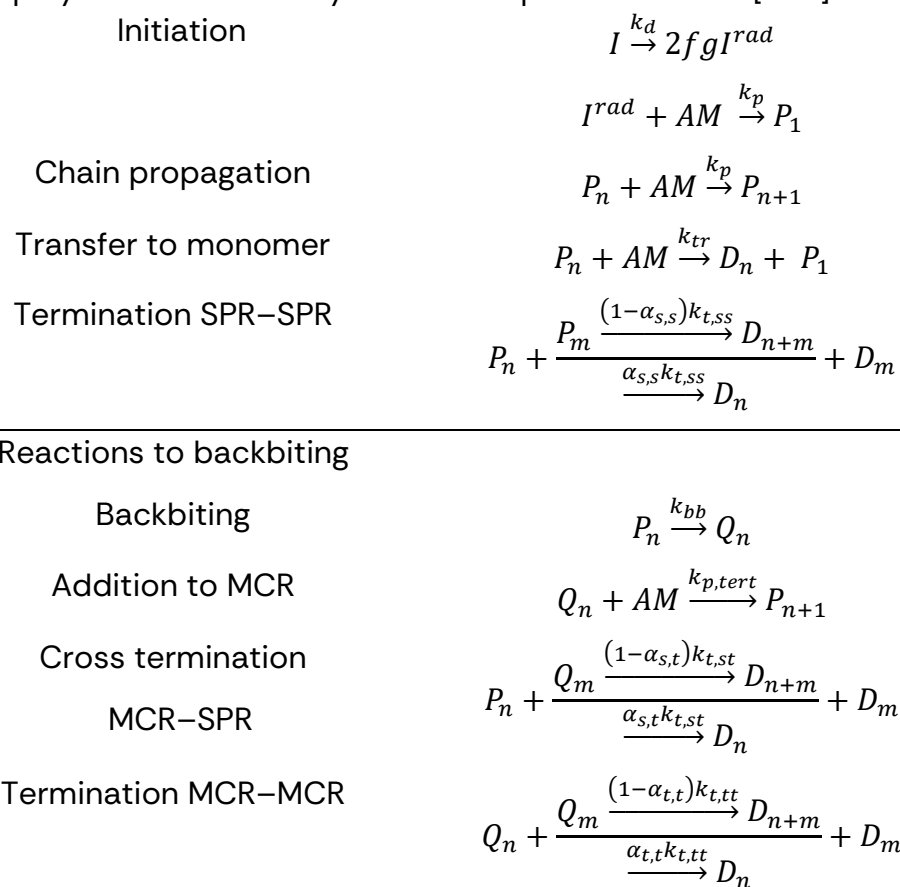
An important aspect in the preparation of polymers employed in flocculation processes is the method of synthesis in solution. Notably, conventional free radical polymerization of acrylamide (AM) in aqueous medium differs substantially from controlled radical approaches such as RAFT polymerization, which allow for improved control over molecular weight distribution and polymer architecture [105].

2.8.1. Free-radical polymerization of acrylamide Free-radical polymerization of acrylamide

Free-radical polymerization of acrylamide, it has been described that the order of the reaction is greater than 1 (1.24 to 1.49) [106], indicating there are certain peculiarities that the conventional free radical polymerization mechanism does not cover.

As can be seen in Equation scheme 1, it summarizes the mechanisms of the free-radical polymerization of acrylamide in aqueous solution.

Equation scheme 1. Kinetic mechanism of the free-radical polymerization of acrylamide in aqueous solution [106].



The path of ideal free radical polymerization kinetics generates a secondary chain-end propagating radicals (SPR) that in the mechanism of Equation scheme 1 is represented by P_n . Being P_n the protagonist of the propagation and termination reactions. While the MCR of the polymerization reactions by free radicals with deviation from ideality are represented as Q_n radicals [106][107].

The batch modeling of the homopolymerization of acrylamide using the aqueous phase was carried out by Preusser C. et al.[105][106], using the method of instantaneous distributions and explaining that the idealized growth by addition of vinyl groups is controlled by parameters such as: temperature, pH, concentration of monomer, the polarity of the solvent, cage effects on the efficiency of the initiator (f), viscosity of the system that ends up influencing the diffusion of the species that participate in the propagation (k_p) and termination (k_t) reactions.

Additionally, aqueous-phase radical polymerization of acrylamide present deviations from the idealized mechanism of free radical polymerization, such as that k_p increased with decreasing monomer concentration. The explanation for this increase in k_p is consistent with the appearance of mid-chain-radicals (MCR), which are medium chain radicals formed by side reactions of the 1,5-hydrogen transfer reaction, also known as backbiting reactions (Figure 8).

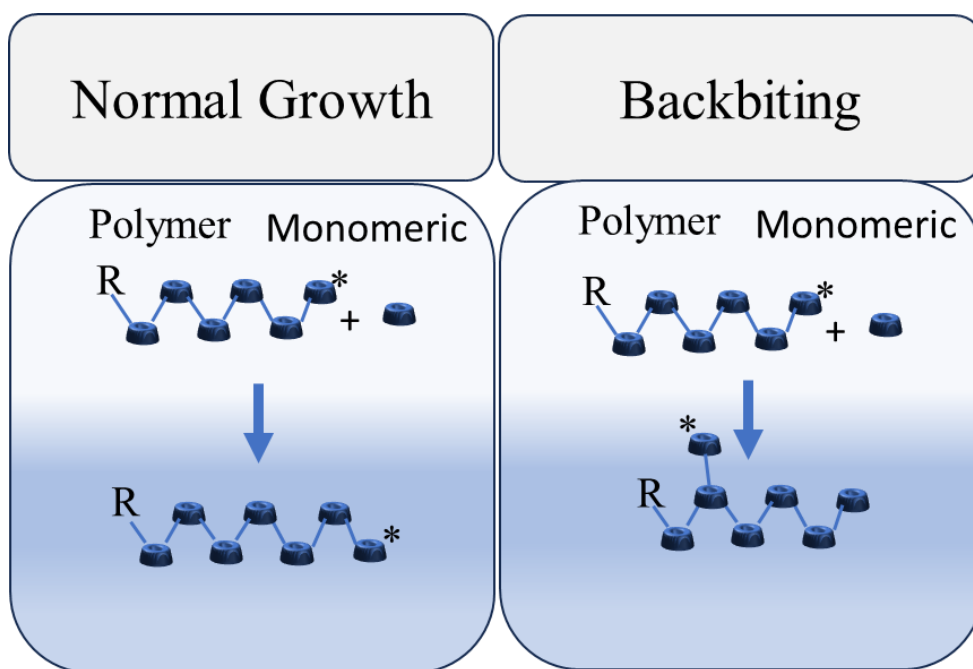


Figure 8. Comparison of Polymerization Mechanisms: Normal Growth vs. Chain Backbiting.

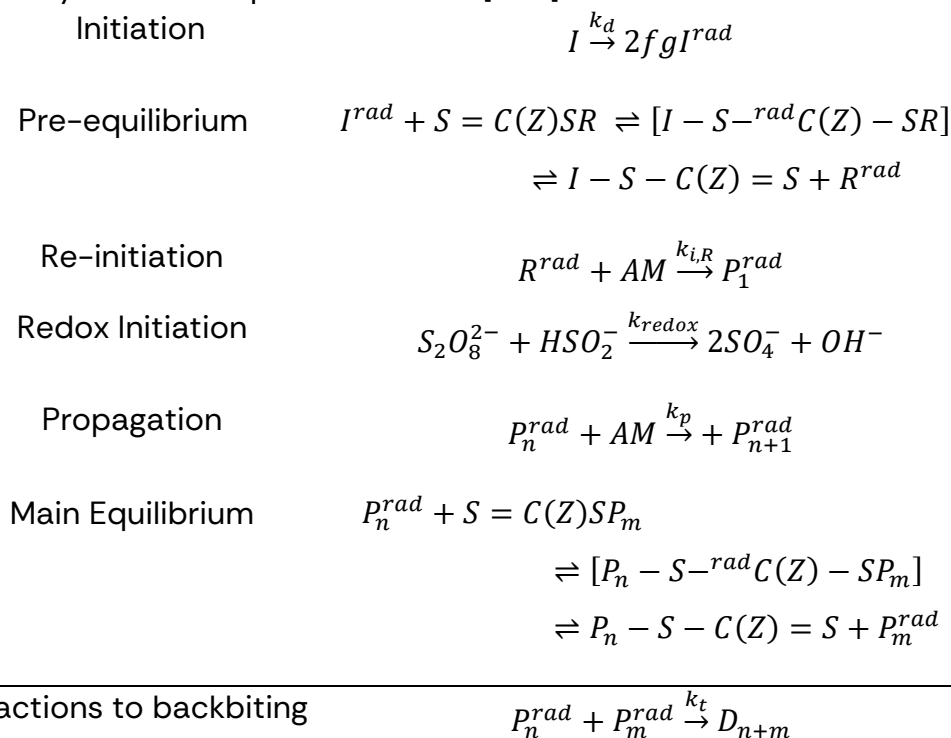
Experimentally the existence of MCR is confirmed by electron parametric resonance (EPR) spectroscopy. In a summary the free radical polymerization reaction of AM is produced by the ideal free radical polymerization plus the free radical polymerization with backbiting [106][107].

Appendix A completes the information of the kinetic model providing the necessary parameters for the calculation of the various variables required for the modeling of the AM polymerization. Preusser C. et al.,[3] reported that these parameters are validated in a temperature range (T) from 40 to 70 °C and monomer concentrations in solution from 5 to 40 wt%.

2.9. Aqueous-Phase RAFT Polymerization of acrylamide

RAFT (Reversible Addition–Fragmentation Chain Transfer) polymerization is described as a technique that enables control over the architecture of the synthesized polymer and its molecular weight, in contrast to conventional free radical polymerization [105].

Equation scheme 2. Kinetic mechanism of the RAFT polymerization of acrylamide in aqueous solution [105].



The synthesis of polyacrylamide polymers via the RAFT method is recognized for producing narrow molecular weight distributions, which translates into precise control over the architecture of the final polymer. This level of control can be interpreted as the result of the RAFT mechanism minimizing or even suppressing side reactions such as backbiting [108][109].

2.10. Reaction kinetic modeling methods

To model the microstructure of the polymer and the various polymer characteristics such as molecular weight distribution, reaction conversion, chain length and other parameters, there is more than one strategy.

2.10.1. Instantaneous Distributions in Polymerization reactions

The instantaneous distribution method is an strategy that can be used to know the progress of free radical polymerization over time, detailing a cumulative distribution of the polymer microstructure through molecular weight calculations, which is provided by knowing the distributions of growth of polymeric chains through consecutive snapshots in time [110]. These snapshots represent a sample in time of the distribution of the chains of assembled macromolecules, considering an approximation to a quasi-steady state because it is a very fast event. The average lifetime of the growing macromolecule is very fast in comparison with other kinetic events within the reaction system.

The great advantage of the instantaneous distribution method is its low computational cost linked to the resolution of analytical expressions. The instantaneous distribution method has particular solutions studied by J. Soares [110] and developed in the analysis of polyolefin free radical polymerization considering three possible objectives: distributions of chain length (CLD1), chemical or comonomer composition (CCD) and long chain branching (LCBD). The calculation of CLD1 is based on molar balances that consider the statistical analysis of the ratio (τ) obtained mathematically from the

frequency of dead chain formation with respect to the frequency of propagation (equation 50). This ratio is kinetically studied by "monomer transfer reaction" and "chain propagation reaction" respectively, which are denoted in Equation scheme 1 and representing the basis of the Shultz-Flory chain size distribution[110]. In addition, this method also allows the evaluation of the chemical composition of the polymer (CLD1 x CCD) of linear binary copolymers.

$$\tau = \frac{\text{frequency of dead chain formation}}{\text{frequency of propagation}} = \frac{R_{tr}}{R_p} \quad (50)$$

$$= \frac{k_{tr}[AM]^a}{k_p[AM]^b} = \frac{1}{r_n}$$

where R is the concentration change in a moment of time for the chain propagation reaction (p) and the monomer transfer reaction (tr). r_n denotes the instantaneous number average chain length of the polymer. Superscripts " a " and " b " are the order of reaction and k is reaction rate coefficient.

For batch reactor, the molar balance of the monomer-free active sites is stated equation 51. The calculation off or living chains with nr monomer units P_n is in equation 52 and dead polymer chains equation 53.

$$\frac{dP_1}{dt} = R_{tr}[AM]^a \sum_{n=1}^{\infty} P_n - k_p[AM]^b P_1 \quad (51)$$

$$\frac{dP_n}{dt} = R_p[AM]^b (P_{n-1} - P_n) - k_{tr}[AM]^a P_n \quad (52)$$

$$\frac{dD_n}{dt} = k_{tr}[AM]^a P_n \quad (53)$$

Considering the steady state approximation, equations 51 and 52 become equations 54 and 55.

$$P_1 = \frac{k_{tr}[AM]^{ap}}{k_p[AM]^{bp}} \sum_{n=1}^{\infty} P_n = \tau Y_0 \quad (54)$$

Representing the total number of moles of polymer in the reactor as Y_0 .

$$P_n = \frac{k_p[AM]^{bp}}{k_p[AM]^{bp} + k_{tr}[AM]^{ap}} P_{n-1} = \left(\frac{1}{1 + \tau} \right) P_{n-1} \quad (55)$$

Soares [110] manipulates these equations until he achieves equation 56

$$P_n = \tau \exp(-nr\tau) Y_0 \quad (56)$$

where nr is the number of moles of living polymer with chain length n and Y_0 is the total number of polymer moles in the reactor. Following the steps of Soares J.[110], it is reported the rate of dead polymer chains (D_n) in equation 57.

$$\frac{dD_n}{dt} = k_p[AM]^{bp} Y_0 \tau^2 \exp(-nr\tau) \quad (57)$$

Managing to calculate the frequency or molar instantaneous chain length distribution (f_n) can be calculated from equation 58 and the weight instantaneous chain length distribution (w_n) is obtained from equation 59.

$$f_n = \frac{\text{rate of } D_n \text{ formation (mol s}^{-1}\text{)}}{\text{total rate dead chain formation (mol s}^{-1}\text{)}} \quad (58)$$

$$= \frac{\frac{dD_n}{dt}}{k_{tr}[AM]^{ap}Y_0} = \tau \exp(-nr\tau)$$

$$w_n = \frac{\text{rate of } D_n \text{ formation (g s}^{-1}\text{)}}{\text{total rate dead chain formation (g s}^{-1}\text{)}} = \quad (59)$$

$$\frac{\frac{d(MWD_n)}{dt}}{MWk_p k_p [AM]^{bp} Y_0} = \tau \exp(-nr\tau)$$

where MW is the molecular weight of the monomer. Soares J.[110] finally defines the dispersity ($\mathfrak{D}$) in equation 60 and shows the molecular weight distribution (MWD) calculation with the equation 61.

$$\mathfrak{D} = \frac{r_w}{r_n} = \frac{\frac{2}{\tau}}{\frac{1}{\tau}} = 2 \quad (60)$$

$$W_{logMW} = 2.3026MW^2 \left(\frac{\tau}{m}\right)^2 \exp\left(-MW\left(\frac{\tau}{m}\right)\right) \quad (61)$$

A clarification that Soares J.[110] included in this article is that τ can be modified to include the sum of all chain transfer reactions in the frequency of dead chain formation. In this case, the addition reactions to *MCR* are included, managing to rewrite τ as equation 62 where the propagation frequency would be modified.

$$\begin{aligned}\tau &= \frac{\text{frequency of dead chain formation}}{\text{frequency of propagation}} & (62) \\ &= \frac{k_{tr}[AM]^{ap}}{k_p[AM]^{bp} + k_{p,tert}[AM]}\end{aligned}$$

Noting that the reactions, from the Addition reaction to MCR, are included by their spread constant ($k_{p,tert}$).

2.10.2. Monte Carlo method

Monte Carlo Simulations (MC) is one of the methods used in the study of global process composed or described by different sub-processes or stages. Stages that occur in parallel and that compete with each other to complete 100 % of the global process studied. In the context of free radical polymerization reactions, mechanistic and kinetic investigations have delineated the overall reaction into distinct sub-reactions, including initiation, propagation, chain transfer, termination, and others. These distinctive features associated with free radical polymerization reactions render them suitable candidates for Monte Carlo simulation. In particular, the stochastic simulation algorithm (SSA)[111] generates random numbers that are used to prioritize one stage at a time. The Monte Carlo simulation combines the mass balance that represents the objective to be achieved and the

generation of random numbers to select the probability of occurrence of one of the individual sub-reactions that are part of the overall reaction process. With this philosophy, small moments of time in which the sub-reactions are carried out in a heuristic way are added consecutively within the global material balance and accumulating each result until the total is covered. SSA's use of random numbers in the decision to prioritize the sub-reaction generates a set of possible outcomes used in polymer reaction engineering modeling to derive the microstructure of macromolecules, degree of polymerization, and many other general details of the reaction[112]. These steps are summarized in the algorithm of Figure 9 where the conversion is used to evaluate the mass value within the system.

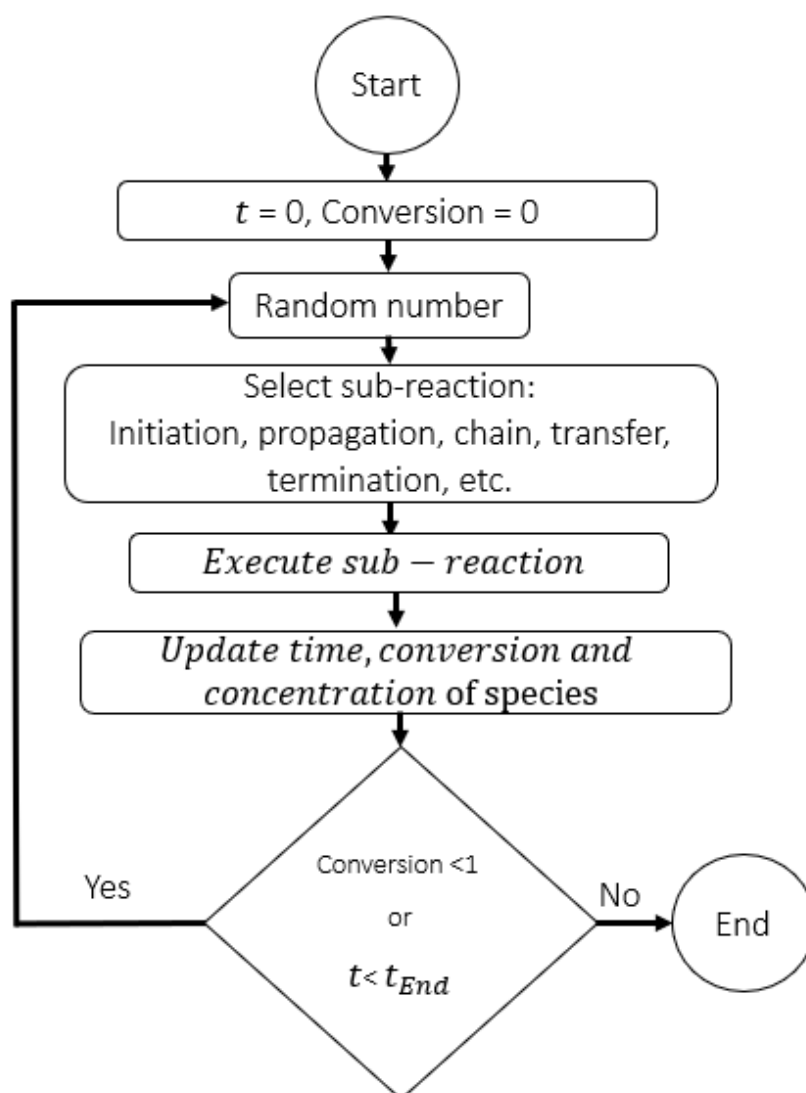


Figure 9. Monte Carlo algorithm proposed for simulate polymerization reactions.

To run a dynamic MC simulation, certain preliminaries must be met, such as establishing a control volume (VC). VC is a space that contains a uniform sample of molecules of the reactor reagents and where the behavior of the polymerization reaction is going to be analyzed. Consequently, the rate constants of macroscopic or experimental origin (k^{exp}) are transformed into equivalent constants within the control volume and are known as Monte Carlo constants or microscopic rate constants (k^{MC}). Another important condition for

transforming the k^{exp} from the constants in k^{MC} is to know the species that are being studied. Therefore, a summary of the calculation of the velocity constants can be found in Equation scheme 3 [111].

Equation scheme 3. Conversion of macroscopic reaction rate constants into microscopic MC rate constants

Unimolecular reactions $k^{MC} = k^{exp}$ (63)

(Initiation)

Bimolecular reactions $k^{MC} = \frac{k^{exp}}{VCN_A}$ (64)
between different
molecules (Propagation)

Bimolecular reactions $k^{MC} = \frac{k^{exp}}{2VCN_A}$ (65)
between equal molecules
(Termination for
disproportion)

Termolecular reactions $k^{MC} = 6 \frac{k^{exp}}{VC^2N_A^2}$ (66)
between equal molecules
(Thermal initiation)

N_A = Avogadro number.

For each sub-reaction in the polymerization, it is considered that there is a reaction rate (R_{sub}^{MC}) that differs from one another when they are identified by the sub-index (R_d^{MC} , R_p^{MC} , etc.). Each R_{sub}^{MC} is calculated by equation 67 where N_c is the number of unique combinations between reactant molecules inside the control volume.

$$R_{sub}^{MC} = k_{sub}^{MC} N_c \quad (67)$$

N_c is calculated by considering the number of reagents involved for each reaction within the control volume as shown in Equation scheme 4 [111].

Equation scheme 4. Definition of NC.

Unimolecular reactions	$N_c = m_i$	(68)
(Initiation)		

Bimolecular reactions	$N_c = m_i m_j$	(69)
between different molecules		
(Propagation)		

Bimolecular reactions	$N_c = \frac{m_i(m_i - 1)}{2}$	(70)
between equal molecules		
(Termination for		
disproportion)		

Termolecular reactions	$N_c = \frac{m_i(m_i - 1)(m_i - 2)}{6}$	(71)
between equal molecules		
(Thermal initiation)		

m_i, m_j = Number of molecules of reactants i and j in the control volume.

Using Monte Carlo simulations, Seyedi A. et al.,[113] modified the initiation feeding policies in free radical polymerizations to analyze the consequences on the dispersity and molecular mass of synthetic polymers.

The various initiator policies were:

1) Number of shots (NOS): The mass of the initiator added to the system is divided into 1, 2, 4, and 8 equal portions of the original mass;

2) Amount of shots (AOS). Four cases (a, b, c, d) were created by varying the initiator addition policy by dividing the original mass of initiator into the concentrations shown below: a:1;2;3;4, b:4; 3; 2;1, c: 1;3;3;1, d: 3;1;1;3 and where each concentration is placed in the reaction medium at 0.25;0.5; 0.75;1 of the conversion;

3) Times of shots (TOS). Four 1/4 mass equivalent portions of the original initiator were added at four different time periods of the standardized reaction per system conversion.

As summarized in Table 1 in each case the initiator addition policies are altered, the MWD suffer modifications, reflected by the weight average molecular weights ($\overline{M}_w$) and the dispersity of the synthesized product. The simplest simulation that corresponds to a single portion of the initiator in the NOS policies is used as a comparison pattern with the other cases studied. For the NOS policies, the $\overline{M}_w$ is increased with the increase in the portions of initiator, appreciating that the also increasing the dispersity increases similarly. The NOS policies, favor propagation reactions avoiding chain transfers to which generates low average molecular weight species. For the TOS policies, where the initiator concentration is the lowest at conversion 0, the formation of long chains and double $\overline{M}_w$ is favored, and dispersity is increased. Finally, the delay of the initiator addition after the thermal initiation begins increases the $\overline{M}_w$ and the dispersity in the case of the TOS policies. From all the cases that studied, the AOS policies with the concentrations a:1;2;3;4 and c: 1;3;3;1, are the best ones to produce polymers with higher Mw increasing the and higher dispersity.

Table 1. *Effects of initiator feeding policies on MWD obtained by Monte Carlo simulation in free radical polymerization [113].*

Policies	Initiator	$\overline{M}_w$ (Da)	Polydispersity	Initiator without reacting (%)
NOS	Portions			
	1	19000	4.2	7.4
	2	24500	5.3	9.5
	4	30000	6.2	11.2
	8	33000	7.0	12.5
AOS	Concentration			
	1;2;3;4	40000	8.4	13.3
	4;3;2;1	25000	6.4	9.5
	1;3;3;1	37000	7.9	11.0
	3;1;1;3	27000	5.7	11.6
TOS	Conversion			
	0;0.05;0.15;0.25	22000	4.9	8.1
	0;0.55;0.65;0.75	29000	6.3	9.9
	0;0.30;0.40;0.50	34000	7.2	13.1
	0;0.13;0.21;0.65	25500	5.5	9.7

In all cases shown in Table 1 it is observed that an excess of unreacted initiator is present in the products. Many producers of the material do not want an excess of the initiator because it can be toxic, change physical properties (color, transparency, etc.) or decrease resistance to UV radiation. The interpretation of the results suggests that an extended duration of placement for the final portion of the initiator and/or an increased concentration of the last fraction of initiator leads to a higher surplus of unreacted initiator. This scenario also has economic implications as initiator placement at high

conversions could be avoided saving reagent usage. The information reproduced by Monte Carlo involves: conversion of monomer and initiators, number- and weight-average molecular weight and molecular weight distribution [113].

2.10.3. Method of moments

One of the deterministic methods that contribute to the calculation of the average molecular weights distribution in free radical polymerizations is the method of moments. Each moment is accounted for by an index k , which is defined as the order of the moment and that it varies from $M_0 = 0$ to $M_0 = i$ th order, representing the initial conditions until the end of the reaction. It is usual to study the growth of the polymer chain through two species: living polymer chains (λ_{M_0}) and dead polymer chains (μ_{M_0}) [114] described as shown in equations 72 and 73.

$$\lambda_{M_0} = \sum_{i=1}^{\infty} i^{M_0} x P_i \quad (72)$$

$$\mu_{M_0} = \sum_{i=1}^{\infty} i^{M_0} x D_i \quad (73)$$

The quasi-steady state approximation (QSSA) is applicable in this case considering that the propagation speed is greater than the speed of the other sub-reactions, indicating that the moments of the propagating radicals are smaller compared to the moments of dead chains.

Important information is obtained from the first order of moments k , such as:

$Mo = 0$ represent the amount of live and dead chains in the studied volume;

$Mo = 1$ represent the concentration of monomeric units that have been incorporated into the polymer chains;

$Mo = 2$ is related to the variation of the distribution of molecular weights.

As it is already known, the dispersity ($\mathfrak{D}$) is calculated as the ratio of the weight average molecular weight (r_w) with the number average molar weight (r_n) molecular weight distribution ratio ($\mathfrak{D} = \frac{r_w}{r_n}$). Equations 74 and 75 show how the average molecular weights can be calculated [115].

$$r_n = MW \frac{\sum_{i=1}^{\infty} i^1 (P_i + D_i)}{\sum_{i=1}^{\infty} (P_i + D_i)} = MW \frac{(\lambda_1 + \mu_1)}{(\lambda_0 + \mu_0)} \quad (74)$$

$$r_w = MW \frac{\sum_{i=1}^{\infty} i^2 (P_i + D_i)}{\sum_{i=1}^{\infty} i(P_i + D_i)} = MW \frac{(\lambda_2 + \mu_2)}{(\lambda_1 + \mu_1)} \quad (75)$$

Frequently equations 74 and 75 are rewritten as equations 76 and 77 considering that the contribution of the moments of the live chains can be neglected since the dead moments are much higher[115].

$$r_n = MW \frac{\mu_1}{\mu_0} \quad (76)$$

$$r_w = MW \frac{\mu_2}{\mu_1} \quad (77)$$

Developing the mass balance on the dead and living polymers adopting the QSSA explicit algebraic expression for the living moments can be obtained[116].

2.10.4. Final considerations

A literature review reveals that the modeling of solid particles dispersed in liquid media, particularly in flocculation processes, has been extensively studied from the perspectives of both system stability and the involved microscopic and macroscopic forces. There is a scientific consensus on using aggregation and breakage balances as the standard mathematical framework to describe this phenomenon.

Although much of the literature reports average particle sizes as the final metric, the underlying foundation lies in the kinetic balances of birth and death rates. Due to the physical and computational complexity of studying systems containing millions of particles, these rates are typically determined using empirical or semi-empirical equations, which enable the modeling of the system's evolution within the stabilizing liquid medium.

Building on extensive studies of particle stability and the forces involved, recent research has focused on modeling suspended solids during flocculation processes. The PBM is recognized as one of the primary tools for predictive calculations in such systems. PBM and related models are grounded in kinetic principles; however, most reported approaches are empirical, semi-empirical, or purely theoretical. There is a scarcity of studies providing direct

measurements and detailed comparisons of aggregation and breakage rates, particularly under conditions that faithfully reproduce flocculation scenarios of suspended particles in aqueous media, such as MFT. Model validation often relies on tracking mean aggregate diameters over time, an approach that may not fully capture the process dynamics. These calculations require advanced mathematical knowledge to interpret the models.

Kaolinite modeling has gained particular relevance, as this material is widely recognized as a standard in many flocculation studies. Its frequent presence in solid–liquid dispersions and the recurring challenges associated with its stability make it a reference system for researchers. With the advantage that kaolinite is widely known as a benchmark material, it has been adopted in numerous studies of flocculation due to its occurrence in a broad variety of solid–liquid dispersions and the frequent reports of stability issues.

The FBRM technique is also well established and has become a state-of-the-art method for reporting particle size distributions in various solid–liquid systems. Furthermore, the development of flocculating agents has been extensively investigated, with a wide range of products available, and synthesis control techniques have enabled significant improvements in their design.

3. Kinetic Proposal

This chapter outlines the steps required to determine models for the aggregation rates of kaolinite particles in the presence of polyacrylamide.

The procedure includes preprocessing of FBRM data, linear model calculations, and basic concepts of particle dispersion, which combined with a traditional kinetic approach, serve as the foundation for the linear models presented.

3.1. Concept of Modeling

In the scientific literature, modeling is described as a conceptual system designed to represent, describe, or explain the behavior of another system. Its purpose is to support understanding and prediction of how the elements of that system interact, using relations, operations, rules, patterns, or mathematical equations. [117] Figure 10 provides a visual synthesis of this conceptual perspective, illustrating the general strategy by which modeling is commonly presented in academic studies. This theoretical overview clarifies the foundations of modeling as described in the literature, offering a conceptual reference that underpins subsequent analyses.

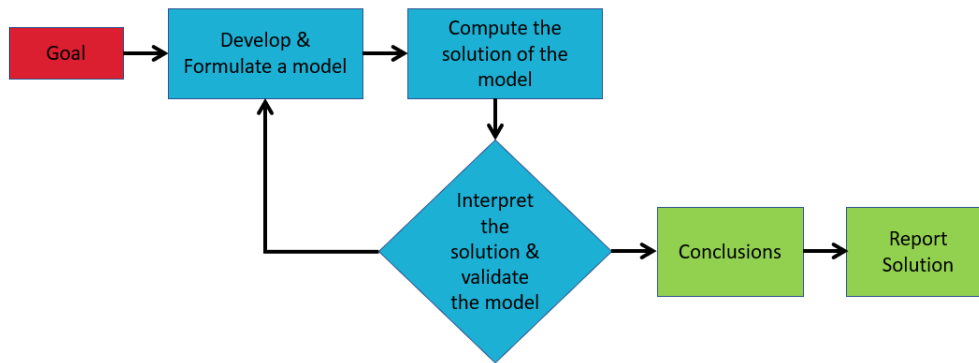


Figure 10. Scheme of the mathematical modeling stage in the present study.

The next subsection presents a conceptual perspective that includes rules and operations, thus illustrating a model that kinetically describes the process of particle aggregation.

3.2. Data Analysis

A summary of the steps followed in the data analysis methodology is presented in Figure 11. The workflow comprises data acquisition with the FBRM probe, preprocessing of the output into chord length distributions, calculation of particle number concentration, estimation of particle conversion, kinetic analysis, and regression modeling.

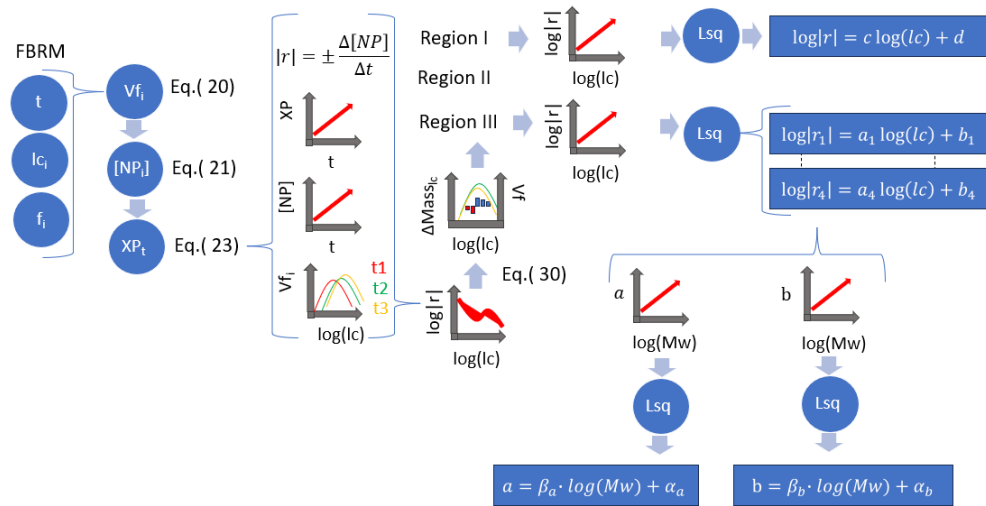


Figure 11. Summary of the steps to follow in this methodology (*Lsq* refers to the method of least square).

As summarized in Figure 11, the proposed methodology for determining rates in the particle aggregation region consists of a sequence of short steps, in which the application of initial equations leads to results that are subsequently analyzed by plotting. This approach enables the identification of trends that ultimately conform to linear models.

To support the development of the analysis, descriptive comments on the samples, technical aspects of the measurement equipment, and references to relevant authors are included, along with the corresponding equations and figures.

The first stage presents the raw FBRM data, primarily used to define the parameters of 200 discrete channels (one channel i for each particle size). Basic equations are then applied to calculate the volume fraction (Vf_i), particle number concentration ($[NP_i]$), and conversion over time (XP_t), thereby initiating the kinetic study of particle-size-

dependent responses through initial plots of Vf_i , XP , and $[NP]$ as functions of time.

These visual results allow for a preliminary calculation of the rate of change in particle concentration, which is plotted as $\log|Rate|$ versus $\log(lc)$. A deeper analysis is then conducted by combining the kinetic response with a variation in particle volume fraction ($\Delta Volume\ fraction$) by particle size, defining domain regions based on specific phenomena, which enables a more logical and structured study of aggregation rates.

These findings lead to the development of linear models, calculated using the least squares method (Lsq), whose responses are further analyzed as a function of the molecular weight of the flocculating agent.

Finally, to validate these models, the obtained aggregation rates were used to invert the calculation and generate simulated values of $[NP]$ and Vf , which were plotted compared with experimental data, demonstrating the predictive capability of the proposed approach.

3.3. FBRM output data preprocessing

The data obtained through FBRM were stored using macro mode and applying the "unweighted" statistical option [81]. The output from the FBRM software is structured as a two-dimensional table, where the horizontal headers represent the midpoint chord lengths (lc_i), in micrometers [μm], corresponding to a total of 200 discrete channels (dch_i). The columns indicate the time intervals (t), thus providing cell coordinates in the format (lc_i, t). Each cell in the table contains a value f_i , which indicates the probability of detecting a particle with a specific chord length at a given time. This configuration enables the calculation

of the mean chord length ($\overline{LM}_0$) based on the total number of unweighted events, according to equation 78 [118].

$$\overline{LM}_0 = \frac{\sum_i f_i lc_i}{\sum_i f_i} = \frac{\sum_i Vf_i lc_i}{\sum_i Vf_i} \quad (78)$$

The fraction volume (Vf_i) at each time (t) and chord length (lc_i) was calculated according to equation 79.

$$Vf_i = \frac{f_i}{\sum_i f_i} \quad (79)$$

To facilitate the calculation, spherical particles will be assumed. The number of particles ($[NP_i]$) in [particle/L] is calculated using equation 80.

$$[NP_i] = Vf_i * \left(\frac{m_k}{\rho_k} \right) * \frac{1 \times 10^{12}}{V_b * \left(\frac{\pi}{6} \right) * lc_i^3} \quad (80)$$

where m_k is the mass of kaolinite in the suspension [g], ρ_k is the density of kaolinite $\left[\frac{g}{mL} \right]$, V_b is the volume of the suspension batch [L].

In the data analysis, a heterogeneous system was studied in terms of particle size, composed of 200 discrete channels obtained through the FBRM (Focused Beam Reflectance Measurement) probe. The discrete channel number (dch_i) is associated with an increase in the chord length (lc_i), following a linear relation described by the equation 81.

$$\log lc_i = 0.018 dch_i + 0.009 \quad (81)$$

3.4. Kinetic approach to the flocculation process

To understand the importance of the kinetic approach to the flocculation process, the particle conversion (XP_t) at each time point can be calculated by equation 82 [119]. This makes the conversion vs. time plot easier to construct.

$$XP_t = \frac{\sum_i^\infty [NP_i]_{t=0} - \sum_i^\infty [NP_i]_t}{\sum_i^\infty [NP_i]_{t=0}} \quad (82)$$

The individual particle conversion per channel can be calculated by 83 [119].

$$XP_{i,t} = \frac{[NP_i]_{t=0} - [NP_i]_t}{[NP_i]_{t=0}} \quad (83)$$

The raw data of $[NP_i]$ were analyzed over the entire duration all the experiment using $[NP_i]$ vs. time (t) plots [120]. By visual inspection, data collected before the onset of the reaction were discarded to obtain real-time reaction tracking. As with any chemical reaction, the approximate initial time point ($t = 0$) was determined based on a notable change in the data slope relative to the baseline. This time point was subsequently validated using the linear region of the rate profile, ensuring that the selected initial time accurately represents the onset of the observed kinetics.[121]. The time course profile of the reaction is

often reported based on the reaction start time which is determined by the kinetic data. According to classical kinetic principles, the reaction rate is expressed as the variation in the concentration of reactants or products with respect to time (equation 84) [122]. Slope analysis of the $[NP_i]$ vs. time (t) plots, combined with estimates from the linear region, formed the basis for calculating reaction rates. To complement this quantitative approach, the same plot-based analysis was conducted using volume fraction (V_f vs t) data, offering a qualitative view of the process under study.

$$r = -\frac{[\Delta_{reactant}]}{\Delta t} = \frac{[\Delta_{product}]}{\Delta t} \quad (84)$$

The conversion versus time data were used to determine the reaction start and end times through graphical methods, specifically the tangent-intercept method (geometric approach) [123][124][125]. The tangent-intercept method is applied to x,y plots (in this case, conversion vs. time), where two segments with different slopes can be observed. The abrupt change in slope reflects a change in the behavior of the system (reaction start or end, regime transition, etc.), which is identified by the intersection point of the tangent lines drawn to each segment [125]

Figure 12 provides a generic representation of the tangent-intercept method, in which the data are grouped into three segments and tangent lines are subsequently drawn. The intersection of these lines identifies the x,y points where a change in the system's behavior occurs.

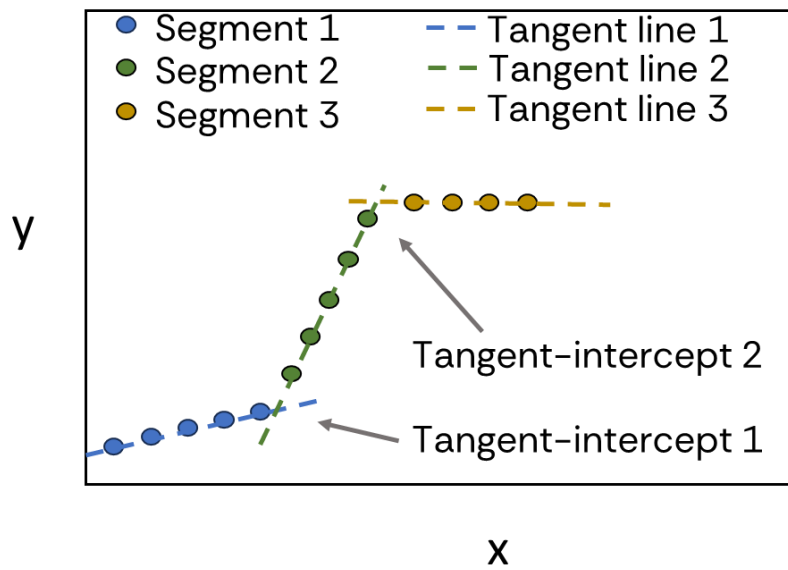


Figure 12 Plot of the tangent intersection method.

3.5. Flocculation from a chemical reaction perspective

The words flocculation reaction are frequently found in literature [126][120][127], to refer to the phenomena of aggregation and breakdown that are in constant contrast in suspensions under flocculation processes[128]. Reversible equations (equation 85) have been proposed to explain the mass balance of these reactions [57][128].



In the direct pathway reaction, P_i and P_j are the particles that react to form the aggregate P_{i+j} . Conversely, in the indirect pathway

reaction, the particle P_{i+j} breaks apart to yield P_i and P_j . A and Q represent the kinetic terms of aggregation and breaking respectively.

PBM models for flocculation take into account mass balance across different types of reactions, which are conceptually introduced here as symmetric and asymmetric reversible reactions [57].

Although this terminology is not standard, it is introduced for explanatory purposes:

In the forward pathway, symmetric reversible reactions (equation 86) involve particles of equal size that combine to form a particle twice the dimensions of the originals. In the reverse pathway, this particle breaks into two equally sized units.



Similarly, asymmetric reversible reactions (equation 85) include, in their forward pathway, particles of different sizes, giving rise to a particle whose size corresponds to the sum of the original magnitudes. If fragmentation occurs, the result is two fragments of unequal dimensions.

Although the distinction between the general equation (equation 85) and the symmetric/asymmetric versions (equations 86–85) may not be immediately apparent, the symmetric and asymmetric equations incorporate four specific kinetic kernels (A , Q) which define the current mechanisms of flocculation through aggregation and rupture event.

3.6. Least squares method.

The least squares method (Lsq) is applied as a key tool for optimizing the determination of experimental results. The data fitting to the linear region of models $y = ax + b$ is performed using the least squares method to obtain the slope (a), the intercept (b), metrics such as the coefficient of determination (R^2). Where y represents the dependent variable and x is the independent variable.

The equations for applying the least squares method [129] represented as deviations from the mean are recalled below.

Averages of x equation 87 and y equation 88:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad (87)$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} \quad (88)$$

The slope (a) equation 89 and the intercept (b) equation 90.

$$a = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (89)$$

$$b = \bar{y} - a\bar{x} \quad (90)$$

The fitted values $\hat{y}$ for the straight-line equation 91.

$$\hat{y} = ax_i + b \quad (91)$$

The residual e_i for each i data equation 92.

$$e_i = y_i - \hat{y} \quad (92)$$

Estimated variance (s^2) equation 93 and standard deviation (s_0) equation 94.

$$s^2 = \frac{\sum_{i=1}^n e_i^2}{n - 2} \quad (93)$$

$$s_0 = \sqrt{s^2} \quad (94)$$

3.7. Sample Dispersion by Particle Size

Cumulative distribution percentiles (DXX) are descriptive statistical measures that indicate the maximum particle size such that a specific percentage (XX) of the total particles have a size equal to or less than that value. That is, in the case of $D10$, it indicates that 10 % of the solid components have a chord length equal to or less than $xx.xx \mu\text{m}$. The same interpretative framework applies to $D50$ and $D90$, as they are likewise cumulative distribution percentiles that reflect the particle size below which a given percentage of the sample is contained [130–132].

The description of polydispersity in a sample containing various particle sizes can be addressed using metrics such as cumulative distribution percentiles ($D10$, $D50$, $D90$), *Span* (equation 95), and the *Geometric amplitude index* (equation 96) [130–132].

$$Span = \frac{D90 - D10}{D50} \quad (95)$$

$$Geometric\ amplitude\ index = \frac{D90}{D10} \quad (96)$$

3.8. Parameter estimation

The different models presented in the kinetic and modeling part of the flocculation process will be written in Python programming language. To perform the estimation of the unknown model parameters can be used: Linear Program (LP), Non-Linear Program (NLP), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Constraint Programming, etc. The model parameters will be estimated by minimizing the objective function ($F(\alpha_{max}, \dots, s_2)$) given in equation 97

$$F(\alpha_{max}, \dots, s_2) = \sum_{t=0}^{max} (d_{exp} - d_{model})^2 \quad (97)$$

where d_{exp} is the average experimental diameter of the particles and d_{model} is calculated by the equation 98

$$d_{model} = \frac{\sum_{i=1}^{max} n_i d_i^4}{\sum_{i=1}^{max} n_i d_i^3} \quad (98)$$

To evaluate the model, it is proposed to use: Coefficient of determination ($R^2, 0 \leq R^2 \leq 1.0$) and Pearson correlation coefficient ($\rho_{prs}, -1.0 \leq \rho_{prs} \leq 1.0$) together with Mean Absolute Percentage Error (MAPE) as described by the equations 99, 100 and 101.

$$R^2 = 1 - \frac{\sum_{j=1}^{j_{max}} (d_{exp,j} - d_{model,j})^2}{\sum_{j=1}^{j_{max}} (d_{exp,j} - \bar{d}_{exp,mean})^2} \quad (99)$$

$$\rho_{prs} = \frac{\sum_{j=1}^{j_{max}} (d_{exp,j} - \bar{d}_{exp,mean})(d_{model,j} - \bar{d}_{model,mean})}{\left(\sum_{j=1}^{j_{max}} (d_{exp,j} - \bar{d}_{exp,mean})^2 \sum_{j=1}^{j_{max}} (d_{model,j} - \bar{d}_{model,mean})^2 \right)^{\frac{1}{2}}} \quad (100)$$

$$MAPE = \frac{100}{j_{max}} \sum_{j=1}^{j_{max}} \left| \frac{d_{exp,j} - d_{model,j}}{d_{exp,j}} \right| \quad (101)$$

where j_{max} is the size of data set studied.

3.9. Final considerations

A concise methodology was presented, based on a traditional kinetic approach widely used in the field of chemistry.

It is proposed that flocculation processes can be described using balance equations analogous to chemical reaction representations, particularly in their atom-to-atom interaction form.

This suggests that the application of such concepts is feasible in the study of particle flocculation.

Additionally, key concepts related to least squares analysis and particle dispersity are introduced, as they are considered essential for a deeper understanding of the system.

4. Materials and Methods

The methodology was organized into short explanatory procedures that explain the methods of polymer synthesis, their mixtures, and characterization. This is followed by a description of the flocculation experiment.

4.1. Materials

Acrylamide, methyl 2-[methyl(4-pyridinyl)carbamoithioylthio] propionate (RAFT agent), 2,2'-azobis(2-methylpropionamidine) dihydrochloride (V-50 initiator), hydroxymethanesulfinic acid monosodium salt dihydrate (NaFS), ammonium persulphate (APS), kaolinite, and ethanol were purchased from Sigma-Aldrich, a chemical supplier..

4.2. Experimental Section

The experimental methodology was described in detail in a previous publication by Dixon and Soares [59][133], , with the exception that the synthesized polymer is a homopolymer of PAM.

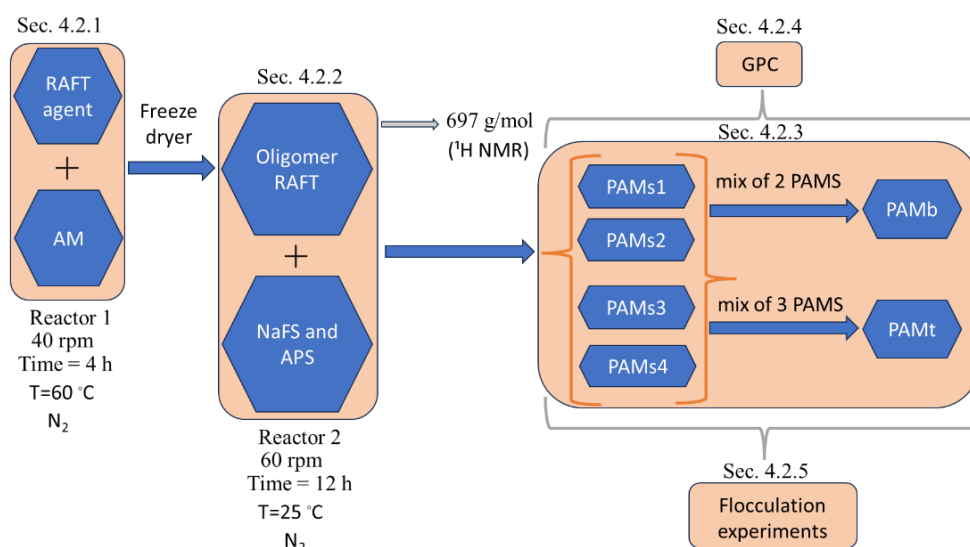


Figure 13. Block Diagram of Experimental Methodology

4.2.1. Synthesis of oligomer RAFT agent

Methyl 2-[methyl(4-pyridinyl)carbamothioylthio]propionate was dissolved in a 1:1 ethanol solution to prepare the RAFT agent. Acrylamide was also dissolved in the solution at a molar ratio of 7:1 (acrylamide to RAFT agent). The PAM oligomer was synthesized under a nitrogen atmosphere while being stirred and heated to 60 °C. V-50 at a concentration of 2.5×10^{-3} M, was injected into the solution. The reaction was terminated after 4 hours. Ethanol was evaporated, and the product was freeze-dried. The structure of the PAM oligomer RAFT agent was confirmed by proton nuclear magnetic resonance (¹H NMR), which showed six acrylamide units per oligomer molecule, with a molecular weight of 697 g/mol calculated from the NMR data [133].

4.2.2. Synthesis of PAM homopolymer

The RAFT oligomer agent was added to a 10 wt% acrylamide solution in water, with the monomer to RAFT agent ratio adjusted to

achieve the desired target molecular weight. Prior to synthesis, the solution was purged with nitrogen for 45 minutes, and the pH was adjusted to 6. Polymerization was carried out at room temperature using equimolar amounts of NaFS and APS as a redox initiator system, a concentration the APS was of 60 ppm relative to acrylamide. The synthesis was carried out at room temperature under stirring overnight. The PAM formed was then precipitated in acetone and freeze-dried. The freeze-dried polymer was then analyzed by gel permeation chromatography (GPC) and prepared for flocculation experiments [133].

4.2.3. PAM blends samples.

Seven PAM blends were prepared by combining equal mass fractions of the PAMs samples. Five binary mixtures are made in 50/50 wt% ratios (PAMb) and same way, two ternary mixtures are made 33/33/33 wt% ratios (PAMt). The indices following the letters "b" (binary) and "t" (ternary) refer to the identifiers of the individual narrow-distribution PAM samples used to create each blend [133].

4.2.4. Polymer Characterization by gel permeation chromatography (GPC)

PAM samples were analyzed using gel permeation chromatography (GPC) with the Agilent 1260 Infinity Multi-Detector Suite, consisting of two TSK-GEL G6000PWxl-CP columns in series, a differential refractive index detector, a dual-angle light scattering detector, and a viscometer. The GPC system was calibrated using

polyethylene oxide standards, and all samples were analyzed in a 0.2 M sodium nitrate solution [133].

4.2.5. Flocculation Experiments

The kaolinite dispersion was prepared at 2.5 wt% in deionized water with pH adjusted to 8.0 using NaOH and 1000 ppm CaCl_2 (0.2 mM). The dispersion was homogenized for no less than 72 hours to achieve stable and reproducible particle suspensions. The solution polymer flocculant, PAM, was prepared as 2 mg/mL solution 24 hours prior to the flocculation experiment to ensure complete dissolution. The volume of flocculant solution added was calculated based on the applied dose, following the proportion expressed in ppm of flocculant, which was defined as milligrams of PAM per kilogram of kaolinite (mg PAM / kg kaolinite). Each experiment was conducted in a 250 mL beaker using 100 mL batches of kaolinite suspension, mixed at 200 rpm with a pitched 4-blade impeller. The particle size was recorded at 2-second intervals using a Focused Beam Reflectance Measurement (FBRM) probe (G400 model, Mettler Toledo). The data collected from the FBRM were stored in macro mode, and the mean chord length was analyzed using the unweighted option [133].

Two doses of the PAM solution were added to the kaolinite suspension. The entire experiment was performed at room temperature (around 20 °C). An initial dose of 1000 ppm PAM was added and monitored for 3 minutes. Subsequently, a second dose of 1500 ppm was added to reach a total concentration of 2500 ppm PAM, which was then monitored for an additional 4 minutes. Four different PAM homopolymer samples, each with distinct molecular weight distributions, were studied as flocculating agents in a kaolinite suspension [133].

5. Results and discussion

The general results are presented, including the characterization of the polyacrylamide polymer, the particle size distribution of kaolinite, and a proposed evaluation of the system using a traditional kinetic approach. In particular, each step is described in detail to clarify how the partial graphical results were interpreted and how they guided the initiation and progression of the kinetic proposal, ultimately leading to the development of the final linear models

5.1. Synthesized PAM homopolymer

Four different PAM homopolymers were synthesized and analyzed using gel permeation chromatography (GPC). The resulting average molecular weight distributions for each sample are reported in Table 2.

Table 2 Molecular weight averages of the PAM samples used in the flocculation experiments.

Sample	M_n (Da)	M_w (Da)	M_z (Da)	$\bar{D}$
PAMs1	3.06×10^5	3.11×10^5	3.16×10^5	1.04
PAMs2	8.79×10^5	9.26×10^5	9.70×10^5	1.05
PAMs3	2.60×10^6	2.86×10^6	3.11×10^6	1.10
PAMs4	2.63×10^6	3.55×10^6	4.39×10^6	1.35

The polymerization using the RAFT synthesis method was successful, considering the degree of polymerization (DP), which ranged approximately from 4,300 to 50,000 acrylamide monomer units. All the PAM homopolymer samples exhibited consistent polydispersity values of $\bar{D} \leq 1.5$ [133]. Sample uniformity was most pronounced in the lower

molecular weight ranges, as the RAFT agent ensures more effective control by acting as a mediator in the polymerization reaction. The RAFT agent regulates monomer incorporation and promotes a narrower molecular weight distribution, leading to reduced polydispersity and a more defined polymer architecture [133].

5.2. Prepared PAM Blend Formulations

Table 3 presents the GPC characterization results of the samples prepared by mixing individual polymer components. As expected, the molecular weight of the blends falls within the range defined by the original PAMs samples. Regarding dispersity, most of the blends are classified as polydisperse, with the exception of PAMb34, which can be considered monodisperse, since its dispersity index ($\mathcal{D}$) is below 1.5, a threshold above which samples are typically classified as polydisperse [134].

Table 3. Molecular weight averages of the PAM blends samples used in the flocculation experiments.

Sample	M_n (Da)	M_w (Da)	M_z (Da)	$\mathcal{D}$
PAMb12	3.78×10^5	6.23×10^5	8.02×10^5	1.65
PAMb13	4.77×10^5	1.58×10^6	2.84×10^6	3.32
PAMt123	5.28×10^5	1.37×10^6	2.42×10^6	2.59
PAMb23	1.06×10^6	1.89×10^6	2.59×10^6	1.78
PAMb24	1.06×10^6	2.24×10^6	3.68×10^6	2.11
PAMt234	1.32×10^6	2.45×10^6	3.47×10^6	1.86
PAMb34	2.54×10^6	3.20×10^6	3.84×10^6	1.26

5.3. Analysis of FBRM Raw Data

As a consequence of the polydispersity of the kaolinite system analyzed in this study, and of the flocculation reaction models proposed in the literature (equations 85 to 86), which consider the simultaneous interaction between two elements, it becomes necessary to consider the hypothesis that reactions occur both in parallel and consecutively [135][136][137]. This hypothesis is essential for understanding part of the complexity involved in the formulation of kinetic models.

In this context, the problem can be effectively addressed through the tracking of individual signals, which is one of the advantages of having channels defined by particle size. The FBRM technique, which reports for each lc_i its corresponding probability of detection, enables the measurement of $d \frac{[NP_i]}{dt}$ as an individual gradient associated with each particle size class.

The calculations of volume fraction and particle number, as presented in equations 79 and 80, are routine and straightforward. These computations ensure accurate unit expression, thereby facilitating the commencement of subsequent analytical procedures. Although the calculations of volume fraction and particle number are not frequently presented in the literature [79], an alternative method for verify these parameters is the calculation of the mean chord length ($\overline{LM}$) according to equations 78. Direct comparison of an output file from the FBRM software with the mean chord length and the reproduced calculation of ($\overline{LM}$) yielded identical results throughout the experimental period studied (Figure 14). These results confirm the methodological robustness of the adopted approach, as the mean chord length ($\overline{LM}$) can be reliably calculated from volume fraction (Vf_i)

values, thereby supporting the accurate interpretation of the raw data obtained via FBRM.

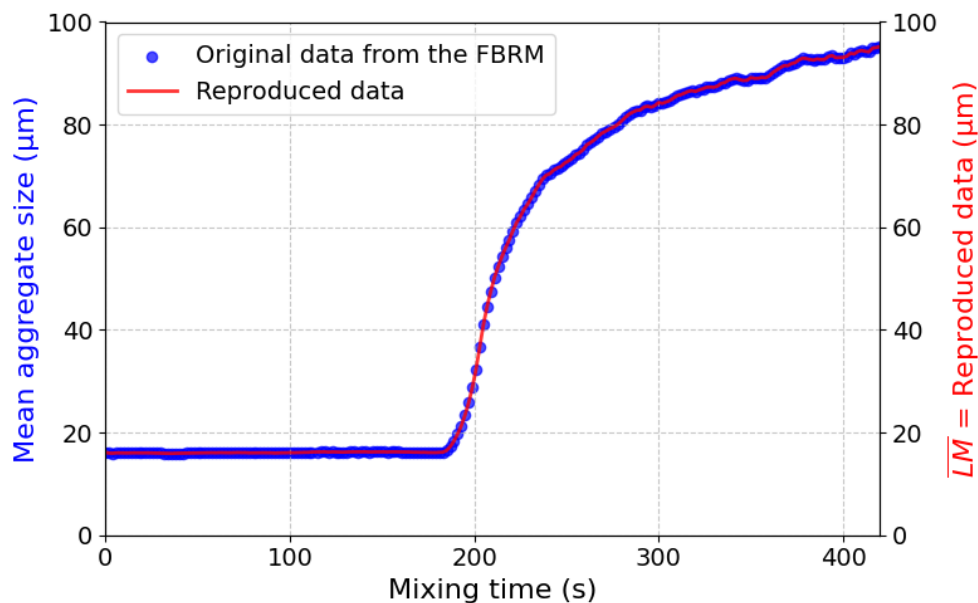


Figure 14. Dynamic analysis of Mean aggregate size of the flocculation process over FBRM modo macro, unweighted option. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1.

Figure 13 serves as a verification that the proposed method for calculating mean particle sizes, derived from the raw frequency data provided by the FBRM probe, reproduces the internal calculations performed by the instrument's software.

5.4. Kaolinite chord length distribution by FBRM.

The kaolinite suspension was evaluated using FBRM at mixing time zero, corresponding to the start of each experiment. The stock solution, prepared at a concentration of 2.5 wt% kaolinite and stabilized over 72 hours, enabled a standardized mixture to be consistently used

throughout all experimental runs. The particle size distribution evaluated between 1.02 μm and 3917.90 μm (Figure 15) revealed that values above 182.09 μm contribute negligible volume fractions, defining an effective range limited to 1.02–182.09 μm . Within this interval, a mean chord length of 16.15 μm , a standard deviation of 16.98 μm , and a variance of 288.38 μm^2 were obtained, values consistent with a highly dispersed distribution. The mode of the distribution was 15.77 μm , and the D10, D50, and D90 values were 2.88 μm , 10.85 μm , and 34.66 μm , respectively, indicating that most particles are fine and partially describing the shape of the distribution curve. However, the presence of an extended tail toward larger diameters, even if infrequent, results in a positive skewness (2.25), as observed in the results obtained with the FBRM probe.

The chord length distribution is dominated by polydisperse particles, which contain a significant proportion of fine particles and exhibit a high standard deviation. One of the advantages offered by the FBRM probe is its ability to simultaneously measure the probability of particle detection in each channel (f_i), which reinforces the interpretation that the volume fraction is directly related to the particle concentration per channel. Furthermore, this advantage enables a more accurate characterization of size distribution in heterogeneous systems, facilitating the kinetic analysis of flocculation reactions occurring simultaneously for each solid element according to its size.

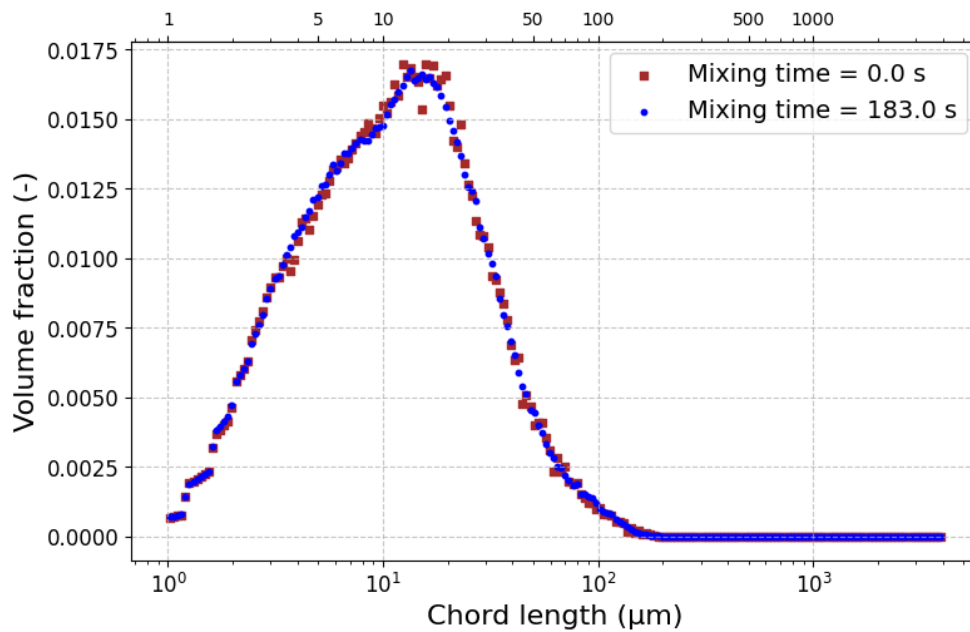


Figure 15. Kaolinite Chord length Distribution.

Figure 15 compares the Cld of kaolinite particles at two different time intervals, 0 s and 183 s. The figure shows that the two Cld scans, evaluated at these intervals, exhibit only minor variations. The evident result is that no significant change occurred in the particle distribution, indicating behavior consistent with a stabilized system in which solids do not aggregate, a process that would otherwise reduce the volume fraction. Furthermore, no evidence of erosion was observed that might increase the number of final particles or lead to higher volume fraction values.

5.5. Particle conversion proposal

Dynamic analysis of the flocculation process over time is traditionally conducted by monitoring changes in average particle size, typically plotted as mean aggregate size versus mixing time (Figure 14). This graphical approach enables visualization of the progressive growth of particulate aggregates and provides a framework for assessing

system stability [23][59]. The proposal to incorporate particle conversion (XP_t), either by considering all channels collectively (equation 82, Figure 16) or through individual channel analysis (equation 83, Figure 17) offers the potential to yield additional insights into the mechanisms underlying the flocculation process. Specifically, it provides information on the variation of kinetic behaviors, such as the actual initiation of the flocculation process, the presence of a steadily increasing linear response, and the onset of a stabilization region in the kinetics. All of these observations allow the formulation of hypotheses regarding the presence of different mechanisms involved in the studied phenomenon.

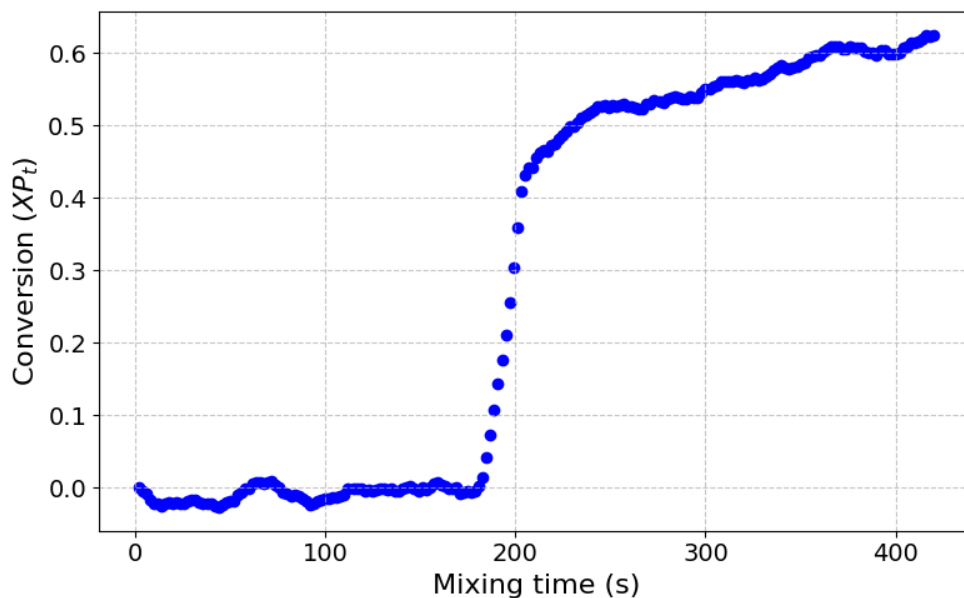


Figure 16 Dynamic analysis of particle conversion (XP_t) of the flocculation process over FBRM modo macro, unweighted option. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1.

Appendix B contains a set of total conversion curves, which include the response of all channels simultaneously, demonstrating the

feasibility of applying the technique to evaluate the flocculation of kaolinite with different PAM polymers used in this study.

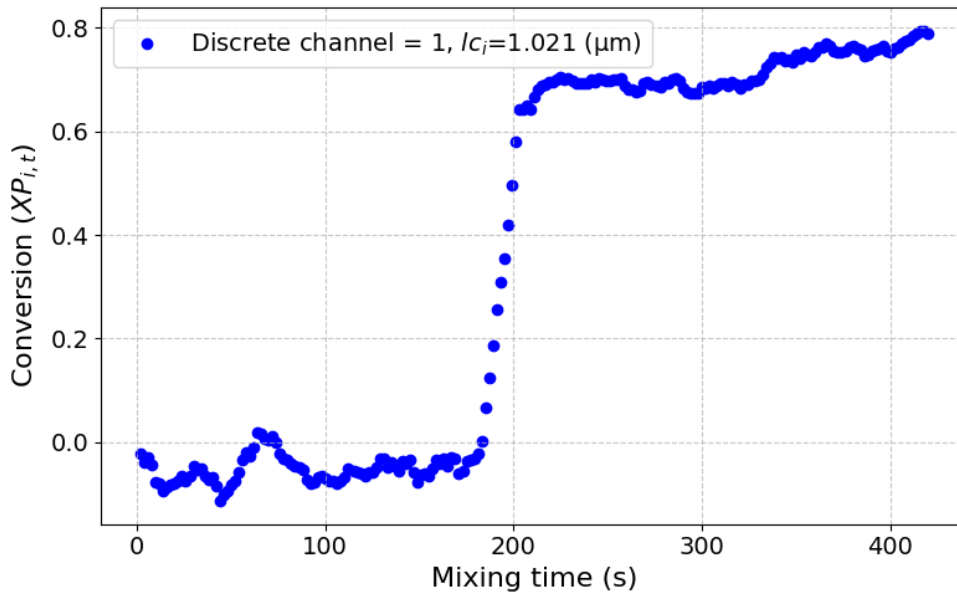


Figure 17. Dynamic analysis of individual particle conversion ($XP_{i,t}$) of the flocculation process over FBRM modo macro, unweighted option. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1.

In comparison with processes in which the conversion curve represents the mass of the reactants or products of a reaction, Figure 16 and Figure 17 illustrate the variation in the number of free particles within the dispersed system in the presence of a flocculating agent. When comparing the readings of the two systems, the importance becomes evident, since time–conversion curves are widely employed in kinetics to differentiate one condition from another (for example, different catalysts, different temperatures, etc.)[138]. The direct readings of the current conversion curves will be presented progressively as the text develops in greater depth.

5.6. Flocculation of kaolinite polymer

To enhance the understanding of the flocculation experiment, allocating individual sections for each polymer dosage facilitates more effective interpretation and comparison of the results.

5.6.1. Dosage of 1000 ppm of flocculant agent

Figure 14 shows how particle diameters change over time under the influence of flocculating agents. In the first section of the plot (up to 183 seconds), no significant changes in particle size are observed. In the second section, a rapid increase in size is noted, eventually reaching apparent stabilization toward the end of the plot. This temporal evolution is reflected in the plot's reading. However, a less obvious aspect arises when aggregates grow in size, leading to the conclusion that there is a decrease in the number of dispersed particles. This analysis occurs in order to maintain the system's mass balance, which starts from a fixed mass of solid particles.

On the other hand, the conversion plot in Figure 16, also provides a direct interpretation, quantitatively showing how the number of particles in suspension decreases over time as a result of the flocculating agent. Likewise, indirect analysis is based on the principle of mass conservation: the reduction in particle count is explained by agglomeration, which by association implies the growth of dispersed elements. Taken together Figure 14 and Figure 16, both time-dependent plots complement each other and offer a more pragmatic and realistic interpretation of the physical behavior being studied.

When comparing Figure 14 and Figure 16, it becomes evident that the conversion data (Figure 16) is more sensitive to changes in the

experimental system due to fluctuations. Equation 82 defines conversion as the fraction of particles remaining in suspension, therefore negative conversions are neither physically possible nor acceptable. The fluctuations toward negative values observed in Figure 16 and Figure 17, within the interval from 0 to 120 seconds, are attributed to disturbances caused by the introduction of the first flocculant dose, generating localized concentration gradients for a short period of time, making the FBRM probe measurement difficult [139]. Another reason for negative values is more mathematical than physical and consists of the selection of the value of $\sum_i^{\infty} [NP_i]_{t=0}$ value, which serves as the denominator in equation 82, and directly affects the magnitude of the conversion output when the numerator fluctuates due to initial disturbances in the system. In the interval from 120 to 183 seconds, determined by the graphical slope-change method (tangent method.) [123][124][125], the data presented in Figure 16 and Figure 17 show greater stability, confirming that the mixture of the two liquids is the reason for the fluctuations in conversion to negative

Observations from Figure 15 at mixing times of 0 and 183 seconds, the particle distribution is virtually identical. This allows us to affirm that the 1000 ppm dose of PAM1s did not act as a flocculating agent, a behavior that is repeated in all PAM samples. Although flocculation studies traditionally exclude results where no aggregation is observed, in this singular case the data provides relevant insights: it confirms that the suspended solids in the aqueous solution are primary particles. Thus, the definition of primary particles is extended to the set of solid elements dispersed in the liquid medium prior to the onset of flocculation, which also do not undergo erosion or breakage under shear conditions induced by mechanical agitation [62][140], remaining unchanged in both size and quantity. It is also interesting to postulate

that, when starting flocculation experiments with primary particles, the aggregation phenomenon must precede floc breakage; otherwise, breakage would not be feasible, as the primary particles possess mechanical stability that shear stress cannot overcome [140].

5.6.2. Concentration of 2500 ppm of flocculant agent

As outlined in the experimental procedure, a concentration of 2500 ppm of PAMs is achieved through the addition of two doses of the flocculating agent, using mixing time as a reference: 1000 ppm at 0 second and 1500 ppm near the 183 second mark. Figure 16 and Figure 17 show a marked change in $\frac{dXP_{i,t}}{dt}$, with a linear behavior. Based on the classical tactic for real-time detection of chemical events, where real-time reaction is a period between on-set and off-set reaction typically refers to the duration of a reaction. The onset of a reactive process is identified by the clear emergence of a signal that exceeds the noise threshold of the monitored variable throughout its temporal evolution (e.g., temperature, mass, color, concentration, etc.) [121][141]. Figure 16 and Figure 17 provide complementary evidence that reinforces the importance of kinetic analysis at this specific stage of the flocculation process.

Figure 16 shows that approximately 40% of the system's particles agglomerate between 183 and 203 seconds. Beyond 203 seconds, the agglomeration efficiency decreases to around 20%. From a kinetic standpoint, the change in slope relative to the linear region highlights a transition in the agglomeration regime. This shift will be addressed in a subsequent analysis. Although this plot Figure 16 stems from a statistical interpretation assuming a single particle class, it reveals a zone of high agglomeration activity. This observation becomes even

more pronounced in Figure 17, where channel 1 displays a stronger behavior: more than 60% of the particles undergo agglomeration within the same time frame. The contrast between a global approach and an individualized one suggests that this time interval represents a critical kinetic region, in which system conditions strongly favor particle transformation. Therefore, this finding not only contributes relevant data but also defines a strategic time window for deepening the understanding of flocculation mechanisms, enabling predictive model optimization and enhancing the system's operational efficiency

Figure 18, a three-dimensional plot, shows the reference adjustment made to account for real-time reaction, projecting the particle concentration behavior $[NP_i]$ at the beginning and end along the left and right walls. Additionally, a partial presentation of the concentration variation over time is included. As the kaolinite sample under study corresponds to a polydisperse system, a partial representation of the channels was chosen to avoid saturating the image. In particular, the trace of channel 1 is highlighted with a thicker continuous line to indicate that the observed slope represents the linear part shown in Figure 17, but expressed as a function of $[NP_i]$.

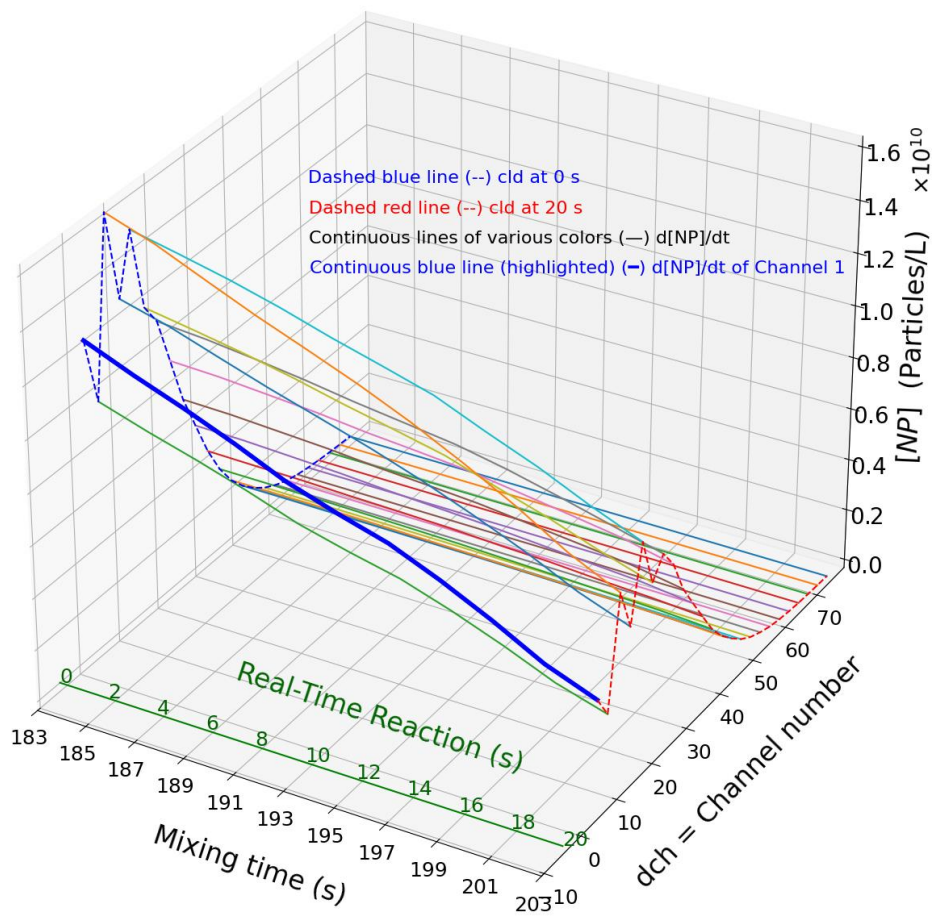


Figure 18. 3D graph showing the reference changes from mixing time to real-time reaction, with projections of particle concentration behavior per channel (remember $\log lc_i = 0.018 dch_i + 0.009$) at the beginning and end, including slope variation in some channels. The trace of channel 1 is highlighted with a thicker continuous line. Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade whit PAMs1

The dependency of $[NP_i]$ on the temporal dynamics within the flocculation reaction region (Figure 18) exhibits a linear behavior, which allows for the calculation of the particle disappearance rate (equation 84) as the slope obtained by applying the least squares method equation 89. Figure 19 presents the R^2 values for different particle sizes. Each point corresponds to the R^2 obtained from plots of $[NP]$ versus

time. Most values exceed 0.9, confirming the predominance of linear models as expected. Within this group, several values are very close to 1, reinforcing the strong alignment with straight-line behavior in certain channels. In contrast, channels near 85 show a marked decrease in R^2 , indicating that these specific channels do not exhibit linear behavior.

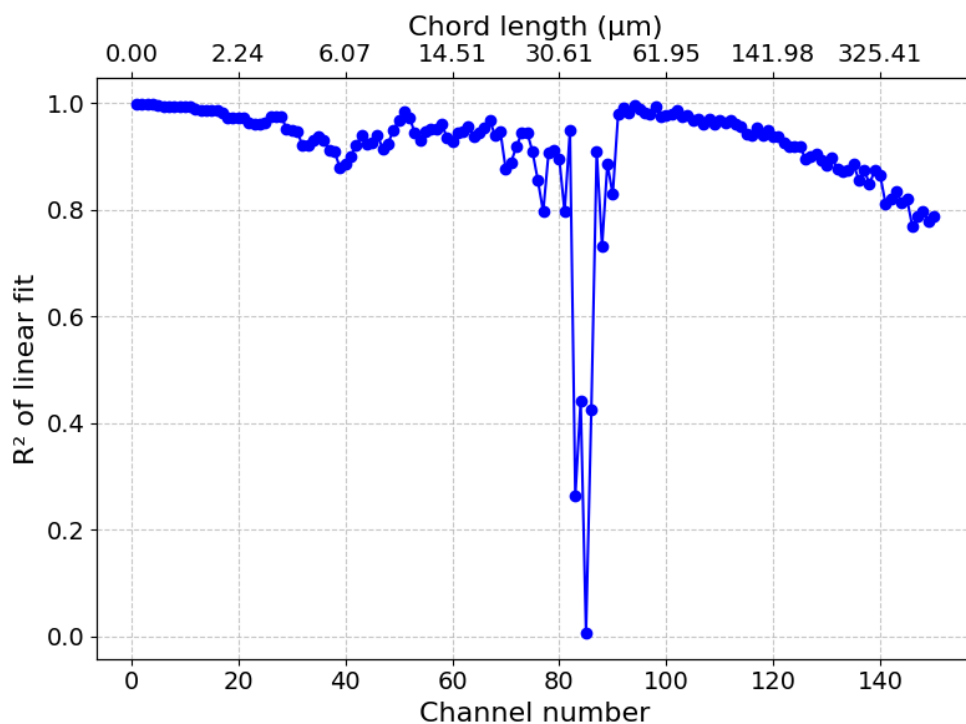


Figure 19. Linear Behavior of [NP] Over Time Assessed by R^2 Across Channels.

The Population Balance Model (PBM) [57] has its historical origins in the works of Smoluchowski (1917) [95] and includes subsequent contributions such as those of Lu et al. (1998) [64], among others. It is evident that the PBM cannot be attributed to a single author, since it remains under continuous development due to ongoing discoveries. This model demonstrates that particle aggregation rates in suspensions under the influence of flocculating agents have long been theorized. The PBM allows the existence of such rates to be expressed through

empirical, semi-empirical, and even phenomenological equations; nevertheless, direct values are difficult to find in the literature. Consequently, the aggregation rate is usually represented only implicitly within broader mathematical models, rather than as an independent value. Additional information regarding the kinetic approach can be found in Appendix C.

As highlighted, the methodology employed and analyzed in this work allows, for the first time, the measurement of the aggregation rate of kaolinite particles in the presence of water-soluble PAM polymers acting as flocculating agents.

5.7. What does linear response mean at the beginning of flocculation?

As shown in Figure 18, the temporal response of particle concentration in individual channels tends to be linear. From a kinetic standpoint, this is only possible if the aggregation or breakup phenomena were occurring separately. Since particle erosion or fragmentation was ruled out, based on the confirmation that the suspended kaolinite solids in these experiments are primary particles, this behavior aligns with the proposal outlined by Lu et al. (1998) [43]. In that work, the author defines a model under the name of hydrophobic aggregation (equations 30–32), where particle growth occurs individually until a critical time is reached, at which point floc disruption events are triggered. The concept that breakage mechanisms are triggered after reaching a critical size under a given agitation intensity is supported by equation 13, proposed by Tambo and Hozumi in 1979 [95]. Regarding kaolinite, several authors have reported that its wettability varies depending on the exposed crystal face. Specifically, it has been observed that the silica face of kaolinite exhibits a moderate

level of hydrophobicity, while the alumina face is predominantly hydrophilic [142]. This surface anisotropy directly influences particle behavior during flocculation. Therefore, the linear behavior observed in the initial stage of the aggregation process aligns with the hydrophobic aggregation model, in which interactions between partially hydrophobic surfaces promote a constant aggregation rate, independent of concentration. This pattern suggests that, in the early stages of the flocculation process, the orientation of kaolinite particles and the exposure of specific crystal faces with greater hydrophobicity play a decisive role in the aggregation kinetics.

The linear decay of particle concentration over time ($[NP_i]_t$) can be described by equation 102 [143], which is used to represent zero-order reactions. In such kinetic processes, the initial particle concentration ($[NP_i]_{t=0}$) decreases at a constant rate (k) over time (t), reflecting a concentration-independent behavior typical of systems where the reaction rate does not depend on the amount of particles present. As a result, the particle concentration decreases with a constant slope over time [143].

$$[NP_i]_t = [NP_i]_{t=0} - kt \quad (102)$$

The time derivative of equation 102 leads to the expression for the particle variation rate, showing that this rate remains constant over time. These results support the hypothesis that the process follows zero-order kinetics, in which the $\frac{d[NP_i]_t}{dt} = -k$ [144]. By definition, the concentration gradient with respect to time is constant and independent of the variation of concentration of reactants [145][146]. In such systems, the reaction rate is influenced by external factors such

as particle size [147] or the saturation of active sites on the catalyst surface [148].

The case of Mature Fine Tailings (MFT) [23], a stable colloidal suspension of solid particles, provides valuable empirical evidence regarding aggregation dynamics in particulate systems. These mining residues originate with a high liquid content, and in an effort to separate the solid from the liquid phase, mining companies spread the tailings over large land areas, thereby promoting the progressive evaporation of water. Over the years, this process leads to a sustained increase in the concentration of solids within the suspension. Nevertheless, MFT maintain their colloidal stability [23], which suggests that particle concentration is not the dominant parameter governing the aggregation rate of suspended elements. This behavior is consistent with zero-order aggregation kinetics, in which the aggregation rate remains constant regardless of the number of particles present in the system [145].

5.8. Kinetic domain regions

The response of the reaction rate behavior with respect to the of chord length (lc) was analyzed in order to establish a functional relationship that enables the study of the chaotic behavior of the kaolinite suspension subjected to flocculation treatment with PAM. As shown in Figure 20, this relationship is illustrated through the application of a logarithmic transformation to the lc data.

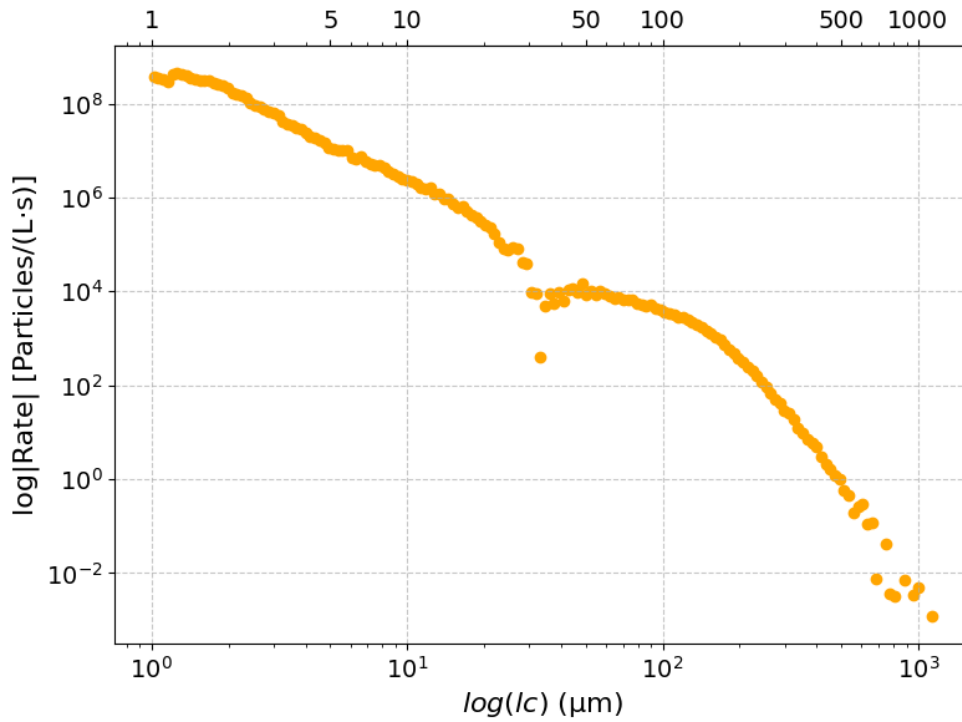


Figure 20. Response of the rate behavior with respect to the logarithm of chord length (lc).

Figure 20 initially shows a decrease in the flocculation rate as the size of kaolinite particles increases. To facilitate the interpretation of these results, Figure 21 displays the variation in volume fraction as a function of particle size (represented as $\log(lc)$) at 20 and 0 seconds, serving as a reference for constructing a variation in particle volume fraction based on the real-time evolution of the process. By applying, for each individual particle size, the difference between the volume fraction data at 20 seconds and 0 seconds, as indicated in equation 103, the variation in particle volume fraction across different size ranges can be determined (Figure 22). It should be noted that the experiment takes place at room temperature, and that the difference in volume fraction in equation 103 is equivalent to the difference in volume fraction, since the consolidated solid particles maintain a constant density over time.

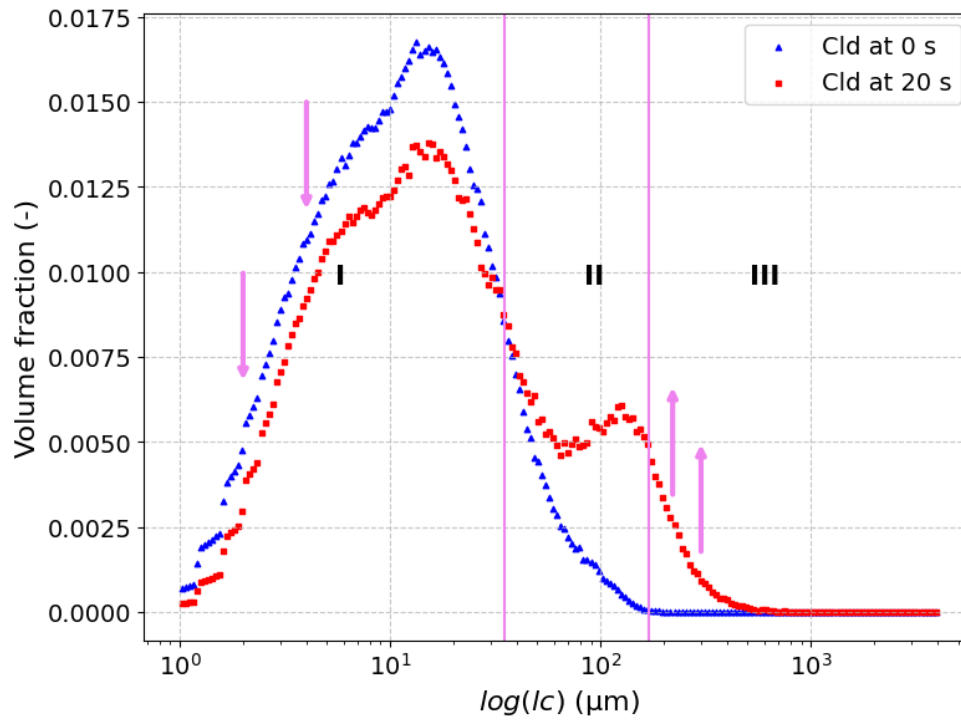


Figure 21. Kaolinite Chord length distribution (*Cld*) at 0 and 20 s in Real-time reaction.

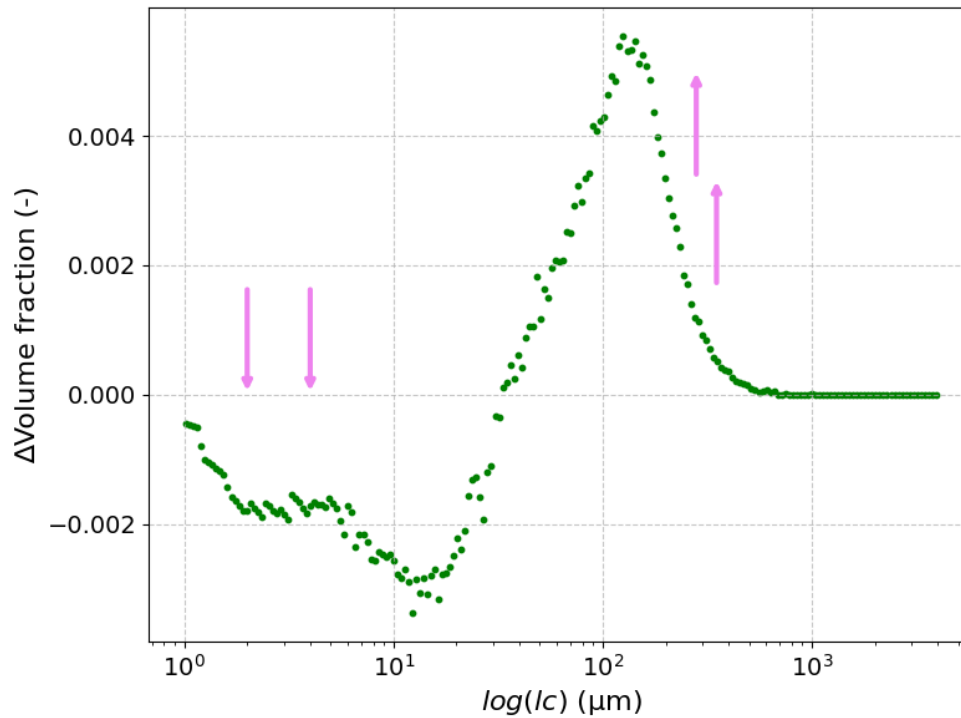


Figure 22 Variation in particle volume fraction ($\Delta Volume Fraction$) between 20 and 0 seconds distributed by particle size.

Figure 22 illustrates the response of the aforementioned Variation in particle volume fraction, where negative values are interpreted as indicators of particle disappearance or death, while positive values reflect the birth or formation of new particles within that size range. Drawing an analogy with a chemical reaction, the reduction in particle volume fraction can be associated with consumed reactants, whereas the increase in volume fraction corresponds to the generated products.

The superposition of Figure 21 and Figure 22 results in Figure 23, in which distinct regions can be identified where particle disappearance (death) or formation (birth) processes predominate, clearly separated by size ranges. This segmentation enables a more structured and precise interpretation of the system's dynamic behavior, facilitating the analysis of transitions between dominant mechanisms. In this context,

Region II (Figure 23) presents an interpretative challenge, as particles are already present at the onset of the flocculation process. However, the volume fraction balance reveals an increase in particle count by size, suggesting that this region, delimited by Regions I and III, may be undergoing simultaneous birth and death phenomena. This overlap of mechanisms precludes a linear model description, requiring a more flexible analytical approach to accurately capture the system's dynamics

Analysis of these data enables the classification of the observed events into three subtypes of phenomena, validated through a simplified volume fraction balance. These regions are visually delimited in the plot by vertical violet lines and are denoted with Roman numerals I, II, and III. Interpreting the increase or decrease in particle number through changes in volume fraction is consistent, since both quantities are directly proportional.

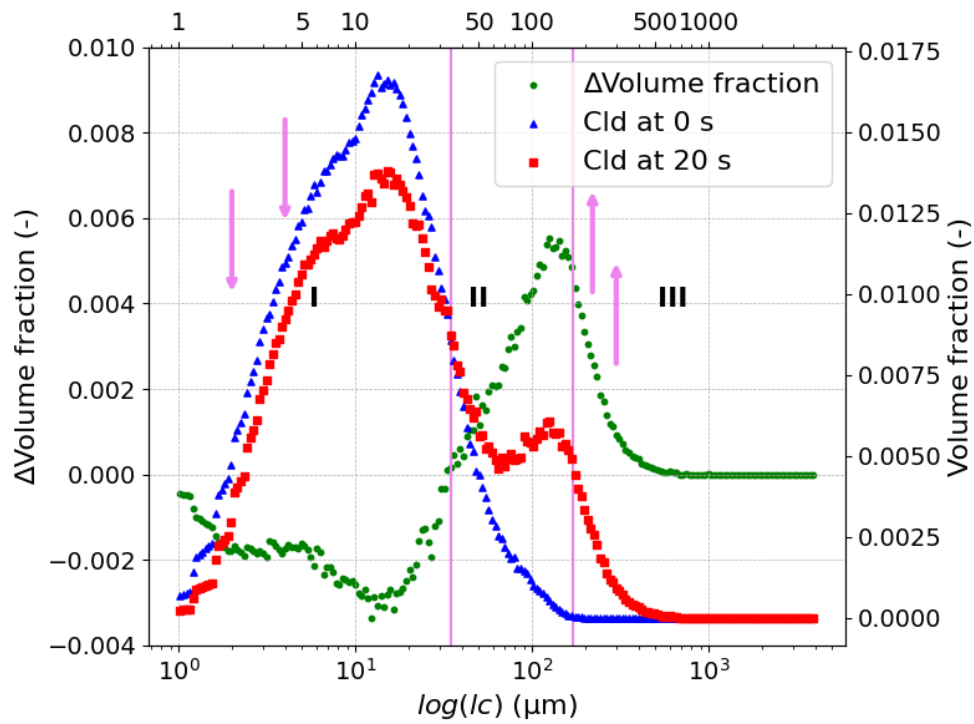


Figure 23. Superposition of Figure 21 and Figure 22, in which regions I, II, III can be identified.

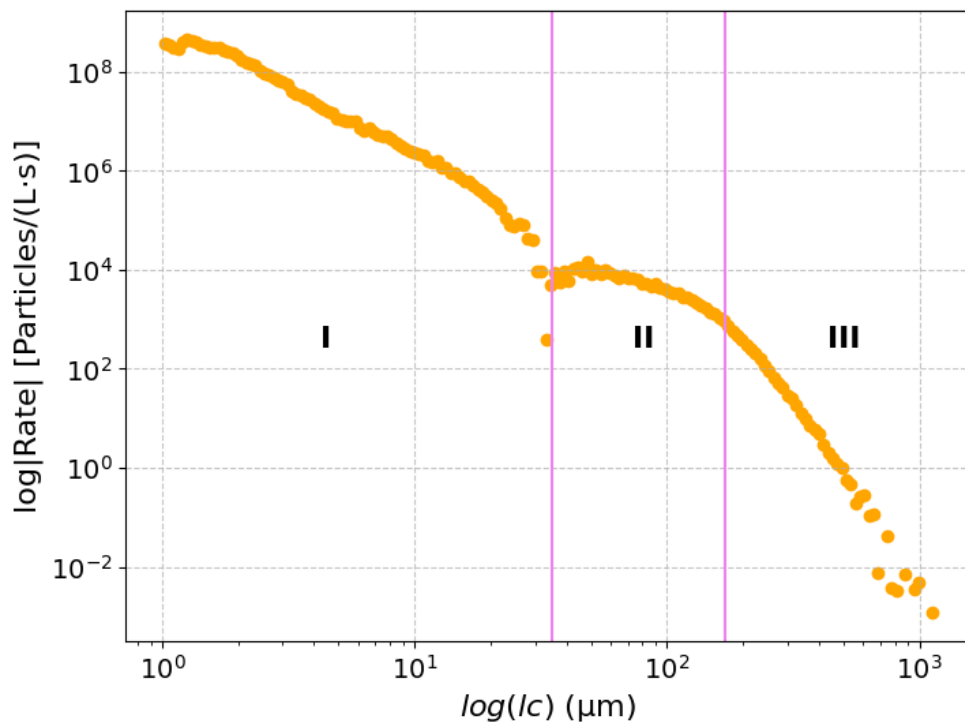


Figure 24. Response of the rate behavior with respect to the logarithm of chord length (lc), including regions I, II, III.

$$\Delta Volume\ fraction_{lc} = Volume\ fraction_{t20,lc} - Volume\ fraction_{t0,lc} \quad (103)$$

5.8.1. Kinetic modeling in region I

In Region I of Figure 23, the concentration of particles decreases between the initial and final times of the process. The downward violet arrows illustrate this reduction in particle number, as derived from the volume fraction balance (equation 103). This region extends from the smallest channels up to the point where the volume fraction curves at 0 and 20 s intersect. At this intersection, the overlap of the data prevents a clear distinction of the prevailing event. By analogy with chemical reactions, the behavior observed in Region I can be interpreted as the consumption of reactants.

This approach enables the analysis to be focused on a single phenomenon, namely the death or disappearance of particles, thereby allowing for a more precise and coherent interpretation of the system's behavior.

Figure 25 presents the particles corresponding to Region I. By analogy with chemical reactions, these particles can be regarded as reactants; accordingly, the associated rate exhibits a negative slope. The analysis of aggregation rates for this set reveals a linear relationship with the logarithm of particle size.

The proposed predictive model for the aggregation rate as a function of particle size constitutes an innovative approach and is shown to be independent of PAM molecular weight within the evaluated range (Figure 25). The model demonstrated an excellent fit to the linear regression (least squares) model, with a coefficient of determination

(R^2) of approximately 0.97 as can be seen in Table 4. Moreover, the 95% confidence interval encompassed all data points, reinforcing the robustness and validity of the proposed model.

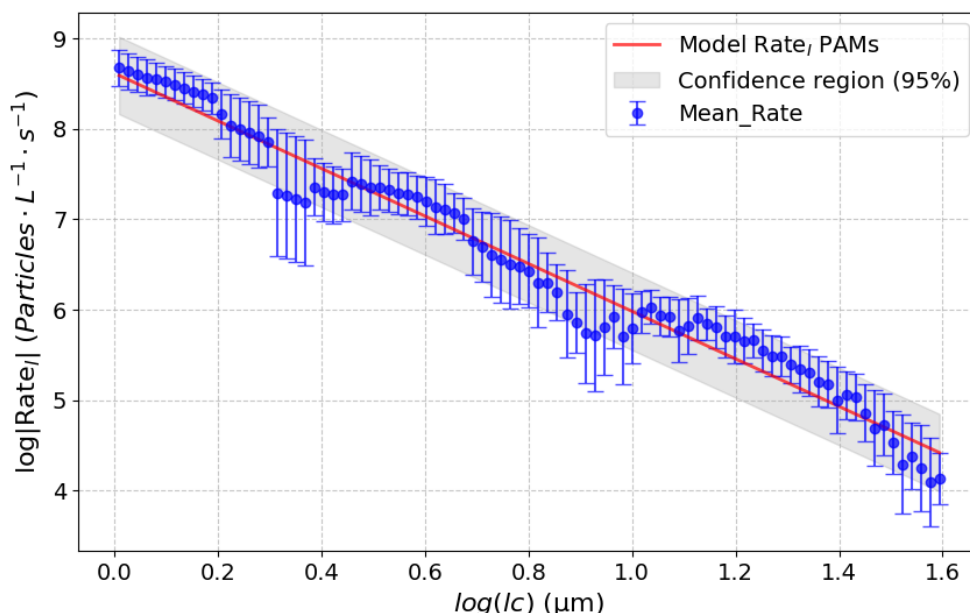


Figure 25 Aggregation rate as a function of the logarithm of chord length (lc_i) for consumed particles (Region I), obtained with PAMs samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.

Table 4. Linear model for aggregation rate in Region I. Particles evaluated as reactants with PAMs.

a	b	R^2	RMSE
-2.63	8.61	0.970	0.22

Equation 104 represents a linear model for estimating the particle aggregation rate, evaluated from the perspective of their behavior as reactants, specifically within the analyzed system of kaolinite particles suspended in water and flocculated with RAFT-synthesized PAM. This result reinforces the hypothesis that the aggregation reaction follows

zero-order kinetics, being independent of concentration variation. Furthermore, it clearly demonstrates that the rate at which particles aggregate depends on their size, which, in this specific case, is represented by the chord length measured using the FBRM technique.

$$\log|Rate_I| = a * \log(lc) + b \quad (104)$$

5.8.2. Kinetic modeling in region III

Similarly, Region III in Figure 23, displays violet arrows pointing upward, indicating an increase in volume fraction that reflects a rise in the number of particles between the two selected time points. The onset of this region is defined by channels that exhibit a volume fraction of zero at time 0 s. Under these conditions, each particle detected can be interpreted as arising from the birth of new solids. By analogy with chemical reactions, the particles in Region III correspond to the products.

Figure 26 displays the particles corresponding to domain region III, in Figure 23 indicated these particles are formed through the successive addition of solid elements that assemble.

Unlike what is observed in region I, the agglomerates formed in region III exhibit distinct behaviors depending on the molecular weight of the polymer used. As shown in Figure 26, an individual linear model has been developed for each polyacrylamide sample, which differ primarily in their molecular weight. The resulting models follow the same structure as that presented in equation 104, with the only difference being that the dependent variable is now $\log|Rate_{III}|$. These models achieved R^2 above 0.95, indicating a statistically significant fit, as shown

in Table 5. Specifically, the literature indicates that kaolinite flocs formed by polyacrylamide (PAM) flocculants tend to increase in size with the polymer's molecular weight [149]. This behavior is attributed to the fact that PAM bridges connecting particles can be absorbed by multiple particles or collide with other bridges, promoting the formation of larger aggregated structures. As the molecular weight of PAM increases, so does the length and flexibility of its chains, enhancing the likelihood of interparticle collisions and the extension of particle-to-particle linkages [150].

In general terms, the agglomeration rate gradient with respect to size presents an inversely proportional response to the average molecular weight of the polyacrylamide (PAM) sample used. This reduction in speed may be associated with the increase in the viscosity of the aqueous medium, which typically increases with the molecular weight of the polymer [151]. On the other hand, the final size of the formed floc shows a direct relationship with the average molecular weight of the polymer [150,152]. This behavior can be attributed to the presence of a greater number of active sites along the polymer chains, which facilitates the attachment of a greater number of free particles in suspension.

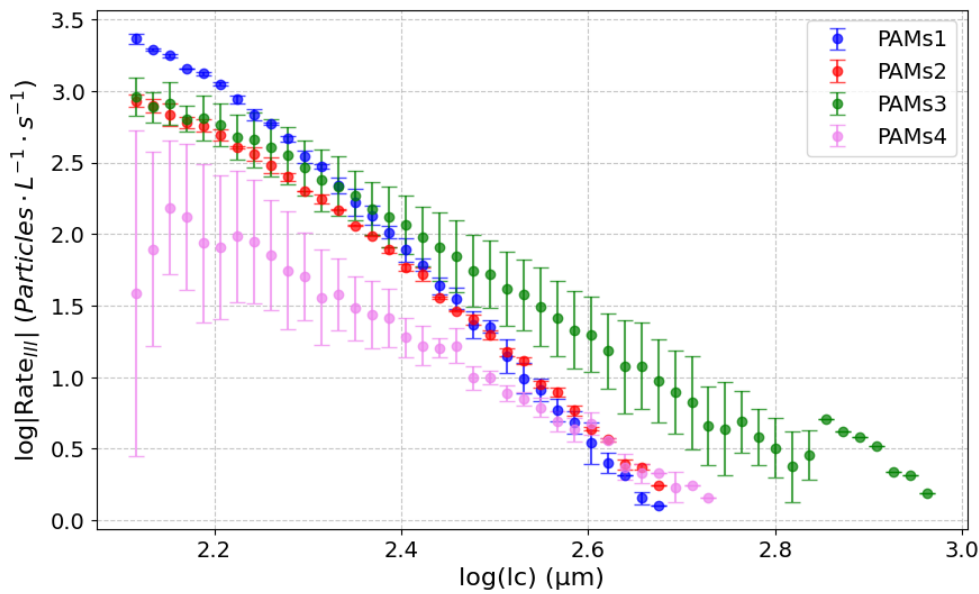


Figure 26 Logarithm of the aggregation rate as a function of the logarithm of chord length for particles in Region III, obtained with PAM samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.

Table 5. Linear model for aggregation rate in Region_{III}.

Sample	Mw (Da)	a	b	R^2
PAMs1	$3.11 \cdot 10^5$	-6.13	16.55	0.993
PAMs2	$9.26 \cdot 10^5$	-4.99	13.70	0.990
PAMs3	$2.09 \cdot 10^6$	-3.46	10.34	0.983
PAMs4	$3.55 \cdot 10^6$	-3.27	9.12	0.952

Conventional approaches, such as population balance models (PBM, equation 16) and velocity balance equations (equation 30), are typically formulated on the basis of aggregation and breakage rates obtained through summation loops and empirical adjustments. Although these frameworks are widely applied, their reliance on parameter fitting often obscures the underlying physical behavior of the

particles, thereby limiting the interpretation of the dynamic patterns involved in flocculation [153]. In contrast, the calculation models based on linear equations equation 104 proposed in this work allow for a clearer segmentation and analysis of the phenomenon. Furthermore, the proposed approach provides predictive capacity that is independent of reactor volume, making it a robust, versatile, and generalizable tool for investigating flocculation systems across different contexts.

5.8.3. Kinetic modeling in region II

The creation of a model that describes the behavior of kinetic region III presents the challenge that it is not possible to separate the rates of aggregation events (whether birth or death by aggregation). As shown in Figure 23, it is not possible to determine what occurs in this region, since the system starts from a non-zero number of particles and the difference in particle counts indicates an increase over time. This leaves open the probabilities that the number of particles may decrease due to death or increase due to birth.

From the outset, it is assumed that both birth and death of particles occur simultaneously in kinetic region II, which is precisely bounded by kinetic region I (dominated by particle death phenomena) and kinetic region III (dominated by particle birth phenomena). Based on this framework, a model was proposed that considers a partial contribution from each rate, whether birth or death, in order to better capture the dynamics of particle aggregation.

It is considered that each rate leading to the birth or death of particles in kinetic region II carries a fractional weight depending on its proximity to the neighboring regions. Thus, at the left boundary of kinetic region II, which borders kinetic region I, the rate is 100% dominated by

particle death phenomena. As the distance from this boundary increases, the weight of the death contribution decreases proportionally until reaching the right boundary of kinetic region II, which connects to kinetic region III, where 100% of the phenomena are described as particle birth.

The coefficient of determination ($R^2 = 0.960$) shows that the simulation (violet line) explains about 96% of the variability observed in the experimental data (yellow points) within the valid range. This reflects an excellent fit between the model and the experiments. The root mean squared error (RMSE = 0.0750) is very low, indicating that the model predictions deviate only minimally from the experimental values. The variance of the differences (0.0049) confirms that the dispersion between simulated and observed data is practically negligible.

Overall, these results suggest that the model very accurately reproduces the experimental behavior in the analyzed region. This reinforces the idea that the proposals for kinetic region II need to be further studied, since such a strong fit in the valid range indicates that simulations capable of reproducing and understanding the phenomena of this region are achievable.

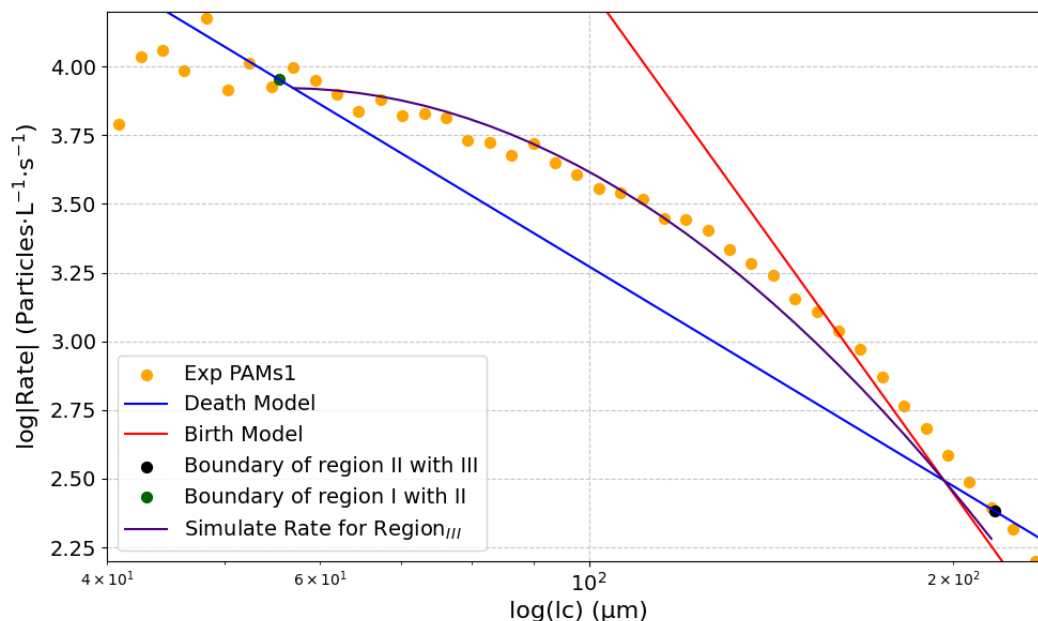


Figure 27. Simulation of particle concentration variation in Regions II and III using PAMs1, based on the death-birth-aggregation rates during the real-time reaction period.

5.9. Modeling the aggregation rates as a function of the polymer average molecular weight.

As previously discussed, the aggregation rate (*Figure 26*) exhibits significant sensitivity to the molecular weight of the water-soluble PAM polymers under study. *Figure 28* illustrates the methodology used to develop linear models for both the slope and the intercept of the aggregation rate as a function of the PAM weight-average molecular weight. As shown in the *Figure 28*, the primary models of aggregation rate versus characteristic length (lc), which represent the baseline behavior of particle flocculation, are integrated to construct a meta-model. This meta-model establishes the correlation between slope and intercept (aggregation rate versus lc) and the polymer molecular weights, expressed as $\log(Mw)$.

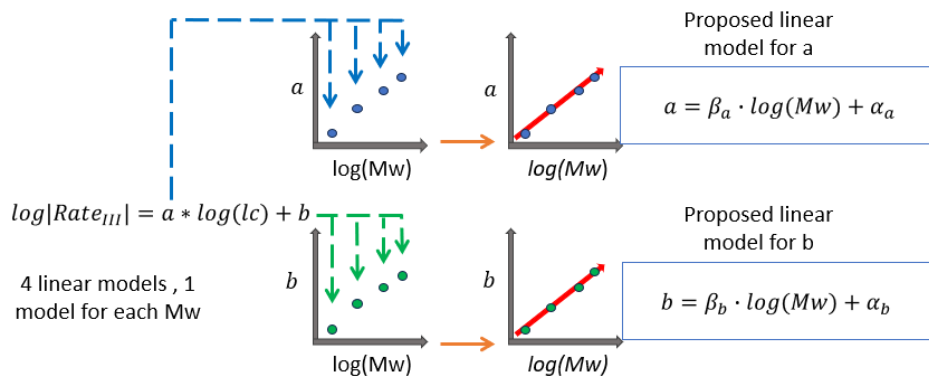


Figure 28. Procedure for developing mathematical models of the slope and intercept of the aggregation rate as a function of the PAM weight-average molecular weight

Figure 29 presents the linear model of equation 105 that predicts the slope of the aggregation rate as a function of the PAM weight-average molecular weight. Notably, the coefficient of determination R^2 reveals that more than 99.6% of the data can be explained by the linear regression model, demonstrating excellent predictive performance. Additionally, the confidence interval analysis confirms that, with 95% certainty, both the intercept and the slope of the regression line lie within the narrow bounds delineated in the Figure 29, reinforcing the statistical robustness of the model. Furthermore, by reorganizing the slopes obtained from individual studies at various PAM concentrations as a function of the logarithm of the PAM molecular weight, it is possible to construct a metamodel (a model of models) that describes and generalizes the behavior of the set of individual models representing the aggregation rate of kaolinite particles in the presence of PAM synthesized by RAFT methods, as demonstrated by the linear trend shown in the Figure 29.

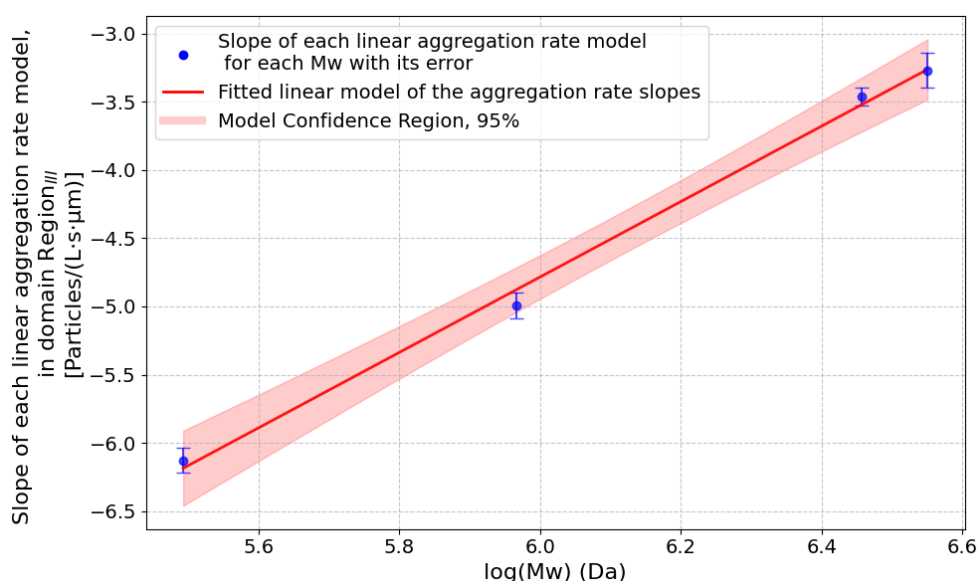


Figure 29. Aggregation rate slope as a function of the PAM weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).

The regression analysis demonstrated excellent predictive performance, with a $R^2 = 0.996$ showing that more than 99.6% of the variability is explained by the model. The slope was estimated at 2.76, bounded within a 95% confidence interval of 2.40 to 3.13, while the intercept was -21.37 , with a 95% confidence interval between -23.61 and -19.13 . Both parameters are statistically significant, as indicated by the very low p-values for the slope (0.00094) and the intercept (0.00059). The RMSE = 0.0717 further highlights the accuracy of the regression fit. Taken together, the narrow confidence intervals, the small RMSE, and the p-values far below the conventional threshold of 0.05 (5%), implying less than a 0.1% probability that the results are due to random variation, provide strong statistical evidence of the robustness and reliability of the linear model in describing the kinetic behavior of the system.

$$a(Mw) = \beta_a * \log(Mw) + \alpha_a \quad (105)$$

Analogously, to complete the analysis of the sensitivity of the aggregation rate in the particle formation region, the intercept values (y-axis) in *Figure 26* are examined for different PAM samples using the model given in This yields , *Figure 30* is analyzed across different PAM samples. This yields which shows that 99.2% of the data of the *plot's intersection of aggregation rate* can be explained by the linear regression model, using the least squares method (Lsq) and with a narrow confidence interval at a 95% certainty level.

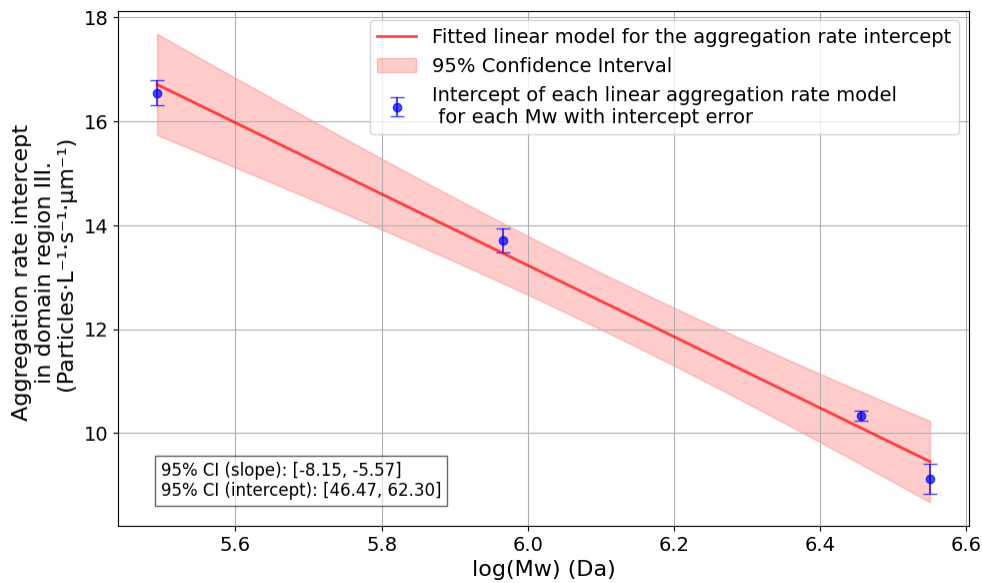


Figure 30 Aggregation rate intercept as a function of the PAM weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).

The regression analysis yielded a slope of -6.86 with a 95% confidence interval ranging from -8.15 to -5.57 , and an intercept of 54.39 with a 95% confidence interval between 46.47 and 62.30 . The R^2

is 0.992 and indicates that more than 99% of the variability in the intercept values can be explained by the linear model, reflecting an excellent fit. The RMSE = 0.25 further confirms the accuracy of the predictions. Importantly, the p-value of the slope (0.00191) and the p-value of the intercept (0.00114) are both far below the conventional significance threshold of 0.05 (5%). This means that the probability of the observed parameters being due to random variation is less than 0.2%, providing strong statistical evidence that both the slope and the intercept are highly significant. The narrow confidence intervals, combined with the very low p-values, reinforce the precision and reliability of the regression parameters, substantiating the robustness of the linear model and its applicability to describe the kinetic behavior of the system.

$$b(Mw) = \beta_b * \log(Mw) + \alpha_b \quad (106)$$

Table 6 summarizes the parameters of the linear models describing the intercept and slope of the aggregation rate in Region III as a function of the PAM weight-average molecular weight.

Table 6. Parameters and regression metrics of the linear models describing the intercept and slope of the aggregation rate in Region III

	β	α	R^2	RMSE
<i>a</i>	2.76	-21.37	0.996	0.07
<i>b</i>	-6.86	54.39	0.992	0.25

5.10. Simulation of particle concentration in Regions I and III from aggregation rates

To demonstrate the validity of the developed models and their predictive capability, Figure 31 presents the original data of the particle concentration projection, starting at 0 seconds and ending at 20 seconds during the real-time reaction. The dynamic evolution at the end of the 20-second period is illustrated with continuous lines, calculated through a simple rearrangement of the output of the rate equation, where the rate is multiplied by the real-time reaction time, resulting in the particle concentration ($[NP]$). These lines, however, only partially represent the simulation.

The first section of the simulation (from 10^0 to $3 \cdot 10^1 \mu\text{m}$) corresponds to Region I, which represents the fraction of smaller dispersed particles and was modelled using the parameters reported in Table 4 for PAMs1. Subsequently, Region III is represented (from $1.8 \cdot 10^2$ to $3.4 \cdot 10^3 \mu\text{m}$), corresponding to the larger aggregates formed during flocculation, modelled according to the parameters in Table 5 for PAMs1. Although Figure 31 provides a good representation of the system's physical behavior, it does not offer detailed information above $10^1 \mu\text{m}$, as the scale magnitude dampens subtle variations.

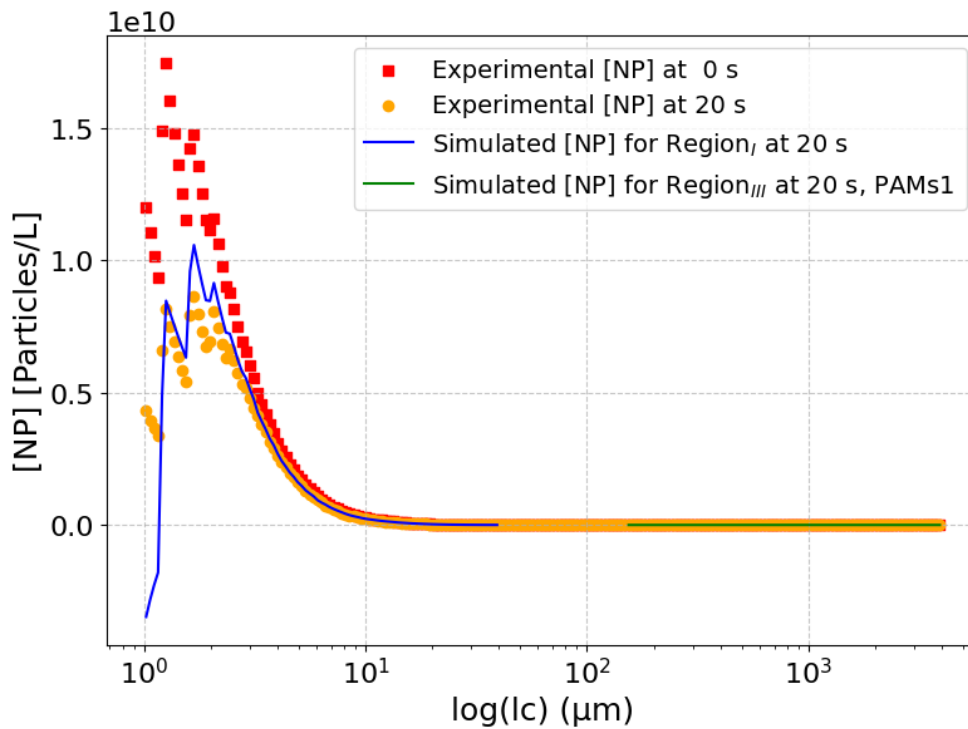


Figure 31 Simulation of particle concentration variation in Regions I and III using PAMs1, based on the aggregation rates during the real-time reaction period.

A more accurate assessment of the simulation results is presented in Figure 32, where the volume fraction is calculated in reverse, starting from particle concentration. Visually, a similar behavior is observed between the simulated curve and the expected outcome for Kinetic Domain Regions I and III. For the simulation in Kinetic Region I, the mean squared error (MSE) is 1.30×10^{-6} , with a variance of 1.29×10^{-5} , R^2 is 0.930 and in Kinetic Region III, the MSE is 8.64×10^{-8} and the variance is 7.13×10^{-8} , R^2 is 0.953. These values suggest that the model closely fits the predicted data, and since the variance is greater than the MSE [154], the model effectively captures the underlying structure of the dataset. The two models for Region I and Region III reproduce 93% and 95% of the variability in the experimental data, respectively.

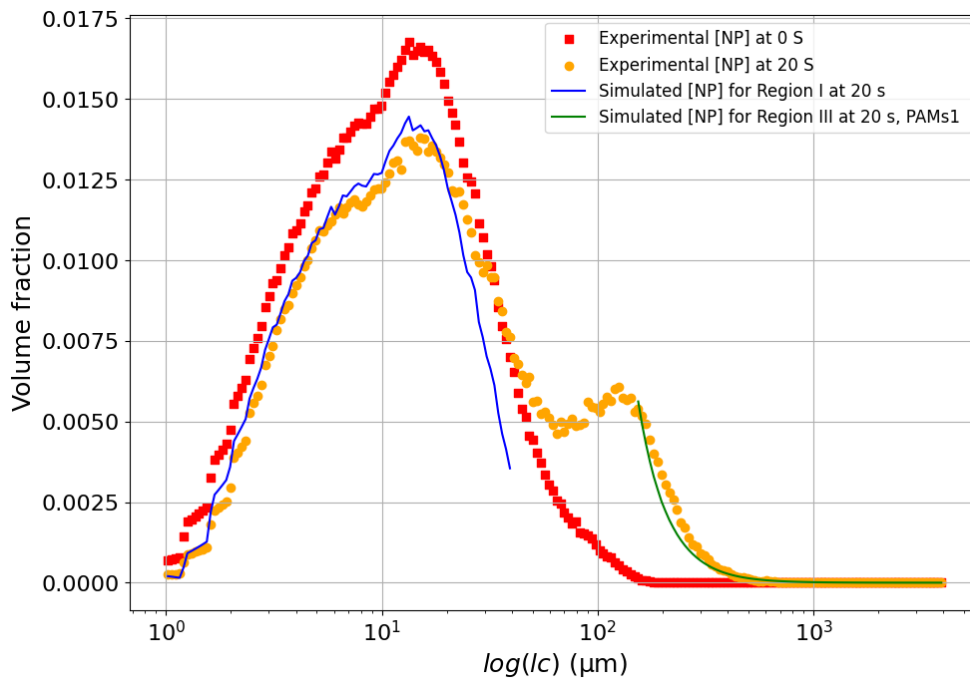


Figure 32 Simulation of particle concentration variation in Regions I and III using PAMs1, based on the aggregation rates during the real-time reaction period.

5.11. Flocculation Results for PAM Blend Samples

Following the methodology applied to the narrow-distribution PAM samples, as described in Figure 11, the results for the binary and ternary blends are presented, aimed at determining the aggregation rates. This analysis enables the evaluation of how polymer combinations influence flocculation kinetics, taking into account both the individual behavior of each component and potential synergistic interactions among them.

5.11.1. PAM blends samples Kinetic region I

Figure 33 shows the response of the logarithm of the aggregation rate of kaolinite particles as a function of the logarithm of chord length.

A linear trend is observed approximately within the range of 1.02 to 39 micrometers, with a linear regression coefficient (R^2) of 0.979, indicating that the model accounts for the majority of the data variability in this region (Table 7 Aggregation rate as a function of the logarithm of chord length (l_{ci}) for consumed particles (Region I),

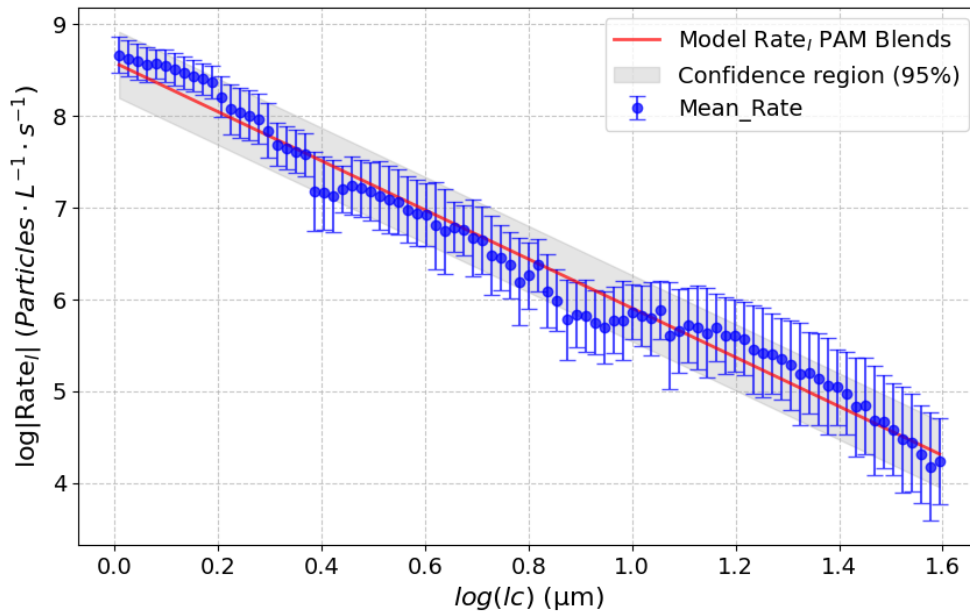


Figure 33. Aggregation rate as a function of the logarithm of chord length (l_{ci}) for consumed particles (Region I), obtained with PAM blends samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.

Table 7. Linear model for aggregation rate in Region I. Particles evaluated as reactants with PAM blends samples.

a	b	R^2	RMSE
-2.68	8.58	0.979	0.18

A comparison of the results shown in Table 4 and Table 7, corresponding to the linear models obtained for narrow-distribution PAM samples ($\log|Rate_I| = -2.63 \log(lc) + 8.61$) and PAM blends ($\log|Rate_I| = -2.68 \log(lc) + 8.58$) reveals a high degree of similarity

between both sets of parameters. This finding confirms that the logarithm of the aggregation rate is linearly dependent on the logarithm of chord length, regardless of whether individual polymers or blends are used. Therefore, it is both intuitive and methodologically sound to construct a unified model using the combined data from both PAM polymer groups, applied to the analysis of kaolinite particles.

The results shown in Figure 34 and Table 8 establish a general model for the variation of the logarithm of the aggregation rate as a function of the logarithm of particle size ($\log(l_c)$). It is observed that, with an increase in data volume, the regression models become more refined, resulting in a higher coefficient of determination (R^2) and a lower root mean square error (RMSE) compared to the individual cases studied for narrow-distribution PAMs and PAM blends. The final model, applied to Kinetic Region I, fits 97.9% of the data, and the 95% confidence region depicted in Figure 34 aligns with the mean errors of the $\log|Rate_I|$ values, thereby confirming the linearity of the proposed model.

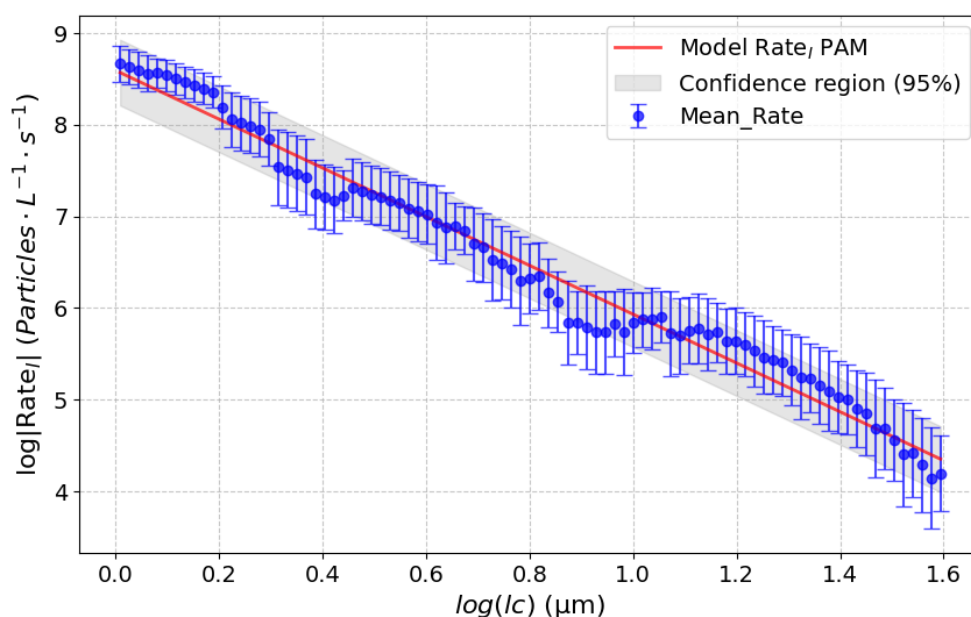


Figure 34. Aggregation rate as a function of the logarithm of chord length (lc_i) for consumed particles (Region I), obtained with PAM samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.

Table 8. Linear model for aggregation rate in Region I. Particles evaluated with PAM samples.

a	b	R^2	RMSE
-2.66	8.59	0.979	0.18

Table 66 presents the rate constants determined for the kaolinite flocculation system with PAMs synthesized via the RAFT method. A total of 45 rate constant values are reported, obtained under the assumption of zero-order reactions, in which the reaction rate is independent of particle concentration. In this regime, the reaction rate is numerically equal to the kinetic constant, reflecting independence from particle concentration.

Table 9 Rate Constants for Kaolinite Flocculation with RAFT-Synthesized PAMs (Zero-Order Kinetics, Units of k : Particles $\cdot$ L $^{-1}\cdot$ s $^{-1}$).

log(l_c) (μ m)	k	log(l_c) (μ m)	k	log(l_c) (μ m)	k
0.8015	6.4290	1.0716	5.9187	1.3418	5.3049
0.8195	6.3000	1.0896	5.7680	1.3598	5.2058
0.8375	6.3013	1.1076	5.8192	1.3778	5.1713
0.8555	6.1948	1.1256	5.9115	1.3958	4.9994
0.8735	5.9512	1.1436	5.8428	1.4138	5.0560
0.8915	5.8549	1.1617	5.8050	1.4318	5.0313
0.9095	5.7367	1.1797	5.7019	1.4498	4.8535
0.9275	5.7147	1.1977	5.7022	1.4678	4.6896
0.9455	5.8014	1.2157	5.6553	1.4858	4.7287
0.9636	5.9221	1.2337	5.6613	1.5039	4.5328
0.9816	5.6991	1.2517	5.5457	1.5219	4.2893
0.9996	5.7912	1.2697	5.4852	1.5399	4.3815
1.0176	5.9735	1.2877	5.4841	1.5579	4.2480
1.0356	6.0301	1.3057	5.3873	1.5759	4.0938
1.0536	5.9286	1.3237	5.3391	1.5939	4.1308

5.11.2. PAM blends samples Kinetic region III

Figure 35 Aggregation rate as a function of the logarithm of chord length (l_{c_i}) for consumed particles (Region III), obtained with PAM blends samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller. and Table 10 compile the results for the PAM blend samples in Region III, including binary and ternary blends, highlighting the sensitivity of the logarithm of the aggregation rate with respect to the logarithm of chord length. These results demonstrate how the flocculation rates of kaolinite particles are affected by the selected flocculant agent, reproducing a response similar to that observed in the flocculation studies conducted with the simple PAM samples. In all cases, as shown in Table 10, the coefficient of determination (R^2) values indicates an excellent fit to a linear model, with the lowest R^2 value being 0.962 and the highest reaching 0.994,

confirming the consistency of the kinetic behavior in this region. These results once again confirm that Region III exhibits a linear behavior when evaluating the aggregation rate of Birth particles.

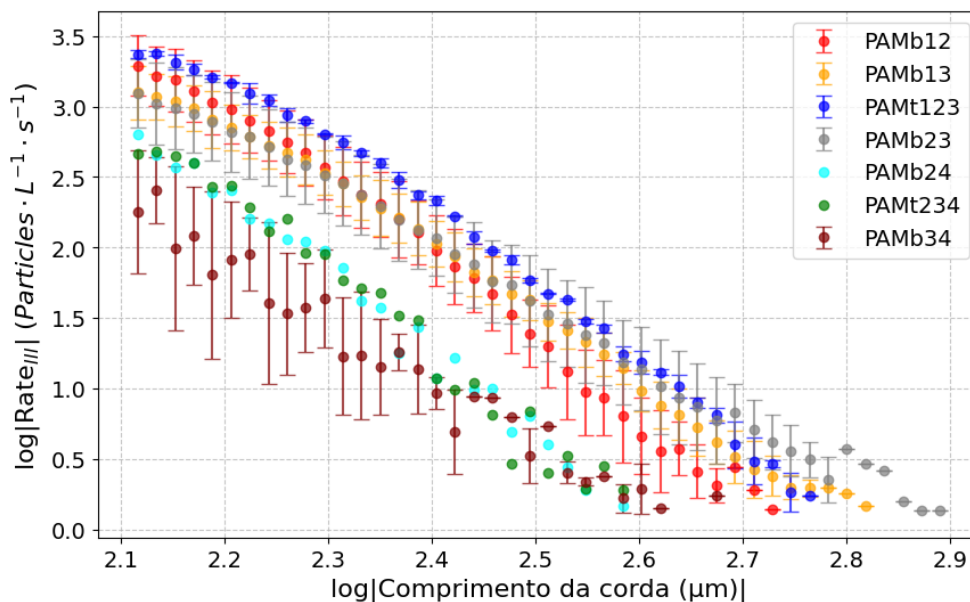


Figure 35 Aggregation rate as a function of the logarithm of chord length (lc_i) for consumed particles (Region III), obtained with PAM blends samples of different molecular weights. Kaolinite suspension (2.5 wt%) mixed at 200 rpm using a pitched four-blade impeller.

Table 10. Molecular weight averages of the PAM blends samples used in the flocculation experiments.

Sample	M_w (Da)	a	b	R^2
PAMb12	6.23×10^5	-5.51	15.14	0.992
PAMb13	1.58×10^6	-4.59	13.00	0.993
PAMt123	1.37×10^6	-5.08	14.40	0.990
PAMb23	1.89×10^6	-4.07	11.78	0.994
PAMb24	2.24×10^6	-5.71	14.95	0.987
PAMt234	2.45×10^6	-5.73	15.00	0.976
PAMb34	3.20×10^6	-4.10	10.90	0.962

5.11.2.1. Response of PAM blend samples with respect to molecular weight.

As previously stated, the PAMb samples exhibit sensitivity to variations in the average molecular weight of the polymer used as a flocculant (equations 105 and 106). In this particular case, a data point outside the confidence region was excluded (Figure 36 and Figure 37), as it was statistically distant from the rest of the dataset. As shown in Table 11, the fitted values for parameters "a" and "b" are 0.887 and 0.971, respectively. The value of "a" reflects a high degree of uncertainty, as illustrated in Figure 29; nonetheless, the linear model remains applicable under these conditions. For parameter "b", the coefficient of determination (R^2) demonstrates strong agreement with a linear trend, supporting the validity of the model within this region.

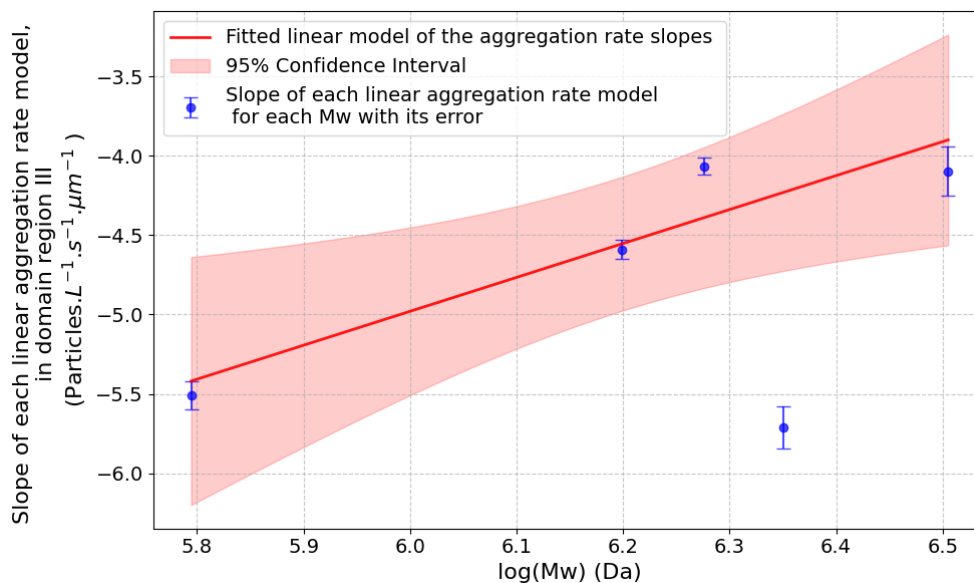


Figure 36. Aggregation rate slope as a function of the PAMb weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).

The regression analysis yielded a slope of 2.14 (Figure 36), with a 95% confidence interval between 0.49 and 3.78, indicating that the true slope lies within this range and, since the interval does not include zero, the relationship between molecular weight and aggregation rate is statistically significant. The intercept was -17.80 , with a 95% confidence interval between -28.00 and -7.61 , also significantly different from zero. The p-values for the slope (0.0305) and the intercept (0.0173) are both below the conventional significance threshold of 0.05 (5%), meaning that the probability of these parameters being due to random variation is less than 5%. The $R^2 = 0.887$ shows that approximately 89% of the variability in the data is explained by the model, reflecting a solid fit, though not as extreme as in previous cases. The mean squared error (MSE = 0.0385) and the root mean squared error (RMSE = 0.1961) confirm that the model's predictions are reasonably accurate.

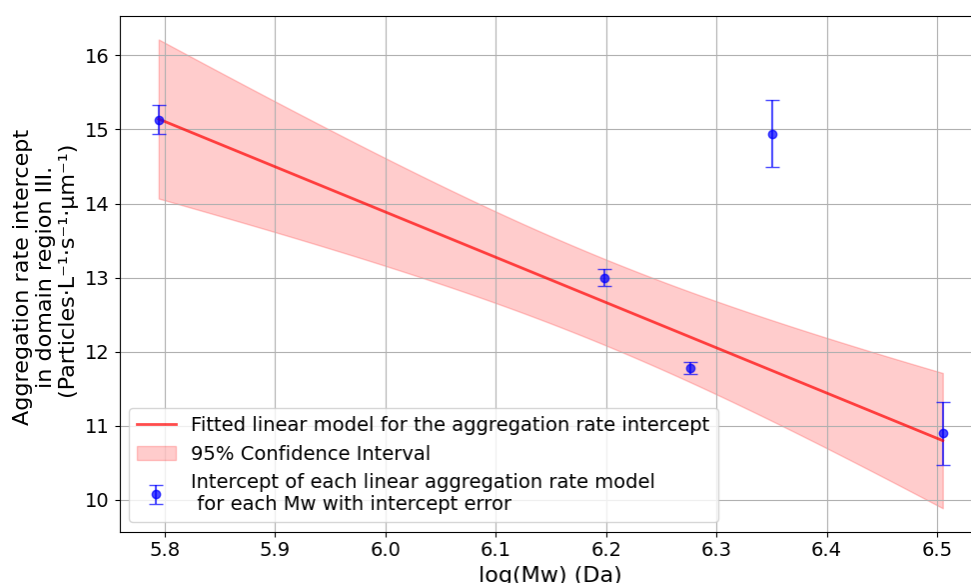


Figure 37. Aggregation rate intercept as a function of the PAMb weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).

Figure 37 shows the regression analysis with a slope of -6.11 (95% CI: -8.37 to -3.85) and an intercept of 50.54 (95% CI: 36.51 to 64.56). The model explains approximately 97% of the variability ($R^2 = 0.971$) and has a prediction error of $RMSE = 0.27$. Both the slope ($p = 0.0073$) and the intercept ($p = 0.0041$) are statistically significant, confirming that the linear model provides a reliable description of the relationship between molecular weight and aggregation rate.

Table 11. Parameters and regression metrics of the linear models describing the intercept and slope of the aggregation rate in Region III for PAMb

	β	α	R^2	RMSE
<i>a</i>	2.14	-17.80	0.887	0.20
<i>b</i>	-6.11	50.54	0.971	0.27

Figure 38 and Figure 39 illustrate the overall behavior of the linear aggregation rate models in Region III, as a function of the molecular weight of the flocculant agent. The predictive performance of the fitted model for PAMs samples appears robust, with a narrow confidence interval closely aligned with the fitted linear model, supporting its reliability. In contrast, the model for PAMb samples shows greater uncertainty, yet it remains suitable for rough estimations. To improve its accuracy, a larger number of experimental data points is required to reduce uncertainty. Finally, for the PAMt samples, the limited number of experimental points prevented the identification of a clear trend. It is therefore recommended to increase the number of data points to enable a more comprehensive assessment of the system's behavior.

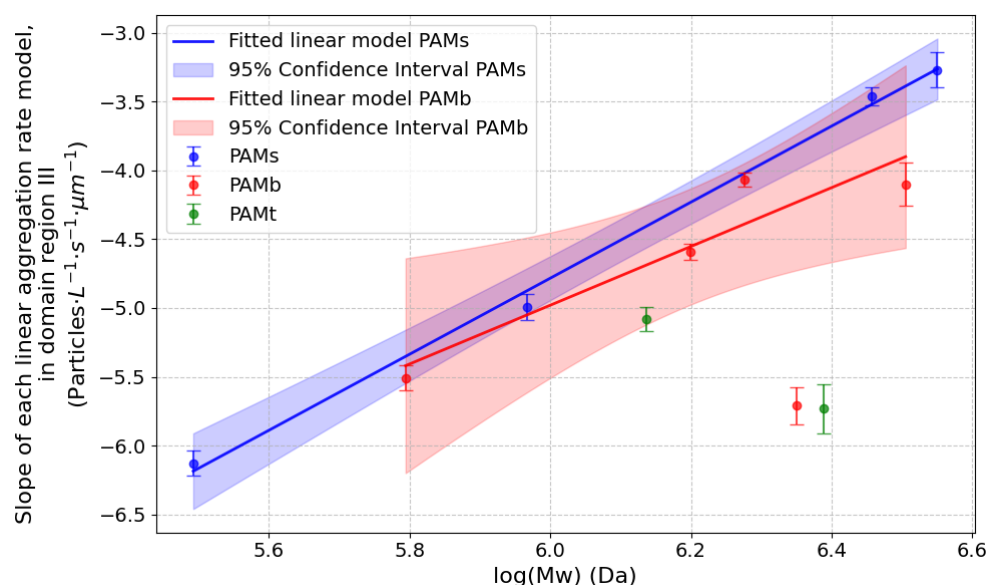


Figure 38. Aggregation rate slope as a function of the PAM weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).

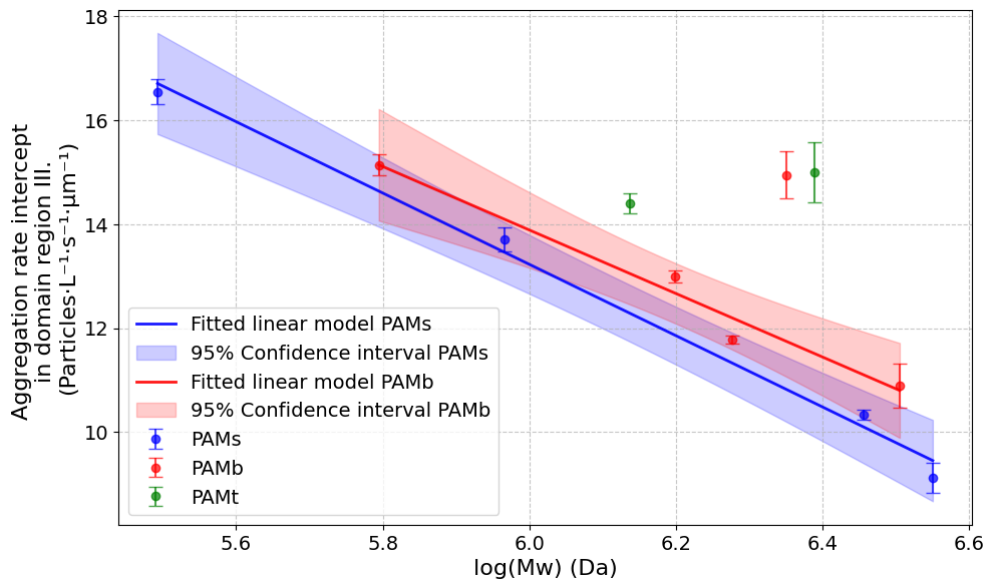


Figure 39. Aggregation rate intercept as a function of the PAMb weight-average molecular weight (Sample kaolinite 2.5 wt% mixed at 200 rpm with a pitched 4-blade).

5.12. Breakage rate

Up to this point, the discussion has focused on the determination of aggregation rates, identified in Figure 16 and Figure 17 through their linear behavior, which helped establish temporal reference points in accordance with the accepted terminology of real-time reaction.

This raises the question: What happens at the end of the linear region represented in Figure 16 and Figure 17.

A possible answer lies in the framework proposed by Tambo and Hozumi (equation 13), which confirms that the breakup process begins after agglomerates reach a maximum size under a critical velocity gradient [95].

This concept is further explored by Lu et al., [64] who report that hydrophobic flocculation processes occur in two distinct stages:

- First stage: involves aggregation only

- Second stage: involves both aggregation and breakage

The previous statements reinforce the hypothesis that the floc aggregation/breakage region occurs after the particles have aggregated. In Figure 14, it is difficult to clearly visualize the boundaries of the aggregation region; however, Figure 16 and Figure 17 show a baseline linear segment prior to flocculation, ranging from 0 to 183 seconds, followed by a sharp and consistent increase that extends up to 203 seconds.

This linear segment below 203 seconds in Figure 16 and Figure 17 corresponds to the region that was evaluated and identified as the particle aggregation region. Subsequently, at the end of the aggregation region, a sudden change in slope is observed, where the inclination of the time-conversion plot begins to decrease, eventually reaching a stable slope, even in the presence of signal noise.

Based on these considerations, a second initiation point for the aggregation/breakage reaction can be established using the data beyond 203 seconds. By repeating the kinetic strategy outlined in Figure 11, Figure 40 illustrates the variation in particle concentration over time, showing a linear decline in the aggregation region from 183 to 203 seconds. Beyond 203 seconds, the variation in particle concentration over time reflects the dynamics of aggregation and breakage phenomena. To proceed with the analysis, the reaction onset is defined at 203 seconds, and all rate evaluations are conducted from this point forward.

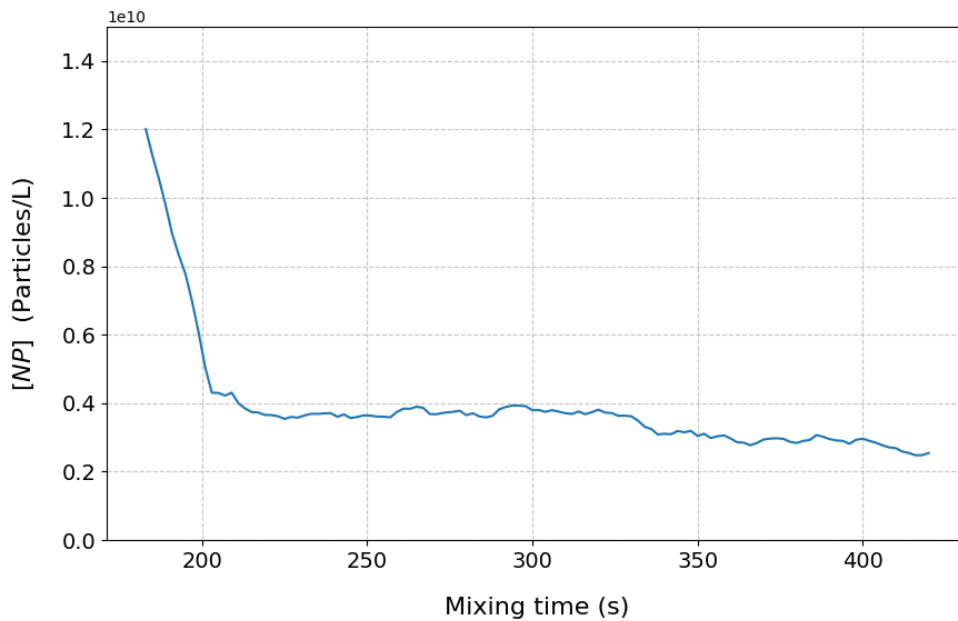


Figure 40. Variation of particle number concentration as a function of mixing time for a dose of 2500 ppm PAMs1, channel 1.

By maintaining the predefined boundaries for Region I and applying Equation 90 to calculate the rate within the time segment from 203 to 420 seconds across all channels, Figure 41 is obtained. This figure demonstrates the linear response of the $-\log|Rate|$ behavior, using PAMs1 polyacrylamide polymers.

Table 12 presents the resulting values from the model, and with an R^2 greater than 1.000, the linear trend of the model is confirmed.

This demonstrates that both the particle aggregation rate and the aggregation/breakage rate can be represented by linear models—specifically, first-degree polynomials that follow the form $y = ax + b$.

Given that breakage and agglomeration rates can be considered two opposing events, it is therefore possible to formulate equation 107 to describe this relationship.

$$-\log|Rate_{aggregation-breakage}| = \log|Rate_{breakage}| - \log|Rate_{aggregation}|$$

107

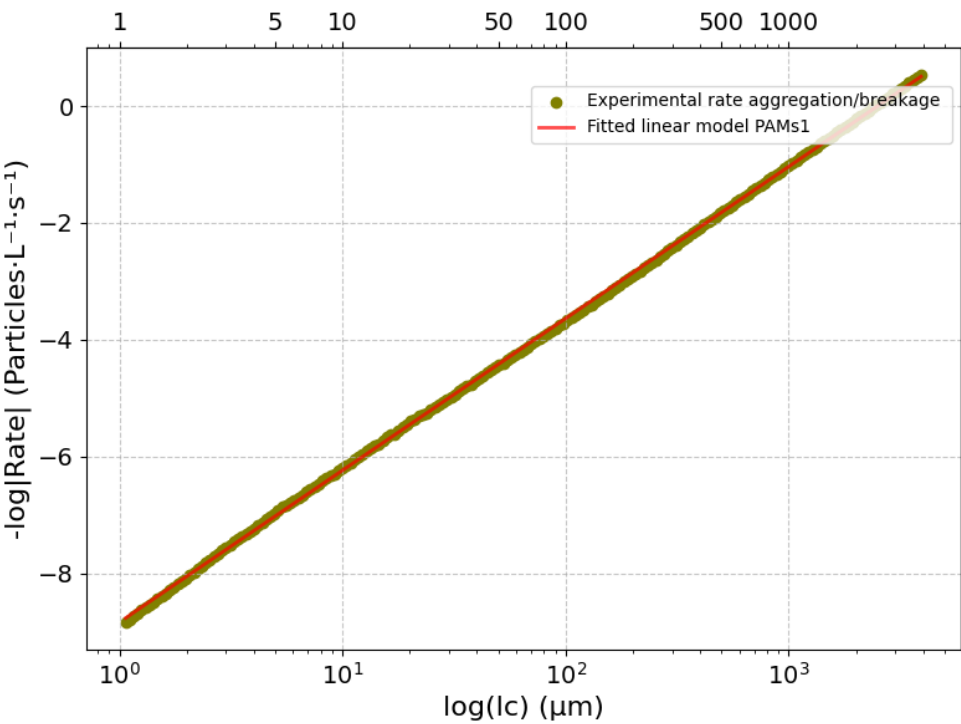


Figure 41 Behavior of $-\log|Rate|$ with $\log(lc)$ in region I for the agglomeration/breakup phenomenon of kaolinite particles, for PAMs1.

Table 12. Linear model for aggregation/breakup phenomenon rate in Region I. Particles evaluated with PAMs1 samples.

<i>a</i>	<i>b</i>	<i>R</i> ²	RMSE
-2.60	8.83	1.000	0.04

6. Conclusion

A methodological framework combining experimental and numerical analysis was developed to study flocculation in kaolinite dispersions using PAMs of different molecular weights. This approach, grounded in a kinetic framework and direct measurement of aggregation rates, provides a robust empirical basis for interpreting flocculation dynamics without relying on strong assumptions or a large number of estimated parameters. In a simple and direct manner, the aggregation rates of particles in this system can be quantified, dispensing with more complex modeling approaches.

Kinetic domains were defined based on variation of volume fraction equations, with the purpose of establishing criteria that differentiate between death particles and births particles during the aggregation process. This separation, achieved through discretization, enables a direct analogy with chemical systems, in which reactants and products are interpreted according to their temporal evolution within the system. Linear models were developed to measure the aggregation rate of kaolinite particles from two perspectives: that of the reactants (consumed particles or death) and that of the products (formed particles or births).

The linear models developed for the kaolinite particles consumed during the flocculation process exhibit behavior that is independent of the molecular weight of the polyacrylamide (PAM) polymers.

The models result to the births particles revealed that the aggregation rate depends strongly on particle size and PAM molecular weight. Linear models were established to describe both the slopes and intercepts of the aggregation rate, showing that higher PAM molecular weights favor the formation of larger aggregates. Additionally, the

particle formation rate was more pronounced as a function of the inverse of the logarithmic of molecular weight, suggesting that the polymer structure directly influences both the size and the rate of floc formation.

The analysis also demonstrated that the system exhibits zero-order kinetics in kinetic region I (particle death) with respect to particle concentration, indicating a regime governed primarily by physical parameters such as particle size and mobility. In kinetic region III, the predominant kinetic order is first-order, consistent with the requirement for a sufficiently large aggregate capable of breaking apart for the shearing phenomenon to occur. In these cases, the reaction rate and the rate constant are of the same magnitude. The different aggregation, breakage, and combined aggregation/breakage rates exhibit linear responses with respect to the logarithm of the chord length and can be represented by first-order polynomials.

The different rates of aggregation, breakage, and combined aggregation/breakage exhibit linear responses with respect to the logarithm of chord length, and can be represented by first-order polynomials.

7. Future Work

Understanding that this procedure for studying various particle/flocculant pairs, the principles established in this work may be extended to build data frameworks that enhance our understanding of the flocculation phenomenon.

Based on the findings, it is considered necessary to investigate the system's behavior in the region where aggregation and breakage coexist, as this zone represents a dynamic equilibrium that is critical to flocculant performance.

Furthermore, it would be of interest to extend these theories to smaller colloidal particles, including those in the nanometric range, in order to uncover aggregation and/or coagulation mechanisms that may differ significantly from those observed in micrometer-scale systems.

Studies examining the effects of temperature variation are essential in any investigation of reaction kinetics, as they enable the determination of activation energy parameters. These parameters are critical for understanding the minimum energy required for the system to initiate a reaction.

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Appendix

Appendix A Supplementary material to support Chapter 2.8.1

Table A1 Rate coefficients and physicochemical information used to model acrylamide aqueous-phase radical polymerization[106].

	Rate expression	Values (**)
Initiation	$k_d(s^{-1}) = 9.24 \times 10^{14} \exp\left(-\frac{14915}{T}\right)$ Initiator efficiency: 0.8	2×10^{-6}
Chain Propagation	$k_p(L \cdot mol^{-1} \cdot s^{-1}) = k_{p,max} \exp(-w'_M(0.0016T + 1.015))$	6.4×10^4
	$k_{p,max}(L \cdot mol^{-1} \cdot s^{-1}) = 9.5 \times 10^7 \exp\left(-\frac{2189}{T}\right)$	8.7×10^4
Transfer to Monomer	$C_{tr} = \frac{k_{tr}}{k_p} = 0.00118 \exp\left(-\frac{1002}{T}\right)$	4.8×10^{-5}
Termination SPR-SPR	$k_{t,ss}(L \cdot mol^{-1} \cdot s^{-1})$ $= 2 \times 10^{10} \exp\left(-\frac{1991 + 1447w'_{M0}}{T}\right)$ $\alpha_{s,s} = 0.1$	1.3×10^7
Reactions related to backbiting		
backbiting	$k_{bb}(s^{-1}) = 3.7 \times 10^9 \exp\left(-\frac{5874}{T}\right)$	26
Addition to MCR	$k_{p,tert}(L \cdot mol^{-1} \cdot s^{-1}) = 0.0155 \exp\left(-\frac{1412}{T}\right) k_p$	11
Cross termination MCR-SPR	$k_{t,st}(L \cdot mol^{-1} \cdot s^{-1}) = 0.27 k_{t,ss}$ $\alpha_{s,t} = 0.7$	3.5×10^6
Termination MCR-MCR	$k_{t,tt}(L \cdot mol^{-1} \cdot s^{-1}) = 0.01 k_{t,ss}$	1.3×10^5
Physicochemical information		
Activation energy	$E_A(K^{-1}) = R_g(1991 + 1477w'_{M0})$	

Preexponential factor	$k_t^0 (L.mol^{-1}.s^{-1})$	1.1×10^{11}
Density	$\rho_{\frac{AM}{H_2O}} (g.mL^{-1}) = (-0.0002T_1^2 + 0.0087T_1 - 0.2493)w_M'^2$ $+ (6 \times 10^{-5}T_1^2 - 0.0023T_1 + 0.1382)w_M'$ $+ (-6 \times 10^{-6}T_1^2 - 0.0003T_1^2 + 1.0048)$	1.0028

*Temperature T [K] or T₁ [°C]

** T₁ = 40 °C & w'_M = 0.2

*** w'_M = mass fraction of AM on a polymer free basis

**** w'_{M0} = initial weight fraction of AM in water

**** R_g = ideal gas constant

Appendix B Supplementary material to support Chapter 5.5

This section contains the time–conversion curves for the different PAM samples used in this study. The samples included are PAMs, PAMb, and PAMt.

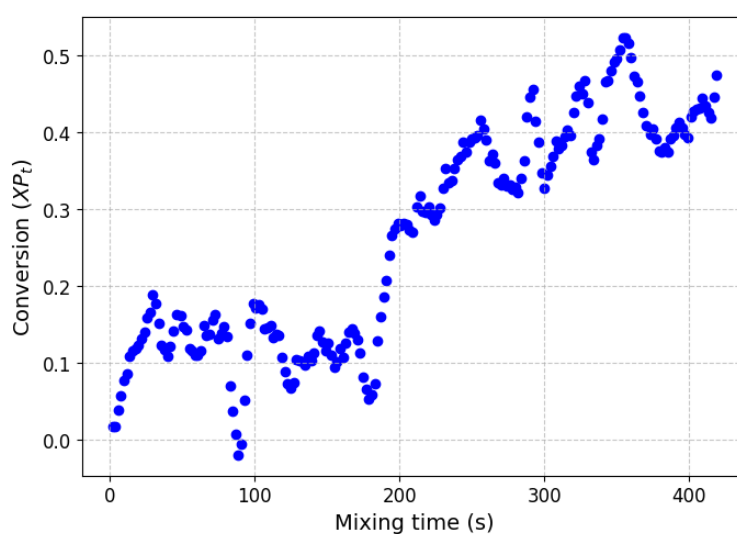


Figure B1 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs1. Experiment 2.

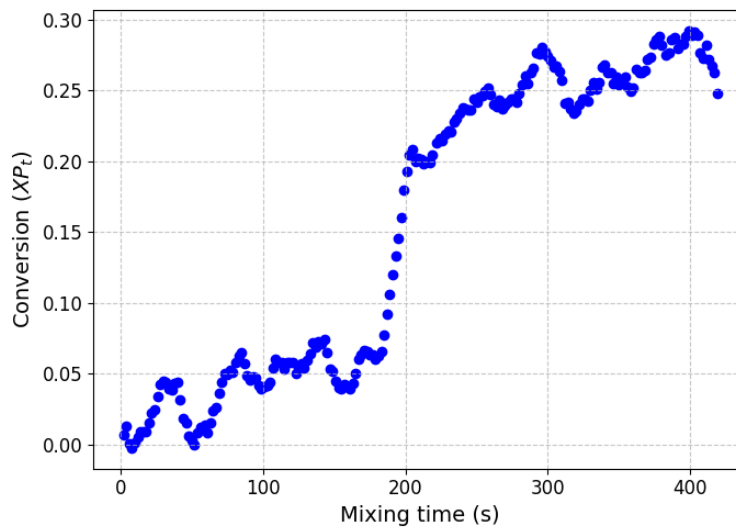


Figure B2 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs2. Experiment 1.

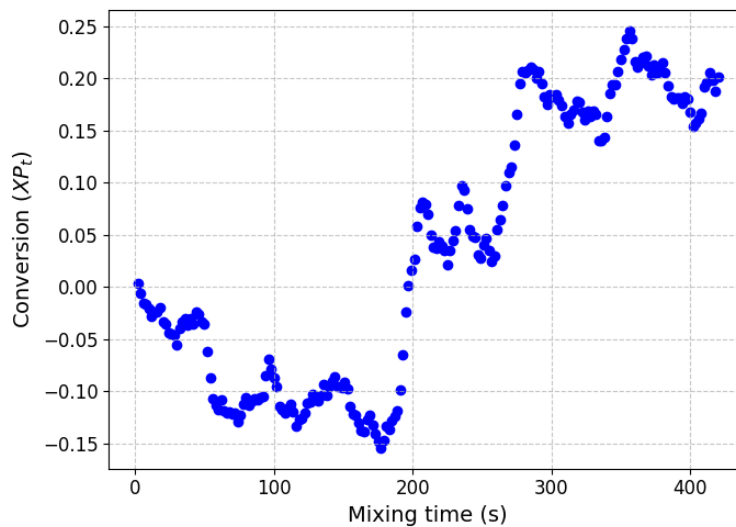


Figure B3 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs2. Experiment 2.

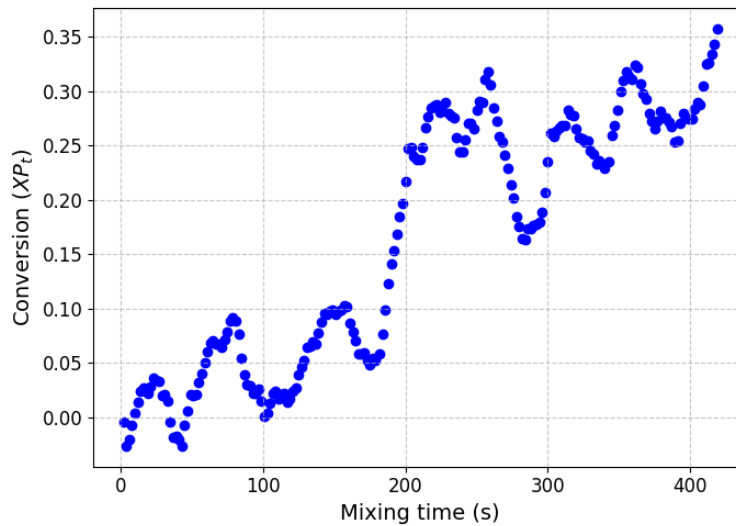


Figure B4 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs3. Experiment 1.

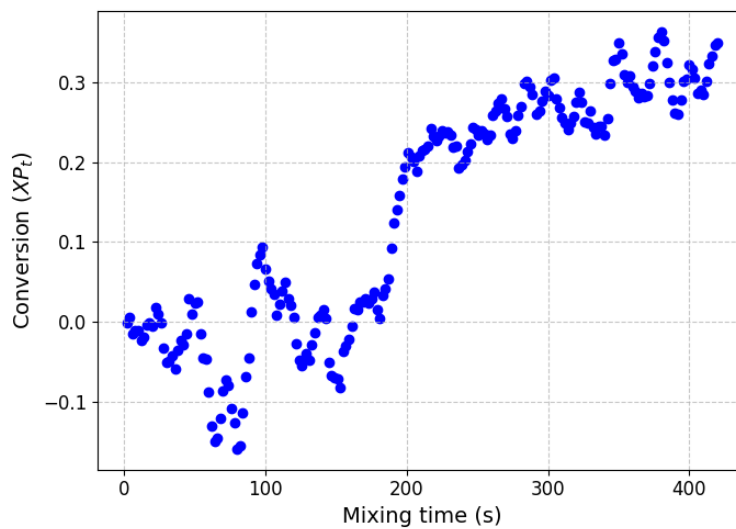


Figure B5 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs3. Experiment 2.

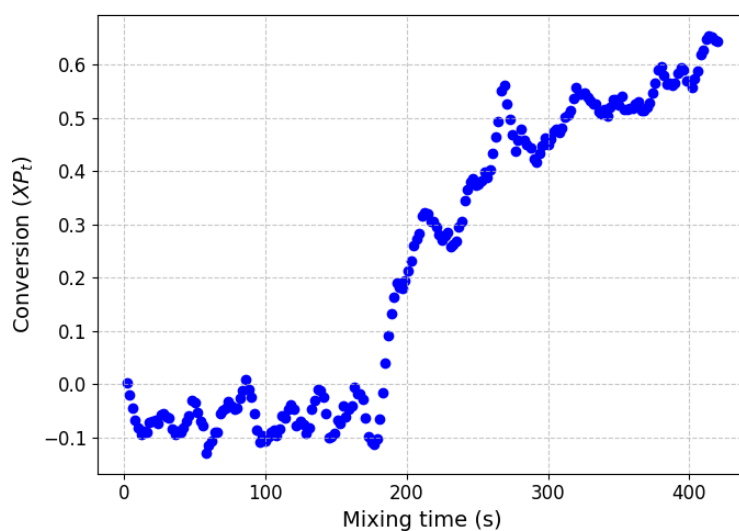


Figure B6 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs3. Experiment 3.

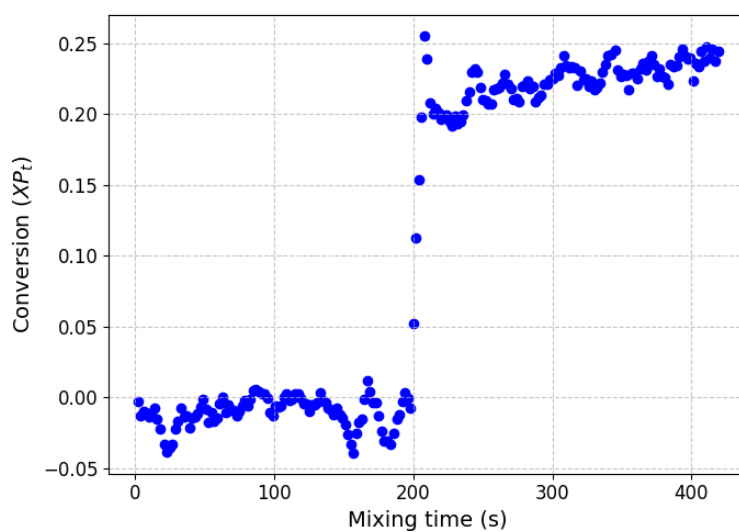


Figure B7 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs4. Experiment 1.

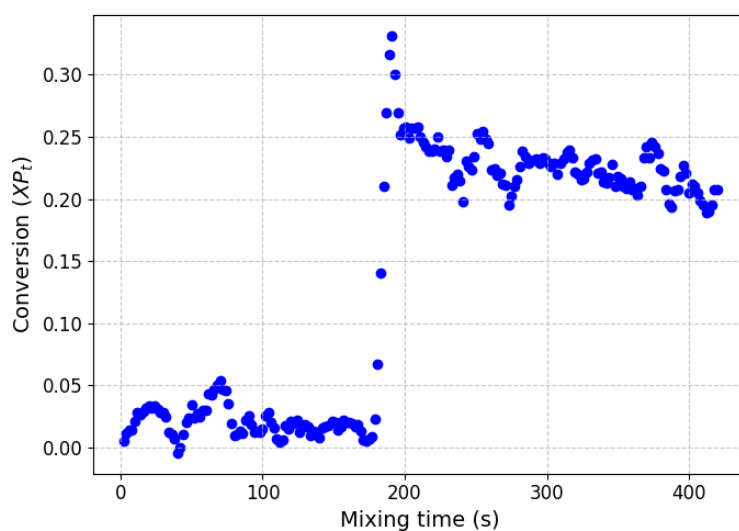


Figure B8 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMs4. Experiment 2.

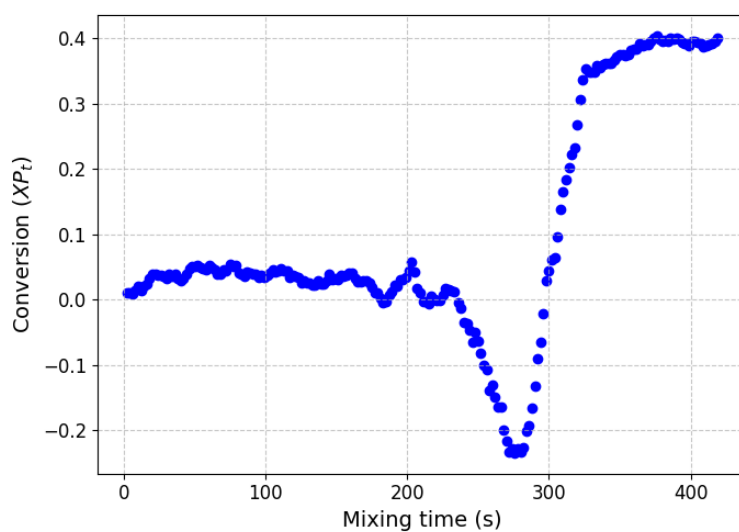


Figure B9 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb12. Experiment 1.

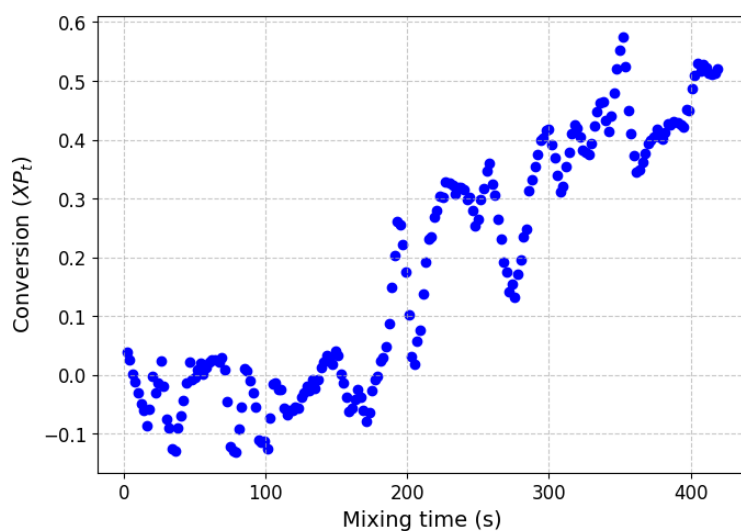


Figure B10 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb12. Experiment 2.

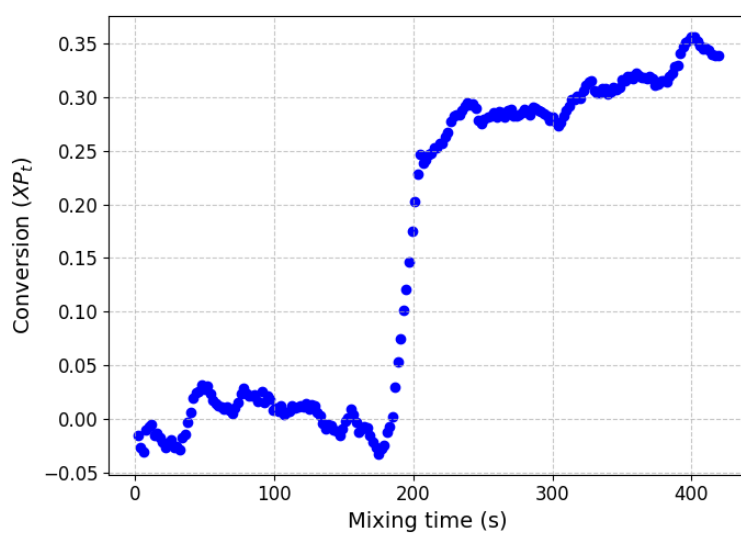


Figure B11 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb13. Experiment 1.

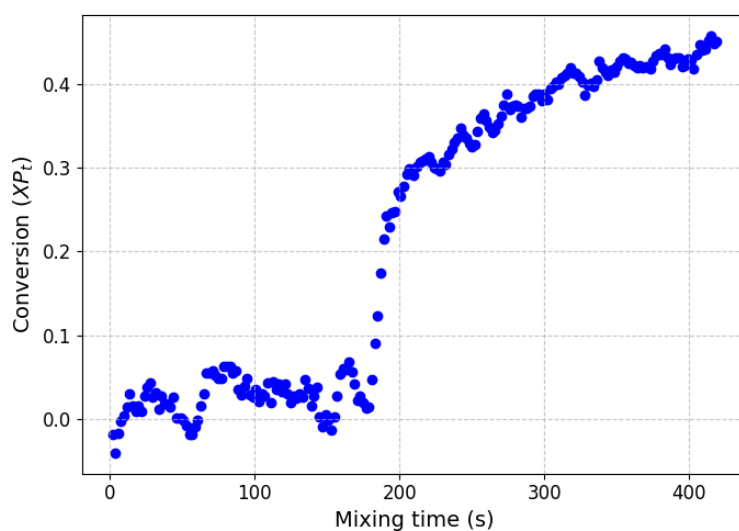


Figure B12 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb13. Experiment 2.

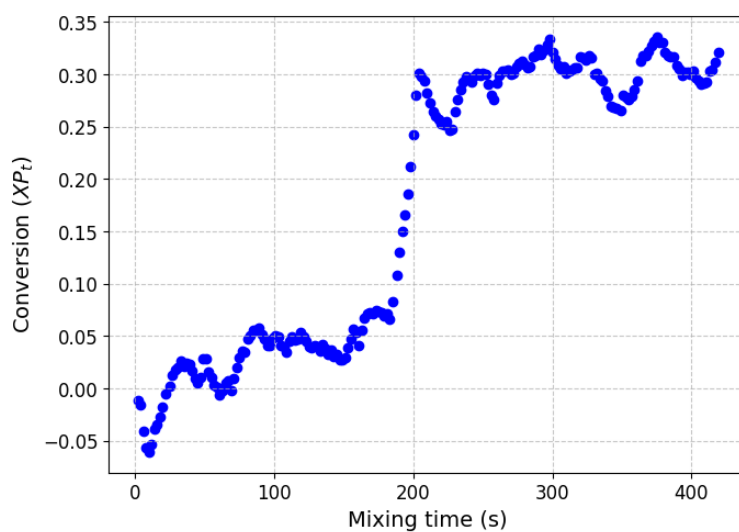


Figure B13 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb23. Experiment 1.

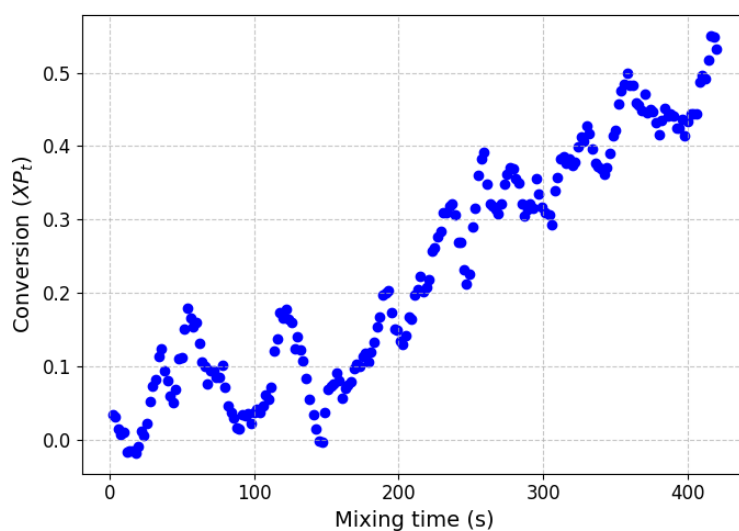


Figure B14 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb23. Experiment 2.

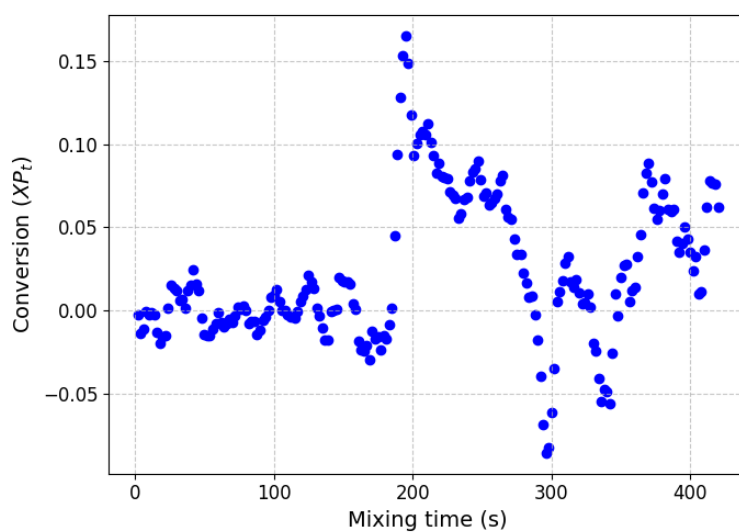


Figure B15 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb34. Experiment 1.

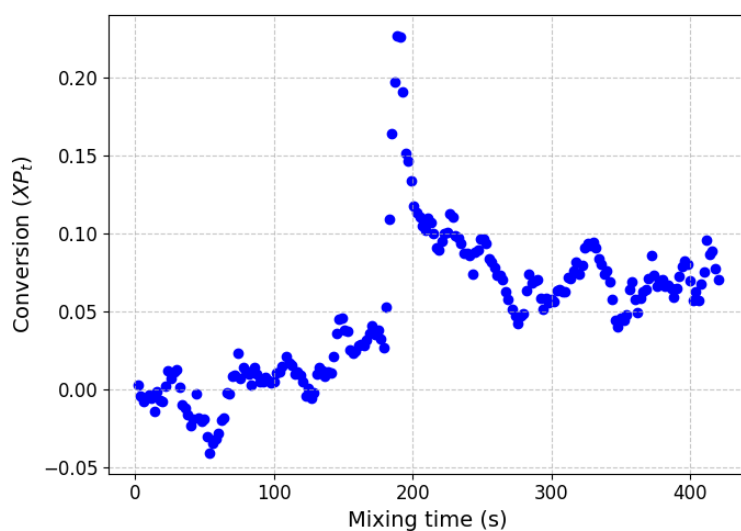


Figure B16 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMb24. Experiment 1.

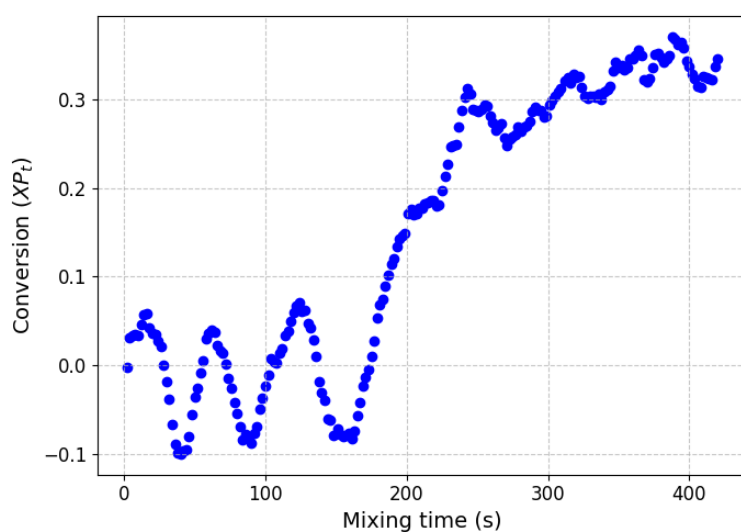


Figure B17 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMt123. Experiment 1.

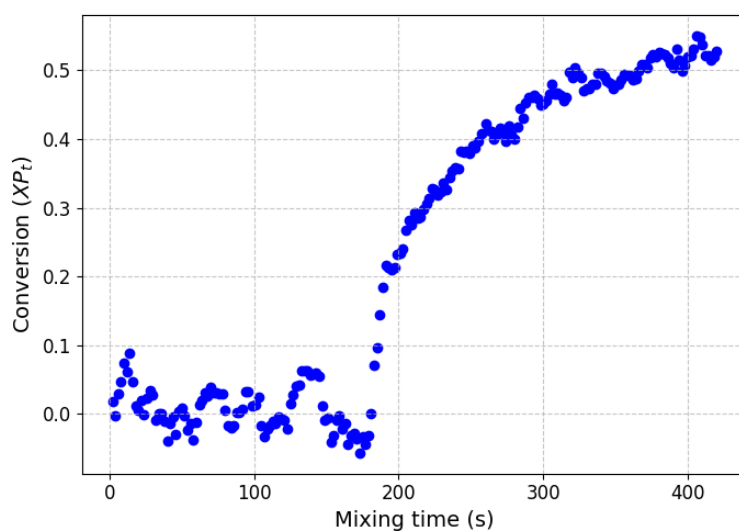


Figure B18 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMt123. Experiment 2.

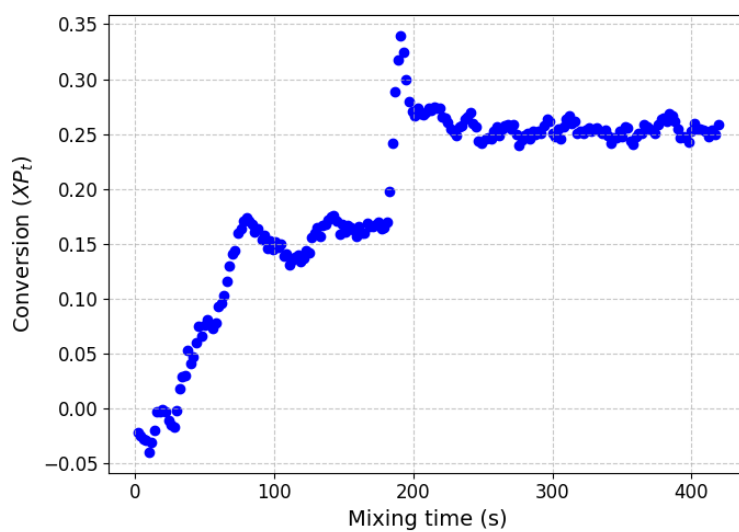


Figure B18 Dynamic analysis of individual particle conversion in the flocculation process using FBRM in macro mode, unweighted option. Kaolinite sample at 2.5 wt% mixed at 200 rpm with a pitched 4-blade impeller and 2500 ppm PAMt234. Experiment 1.

Appendix C Supplementary material to support Chapter 5.6.2

Abstract

The supplementary material presented here includes the classic approach to kinetic data treatment by the integral method, which is used to identify the kinetic pseudo-order of aggregation phenomena in kaolinite particle flocculation processes.

1.Introduction

The kinetic approach has previously been applied to evaluate reaction dynamics in coagulation-flocculation processes. Igwegbe C. et al.[1] studied the coagulation-flocculation of contaminants in the waters of Agu-Awka, Nigeria, using a coagulant derived from *Garcinia kola* seeds (GKC). The progress of the separation process was assessed through measurements of total dissolved and suspended solids (TDSP). In this study, the best kinetic fit was obtained with the second-order model, indicating that the reaction rate depends quadratically on the concentration of the species involved.

Similarly, Ovuoraye P. et al.[2] employed turbidimetric techniques (standard nephelometric test) to analyze the kinetics of coagulation-flocculation in cosmetic waste, also reporting an optimal fit to the second-order model.

In both cases, the rate law is expressed eq 1:

	$v = k[A]^2$	(1)
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where v is the reaction rate, k the kinetic constant, and $[A]$ the particle concentration. This behavior can be interpreted as a process in which two particles must collide simultaneously to form a stable aggregate. In other words, the probability of floc formation depends on the interaction between particle pairs, which explains the quadratic nature of the relationship.

The data obtained through Focused Beam Reflectance Measurement (FBRM) are employed to characterize the particle size distribution of kaolinite dispersed in an aqueous medium. This procedure enables the determination of solid sizes expressed in terms of chord length, the quantification of particles within each size range, and the analysis of their temporal variation during the flocculation process in the presence of polyacrylamide polymers (PAMs) with controlled polydispersity, synthesized via the Reversible Addition–Fragmentation Chain Transfer (RAFT) method, with a molecular weight (M_w) of 3.11×10^5 Da and a dispersity ($\mathcal{D}$) of 1.04.[3] The initial conditions establish the presence of a 2.5 wt% concentration of kaolinite, with stirring at 200 rpm. Unlike TDSP techniques and turbidimetric tests, FBRM provides information on the variation of each particle size in the flocculation process.

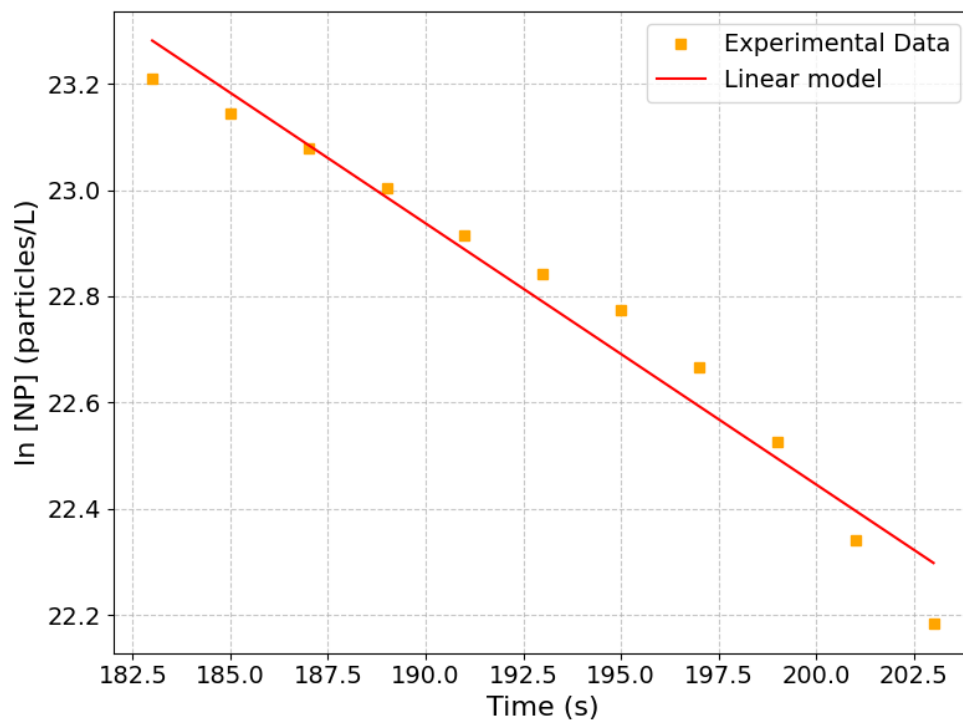
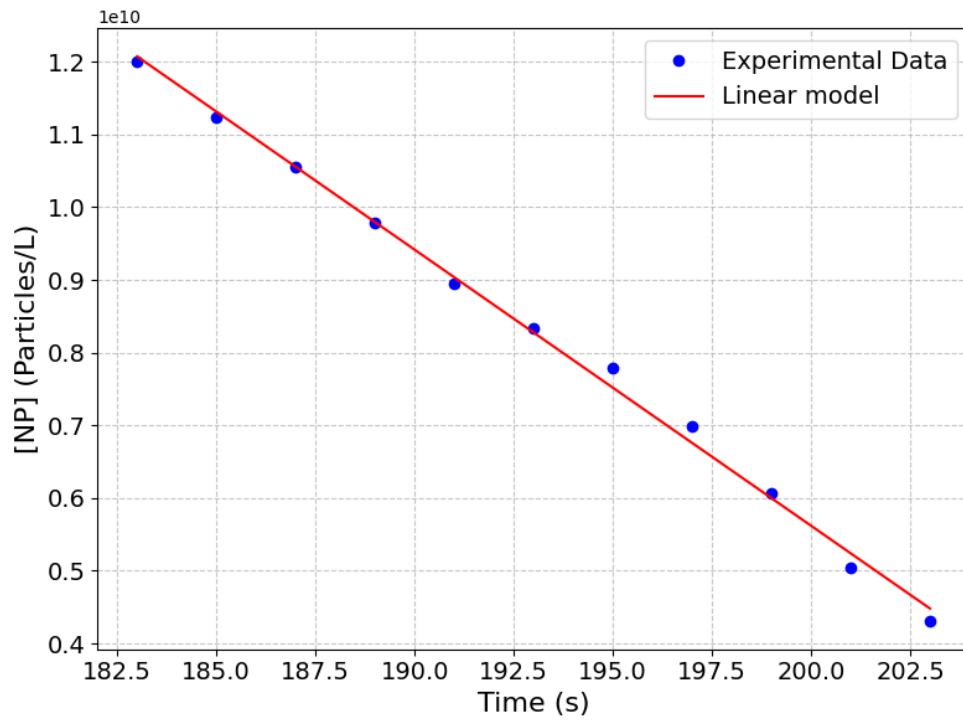
The classical approach to kinetic analysis of reactions using the integral method involves constructing the plots corresponding to the models proposed in eq 2, 3, and 4 (Table 1). The selection of the best-fitting model is carried out by comparing the linear regression coefficients (R^2) of each graphical representation. From this procedure, the determination of the reaction pseudo-order that most accurately describes the kinetics is achieved by selecting the model associated with the highest R^2 value[4].

Table 1. Models for determining the reaction order by the integral method

Orde	Model	Equation
0	$[A] = -kt + [A_0]$	(2)
1	$\ln[A] = -kt + \ln [A_0]$	(3)
2	$\frac{1}{[A]} = kt + \frac{1}{[A_0]}$	(4)

2.Results and discussion

Figure 1 presents how the integral method models are fitted to the data from channel 1 (chord length = 1.0210 μm) of the FBRM, obtained during the first 20 seconds of the real-time reaction, where approximately 40% of the particles are consumed to form aggregates in the flocculation process in the presence of PAM. As can be visually observed, the zero-order model provides a better fit than the first- and second-order models. This observation is corroborated in Table 1 by comparing the R^2 values for each reaction order. For channel 1, the R^2 indicates a 99.67% agreement of the data with the linear model deduced by the integral method, which exceeds the 96.35% obtained for first order and the 89.12% for second order. Thus, only the zero-order option remains valid for channel 1.



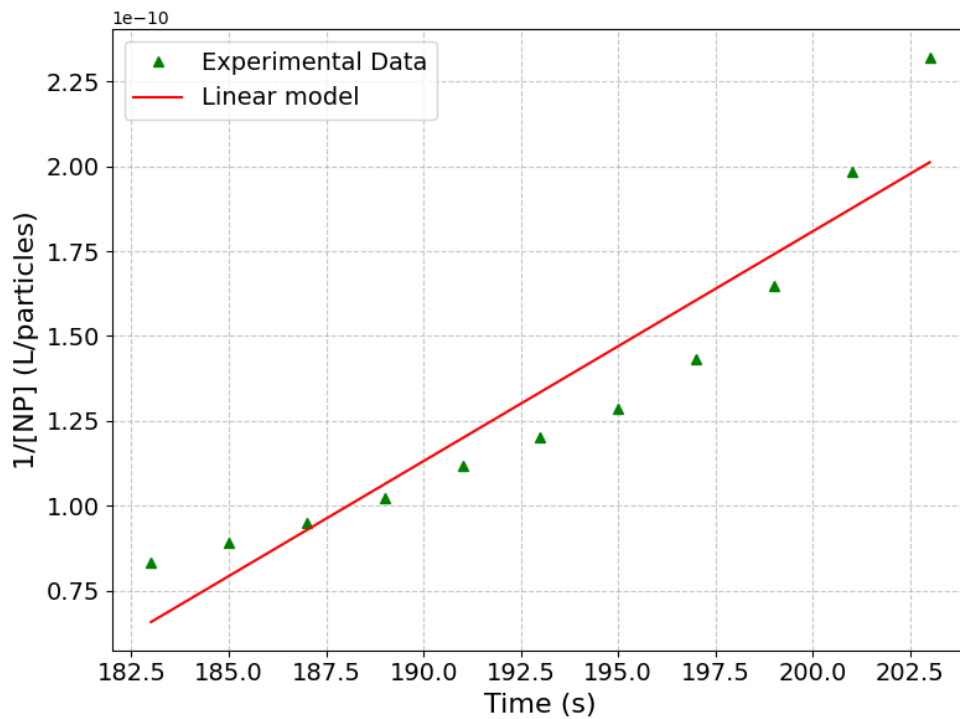


Figure 1. (a) The plot of $1/[NP]$ vs. time corresponds to a zero pseudo-order reaction; (b) the plot of $\ln[NP]$ vs. time corresponds to a first pseudo-order reaction; and (c) the plot of $\ln(1/[NP])$ vs. time corresponds to a second pseudo-order reaction. These representations are used to determine the pseudo-order of reaction in the aggregation of kaolinite particles in channel 1, based on the data obtained through FBRM.

Table 2. Results of the linear adjustments of the integral models for the kaolinite particles of channel 1 subjected to the flocculation process.

Orde	Slope	Intercept	R^2
0	-379682607.2	81556318219.2	0.9967
1	$-4.9 \cdot 10^{-2}$	32.3	0.9635
2	$-1.2 \cdot 10^{-9}$	$6.8 \cdot 10^{-12}$	0.8912

The FBRM data indicate that although the instrument is equipped with 200 channels, only the first 150 channels contain particle size information. The remaining 50 channels show no data, as the volume fraction contributions are zero, meaning that the number of particles in these channels is below the detection limit of the technique (Figure 2).

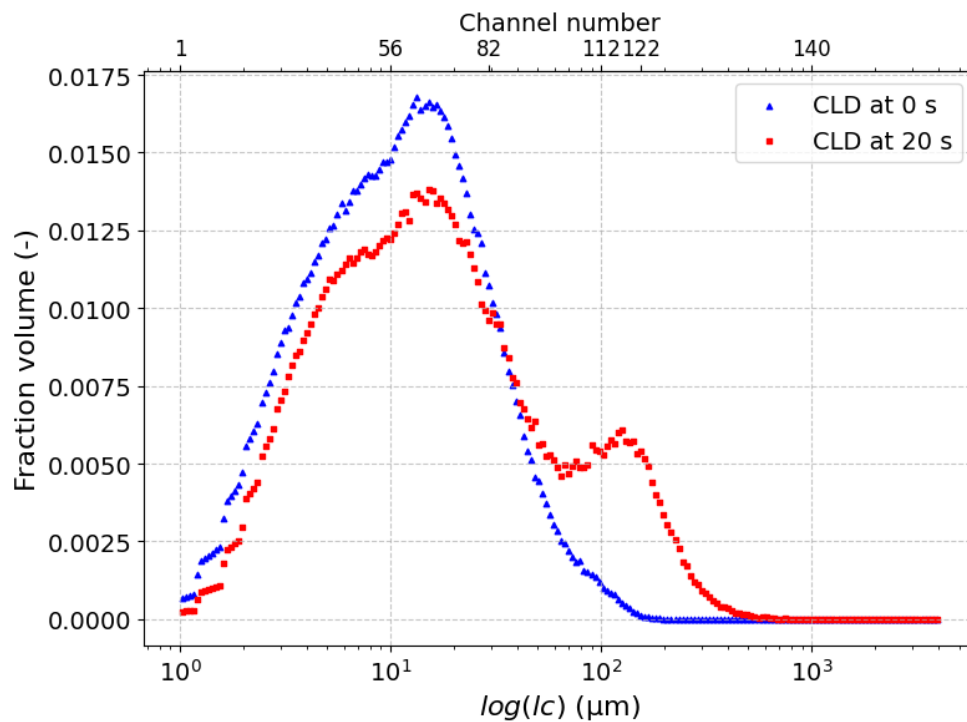


Figure 2. Kaolinite Chord length Distribution (CLD) at 0 and 20 s in Real-time reaction.

Figure 3 shows that there is no single R^2 value for all particle sizes, since each size can be considered as an individual reaction occurring in parallel. Recording particle concentration separately with the FBRM probe allows the response of individual channels to be obtained as a

function of particle size over time. In the smaller channels, the zero-pseudo-order model provides a better fit, while increasing particle size leads to fluctuations in R^2 values, possibly due to entropic effects in the system. Nevertheless, the zero-pseudo-order model remains superior to the first- and second-pseudo-order models up to approximately channel 82, where a sharp drop in R^2 values occur. The R^2 drop can be explained by Figure 2, when the particle concentrations under study are very close or show little variation within the time interval, it is not possible to distinguish between particle consumption events or the appearance of new particles.

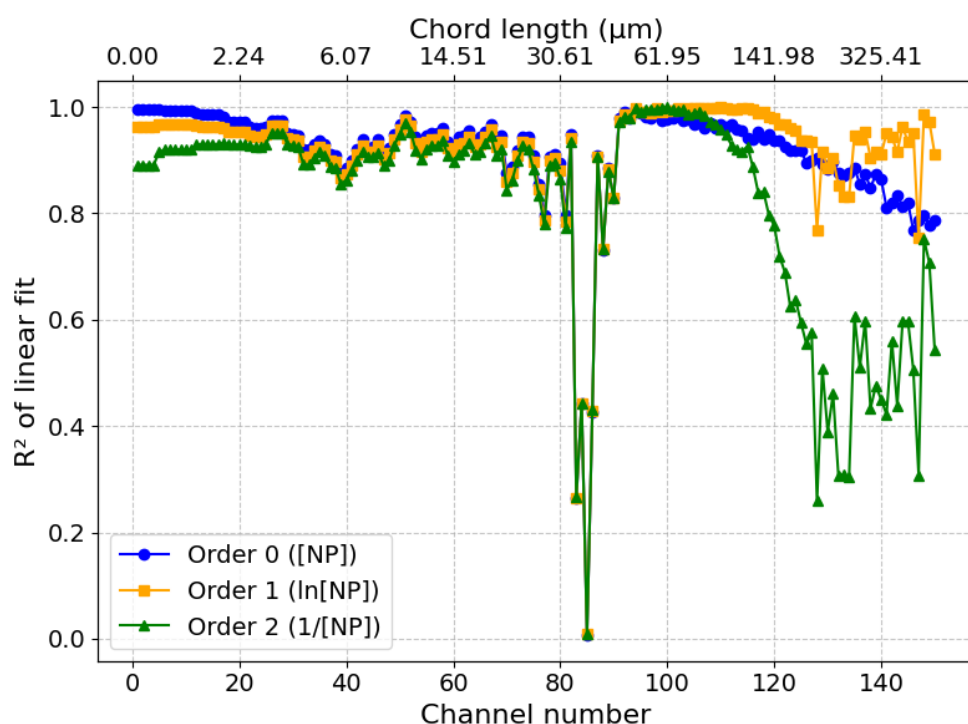


Figure 3 Comparison of R^2 values for the kinetic models derived from the integral method of zero, first, and second order.

When analyzing R^2 data in kinetic plots using the integral method, it should be remembered that values close to 1 indicate a better fit of the model to the experimental data. However, the literature shows that values such as 0.822 are also accepted to identify the most suitable model under this approach.[5]

Once the region of particle concentration overlap (channels near 82) is surpassed, the R^2 values recover and remain close to the maximum. In this range, the pseudo-second-order kinetic model outperforms the zero- and first-order models until approximately channel 108, where the first-order model begins to predominate. These variations in pseudo-order are significant because, in chemical terms, they may indicate changes in the reaction mechanism or the emergence of parallel reactions. Since the system involves kaolinite particles in the presence of PAM, and the chemical composition does not vary, it is reasonable to consider modifications in the mechanism.[6] This suggests the existence of two distinct flocculation mechanisms: below channel 80, particle consumption occurs through aggregation, whereas above channel 90, the formation of larger particles is attributed to aggregation. Such behavior is consistent with the population balance framework proposed by Jeldres, R. et al., [7] which identifies four rate processes within the PBM model.

3. Conclusions

Two distinct aggregation mechanisms were identified for kaolinite particles within the studied size interval in the presence of polyacrylamides synthesized by RAFT methods. The first mechanism corresponds to particle consumption through aggregation, while the

second involves the formation of larger aggregates, highlighting the complex flocculation behavior under these conditions.

Particle disappearance through aggregation occurs below channel 80 and follows a zero-order reaction behavior, whereas particle formation through aggregation above channel 90 complies with a first-order reaction law.

The identification of two regions of particle appearance or disappearance through aggregation enables the estimation of kinetic rates across different particle size ranges.

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